

RANDOM PROCESSES

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LECTURE 1. RANDOM PROCESSES AND MEASURES

Definition 1. ($\mathbb{R}$ -valued) random variable is a measurable function from $X : (\Omega, A, \mathbb{P}) \rightarrow \mathbb{R}$.

Definition 2. ($\mathbb{R}$ -valued) random process is a set of random variables $\{X_t : (\Omega, A, \mathbb{P}) \rightarrow \mathbb{R}\}_{t \in T}$.

We'll assume T is a metric space (that gives us natural topology and σ -algebra on it).

Let $\omega \in \Omega$. Consider $X_\omega : T \rightarrow \mathbb{R}$, $X_\omega(t) = X_t(\omega)$. This is called a version (trajectory, sample path) of the process.

Our purpose is to define a measure on the set of processes.

Start with establishing a sigma-algebra on it. Let $(t_1, t_2, \dots, t_n) \subset T$ be n distinct points in T . Let $A \subset \mathbb{R}^n$ be a Borel set. Define a cylinder $C_{t_1, \dots, t_n}(A)$ by

$$C_{t_1, \dots, t_n}(A) = \{X : T \rightarrow \mathbb{R} \mid (X(t_1), \dots, X(t_n)) \in A\}.$$

Cylinders in $\mathbb{R}^T$ form an algebra. Note that we can define operations on different cylinders as

$$C_{t_1, \dots, t_n}(A) = C_{t_1, \dots, t_n, s_1, \dots, s_m}(A \times \mathbb{R}^m)$$

and

$$C_{s_1, \dots, s_m}(B) = C_{t_1, \dots, t_n, s_1, \dots, s_m}(\mathbb{R}^n \times B).$$

Let $\{X_t : \Omega \rightarrow \mathbb{R}\}_{t \in T}$ be a random process. Define a measure μ on the algebra of cylinders:

$$\mu(C_{t_1, \dots, t_n}(A)) = \mathbb{P}((X_{t_1}(\omega), \dots, X_{t_n}(\omega)) \in A).$$

Definition 3 (equidistributed processes). Processes $\{X_t : (\Omega, A, \mathbb{P}) \rightarrow \mathbb{R}\}_{t \in T}$ and $\{Y_t : (\Omega', A', \mathbb{P}') \rightarrow \mathbb{R}\}_{t \in T}$ are equidistributed, if their finite dimensional distributions are the same.

To extend measure from the set of cylinders (from finite dimensional distributions to infinite), we need Kolmogorov extension theorem.

Example. Kolmogorov extension theorem is needed to guarantee the existence of a sequence of independent random variables, however sometimes we can construct them explicitly.

For example, the sequence of independent symmetric $\{-1, 1\}$ variables $\{X_n\}_{n=1}^\infty$ can be defined as a sequence of Rademacher functions on the space $(\Omega, A, \mathbb{P}) = ([0, 1], B, m)$ (m means Lebesgue measure).

In Hilbert space we can also explicitly construct sequence of normal variables, but in most cases these constructions are too hard.

So, let μ be a measure generated by a random process $(X_t)_{t \in T}$. For any $(t_1, \dots, t_n) \in T$ we can define a measure

$$\mu_{t_1, \dots, t_n}(A) = \mu(C_{t_1, \dots, t_n}(A)).$$

And then

$$\mu_{t_1, \dots, t_n, t_{n+1}}(A \times \mathbb{R}) = \mu_{t_1, \dots, t_n}(A)$$

(this is called *Kolmogorov consistency condition*).

Theorem 1 (Kolmogorov extension theorem). *Let T be a set and assume that for every finite set $(t_1, \dots, t_n) \in T$ we have a Borel measure $(\mu_{t_1, \dots, t_n})_{\{t_1, \dots, t_n\} \subset T}$ on $\mathbb{R}^n$ and the consistency condition above is satisfied, then the measure μ on the algebra of cylinders defined by*

$$\mu(C_{t_1, \dots, t_n}(A)) = \mu_{t_1, \dots, t_n}(A)$$

is σ -additive. (And so it can be extended to σ -algebra, generated by cylinders.)

Example. (How the theorem helps us to construct infinite sequence of independent identically distributed random variables.)

Let $T = \mathbb{N}$, ν – probability measure on $\mathbb{R}$. Set

$$\mu_{1, \dots, n}(A_1 \times A_2 \dots \times A_n) = \prod_{j=1}^n \nu(A_j).$$

Clearly, the consistency condition is hold. By the theorem, there exists a measure μ in the set of all sequences, such that

$$\mu(\{X_n\}_{n=1}^\infty | X_1 \in A_1, \dots, X_n \in A_n) = \prod_{j=1}^\infty \nu(A_j).$$

Proof. Let μ be finitely additive measure on algebra C . Assume that $A_1, \dots, A_n \in C$, $A = \coprod_{n=1}^\infty A_n$. Then

$$\begin{aligned} A &= \coprod_{n=1}^N A_n \cup \coprod_{n=N+1}^\infty A_n \\ \mu(A) &= \sum_{n=1}^N \mu(A_n) + \mu\left(\coprod_{n=N+1}^\infty A_n\right) \geq \sum_{n=1}^N \mu(A_n) \end{aligned}$$

Thus

$$\sum_{n=1}^\infty \mu(A_n) \leq \mu(A).$$

To prove the σ -additivity, we have to show the opposite inequality. It can be done similarly with the proof of σ -additivity of Lebesgue measure, defined on the intervals (there is a compact set inside A with the close measure, we can extend sets A_n a bit to make them open, then take finite subcover, apply finite additivity and check that measure was perturbed infinitely small).

LECTURE 2. KOLMOGOROV EXTENSION THEOREM

Theorem 2 (Kolmogorov extension theorem). *Let T be a set and assume that for every finite set $(t_1, \dots, t_n) \in T$ we have a Borel measure $(\mu_{t_1, \dots, t_n})_{\{t_1, \dots, t_n\} \subset T}$ on $\mathbb{R}^n$ such that Kolmogorov consistency condition*

$$\mu_{t_1, \dots, t_n, t_{n+1}}(A \times \mathbb{R}) = \mu_{t_1, \dots, t_n}(A)$$

is satisfied, then the measure μ on the algebra of cylinders defined by

$$\mu(C_{t_1, \dots, t_n}(A)) = \mu_{t_1, \dots, t_n}(A)$$

is σ -additive. (And so it can be extended to σ -algebra, generated by cylinders.)

Proof. Let μ be finitely additive measure on algebra C . Assume that $A_1, \dots, A_n \in C$, $A = \coprod_{n=1}^{\infty} A_n$. Then

$$A = \coprod_{n=1}^N A_n \cup \coprod_{n=N+1}^{\infty} A_n$$

$$\mu(A) = \sum_{n=1}^N \mu(A_n) + \mu\left(\coprod_{n=N+1}^{\infty} A_n\right) \geq \sum_{n=1}^N \mu(A_n)$$

Thus

$$\sum_{n=1}^{\infty} \mu(A_n) \leq \mu(A).$$

To prove the σ -additivity, we have to show the opposite inequality. It can be done similarly with the proof of σ -additivity of Lebesgue measure, defined on the intervals. Let

$$C_{s_1, \dots, s_n}(A) = \coprod_{m=1}^{\infty} C_{s_1^m, \dots, s_{n_m}^m}(A_m)$$

We have to show that

$$\mu(C_{s_1, \dots, s_n}(A)) \leq \sum_{m=1}^{\infty} \mu(C_{s_1^m, \dots, s_{n_m}^m}(A_m))$$

Assume the opposite

$$\mu(C_s(A)) \geq \sum_{m=1}^{\infty} \mu(C_{s_m}(A_m)) + \varepsilon$$

for some $\varepsilon > 0$.

We will introduce the product topology on $\mathbb{R}^T$. This topology is generated by the cylinders $C_{t_1, \dots, t_n}(U)$, where $U \subset \mathbb{R}^k$ is an open set (smallest topology such that coordinate functions are continuous).

$$C_s(A) = \prod_{m=1}^{\infty} (C_{s_m}(A_m))$$

There exists a compact set $K \subset A \subset \mathbb{R}^n$ such that $\mu_s(A) - \frac{\varepsilon}{4}$ (μ_s is finite, therefore regular). Similarly, for any $m \in \mathbb{N}$ there exists an open set U such that $A_m \subset U_m \subset \mathbb{R}^{n_m}$ and $\mu_{s_m}(U_m) < \mu_{s_m}(A_m) + \frac{\varepsilon}{2^{m+2}}$. Then

$$\mu_s(K) > \sum_{m=1}^{\infty} \mu_{s_m}(U_m) + \frac{\varepsilon}{4}.$$

Therefore,

$$\begin{aligned} \mu(C_s(K)) &> \sum_{m=1}^{\infty} \mu(C_{s_m}(U_m)) + \frac{\varepsilon}{4} \\ C_s(K) &\subset \bigcup_{m=1}^{\infty} C_{s_m}(U_m) \end{aligned}$$

There exists a finite subcover

$$C_s(K) \subset \bigcup_{j=1}^J C_{s_{m_j}}(U_{m_j}).$$

We have desired inequality

$$\mu(C_s(K)) \leq \sum_{j=1}^J \mu(C_{s_{m_j}}(U_{m_j})) \leq \sum_{m=1}^{\infty} \mu(C_m)(U_m).$$

But, the proof has a flaw now: K is compact in $\mathbb{R}^n$, but the cylinder $C_s(K)$ has no reason to be compact in the space $\mathbb{R}^T$. However, $C_s(K)$ is closed as the preimage of the closed set K , and $C_{s_m}(U_m)$ are open as preimages of U_m under the corresponding coordinate projections.

To deal with the flaw, consider any compactification of $\mathbb{R}$. For instance, set $\overline{\mathbb{R}} = [-\infty, \infty]$. The cylinders are subsets of $\overline{\mathbb{R}}^T$.

Now use Tychonoff compactness theorem: if $(X_t)_{t \in T}$ be a family of compact topological spaces, then the product space with the product

topology is also compact. So $\overline{\mathbb{R}}^T = \prod_{t \in T} \overline{\mathbb{R}}$ is compact, $C_s(K)$ is its closed subset, therefore compact in $\overline{\mathbb{R}}^T$. And the proof works. ■

Remark 1. *In the case, when T is countable, we prove Tychonoff compactness theorem and Kolmogorov extension theorem by elementary means using Cantor diagonal process (see Durrett).*

Remark 2. *It's enough to check the Kolmogorov consistency condition for the sets*

$$A = A_1 \times A_2 \times \dots \times A_n,$$

where $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})$ (and even for $A = (-\infty, a_1] \times \dots \times (-\infty, a_n]$).

Consider the case $n = 2$ (the other cases are treated the same way).

$$\Lambda = \{A_1 \times A_2 \mid A_1, A_2 \in \mathcal{B}(\mathbb{R})\}$$

is the semi-ring of rectangles in $\mathbb{R}^2$. Any σ -additive measure on Λ has a *unique* extension to the σ -algebra generated by Λ , which is $\mathcal{B}(\mathbb{R}^2)$. Let $t_1, t_2, t_3 \in T$. Consider measures η, ν on Λ defined by

$$\eta(A_1 \times A_2) = \mu_{t_1, t_2}(A_1 \times A_2)$$

$$\nu(A_1 \times A_2) = \mu_{t_1, t_2, t_3}(A_1 \times A_2 \times \mathbb{R})$$

Since $\eta|_{\Lambda} = \nu|_{\Lambda}$, the extensions of η and ν to $\mathcal{B}(\mathbb{R}^2)$ are the same.

Example. Let $(\Omega, \mathcal{A})$ be a measurable space, and let $M_1(\Omega)$ be the set of all probability measures on Ω . For $n \in \mathbb{N}$ let $\mu : \Omega \rightarrow M_1(\Omega)$ be a function such that $\mu_{\omega}(A)$ is a measurable function for any measurable $A \in \mathcal{A}$. Let $\mu_0 = M_1(\Omega)$ (also a probability measure).

For $A_0, A_1, \dots, A_n \in \mathcal{A}$ define

$$\mu_{0, \dots, n}(A_0 \times \dots \times A_n) = \int_{A_0} d\mu_0(\omega_0) \int_{A_1} d\mu_{\omega_0}(\omega_1) \dots \int_{A_n} d\mu_{\omega_{n-1}}(\omega_n)$$

These measures satisfy Kolmogorov consistency condition (integral by the whole line is 1). This is a model of Markov chain.

Example. Let $T = [0, +\infty)$. For $0 \leq t_1 < \dots < t_n$ let $\mu_{t_1, \dots, t_n}$ be the distribution of the n -dimensional Gaussian vector

$$\mu_{t_1, \dots, t_n}(A_1 \times \dots \times A_n) = \mathbb{P}(g_{t_1}, \dots, g_{t_n}) \in (A_1 \times \dots \times A_n),$$

such that mean $\mathbb{E}g_{t_j} = 0$ and covariance $\mathbb{E}g_{t_j}g_{t_{j+1}} = t_{j+1} - t_j$. Kolmogorov consistency condition can be also checked. This example looks very much like model for the Brownian motion, but it's not it. Indeed, all trajectories of the Brownian motion are continuous. This means that the set of continuous functions should be measurable. However, the set of continuous functions is not contained in the σ -algebra of

cylinders on the set $\mathbb{R}^{\mathbb{R}}$ (prove it!) Despite that, there exists a *modification* of this process with continuous trajectories. This idea is used in Kolmogorov's construction of the Brownian motion.

Definition 4 (modification of the process). *Let $(X_t, (\Omega, \mathcal{A}, \mathbb{P}) \rightarrow \mathbb{R})_{t \in T}$ and $(Y_t, (\Omega, \mathcal{A}, \mathbb{P}) \rightarrow \mathbb{R})_{t \in T}$ be two random processes (defined on the same probability space!). The process (Y_t) is called a modification of the process (X_t) , if*

$$\forall t \in T \quad \mathbb{P}(X_t \neq Y_t) = 0.$$

Definition 5 (indistinguishable processes). *$(X_t, (\Omega, \mathcal{A}, \mathbb{P}) \rightarrow \mathbb{R})_{t \in T}$ and $(Y_t, (\Omega, \mathcal{A}, \mathbb{P}) \rightarrow \mathbb{R})_{t \in T}$ are called indistinguishable if there exists a set $\Omega_0 \subset \Omega$, such that $P(\Omega_0) = 1$ and*

$$\forall \omega \in \Omega_0 \quad \forall t \in T \quad X_t = Y_t.$$

Note, that the second notion is stronger (for all t the sets $\{X_t \neq Y_t\}$ should coincide) (see homework 1 for the examples). But if T is a countable set, then modifications are indistinguishable.

Discrete time (strongly) stationary random processes. (We will always consider discrete time in this course.)

Definition 6 (stationary process). *A random process $(X_n)_{n \in \mathbb{N}}$ is called stationary process, if for any $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})$ and for any $m \in \mathbb{N}$,*

$$\mathbb{P}(X_0 \in A_0, \dots, X_n \in A_n) = \mathbb{P}(X_m \in A_0, \dots, X_{n+m} \in A_n).$$

Remark 3. *Instead of $n \in \mathbb{N}$ in $\{X_n\}$ it can be $n \in \{0 \cup \mathbb{N}\}$ or $n \in \mathbb{Z}$.*

Definition 7 (measure preserving transformations). *Let $(\Omega, \mathcal{A}, \mu)$ be a probability space and let $T : \Omega \rightarrow \Omega$ be a measurable function. T is called measure preserving if*

$$\forall A \in \mathcal{A} \quad \mu(T^{-1}A) = \mu(A).$$

Let $f : \Omega \rightarrow \mathbb{R}$ be a measurable function. Define a process $(X_n)_{n \in \{0\} \cup \mathbb{N}}$ by $X_n(\omega) \in f(T^n(\omega))$.

Claim 1. *$(X_n)_{n \in \{0\} \cup \mathbb{N}}$ is a stationary process.*

Next time: show that essentially measure preserving transformations are the only case of stationary processes.

LECTURE 3. MEASURE PRESERVING TRANSFORMATIONS

Recall the definitions.

Definition 8 (stationary process). *A random process $(X_n)_{n \in \mathbb{N}}$ is called stationary process, if for any $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})$ and for any $m \in \mathbb{N}$,*

$$\mathbb{P}(X_0 \in A_0, \dots, X_n \in A_n) = \mathbb{P}(X_m \in A_0, \dots, X_{n+m} \in A_n).$$

Remark 4. *Clearly, it's enough to check equality only for $m = 1$.*

Definition 9 (measure preserving transformations). *Let $(\Omega, \mathcal{A}, \mu)$ be a probability space and let $T : \Omega \rightarrow \Omega$ be a measurable function. T is called measure preserving if $\forall A \in \mathcal{A} \mu(T^{-1}A) = \mu(A)$.*

Example 1. Let $T : \Omega \rightarrow \Omega$ be a measure preserving transformation and let $f : \Omega \rightarrow \mathbb{R}$ be a measurable function. Then the random process $\{X_n\}$ defined by

$$X_n(\omega) = f(T^n \omega)$$

is a stationary process.

Obviously, it is a random process (composition of two measurable functions is measurable), and we need to check its stationarity.

Let $A_0, \dots, A_n \in \mathcal{B}(\mathbb{R})$. Then $X_1 \in A_0, \dots, X_{n+1} \in A_n$ means that

$$f(T\omega) \in A_0, \dots, f(T^{n+1}\omega) \in A_n, \text{ or}$$

$$\omega \in T^{-1}(f^{-1}(A_0)), \dots, \omega \in T^{-n-1}(f^{-1}(A_n)), \text{ or}$$

$$\omega \in \bigcap_{j=0}^n T^{-j-1}(f^{-1}(A_j)) = T^{-1}\left(\bigcap_{j=0}^n T^{-j}(f^{-1}(A_j))\right).$$

(Note that due to the definition of measure preserving transformations we deal with preimages, not images. And it's important here as we use the property $T^{-1}(A \cap B) = T^{-1}(A) \cap T^{-1}(B)$.) Then, using measure preserving condition in the second equality, we have

$$\begin{aligned} \mathbb{P}(X_1 \in A_0, \dots, X_{n+1} \in A_n) &= \mu(T^{-1}\left(\bigcap_{j=0}^n T^{-j}(f^{-1}(A_j))\right)) = \\ &= \mu\left(\bigcap_{j=0}^n T^{-j}(f^{-1}(A_j))\right) = \mathbb{P}(X_0 \in A_0, \dots, X_n \in A_n). \blacksquare \end{aligned}$$

Next theorem shows that it is essentially the only possible model of the stationary process.

Theorem 3. *Let $(X_n)_{n \in \mathbb{Z}_+}$ be a stationary random process. Then there exists a probability space $(\Omega, \mathcal{A}, \mu)$, a measure preserving transformation $T : \Omega \rightarrow \Omega$ and a measurable function $f : \Omega \rightarrow \mathbb{R}$ such that the processes $(X_n)_{n \in \mathbb{Z}_+}$ and $(f(T^n \omega))_{n \in \mathbb{Z}_+}$ are equidistributed.*

Proof. We need to construct a new process, let's do this using Kolmogorov extension theorem. So initially we need to define a sequence of measures: for $n \in \mathbb{Z}_+$ let $\mu_{0,\dots,n}$ be

$$\mu_{0,\dots,n}(A_0 \times \dots \times A_n) = \mathbb{P}(X_0 \in A_0, \dots, X_n \in A_n)$$

for $\forall A_0, \dots, A_n \in \mathcal{B}(\mathbb{R})$.

Note that consistency condition is automatically satisfied, so by Kolmogorov extension theorem there exists a measure μ on $\mathbb{R}^{\mathbb{Z}}$ equipped with the product σ -algebra, such that $\forall A_0, \dots, A_n \in \mathcal{B}(\mathbb{R})$

$$\mu(C_{0,\dots,n}(A_0 \times \dots \times A_n)) = \mu_{0,\dots,n}(A_0 \times \dots \times A_n).$$

Recall the definition of a cylinder:

$$C_{0,\dots,n}(A_0 \times \dots \times A_n) = \{\{\omega_j\}_{j \in \mathbb{Z}_+} : \omega_0 \in A_0, \dots, \omega_n \in A_n\}.$$

So, we have

$$\mathbb{P}(X_0 \in A_0, \dots, X_n \in A_n) = \mu(\{\omega : \omega_0 \in A_0, \dots, \omega_n \in A_n\}).$$

Now set $f(\{\omega_j\}_{j \in \mathbb{Z}_+}) = \omega_0$ and $T : \mathbb{R}^{\mathbb{Z}_+} \rightarrow \mathbb{R}^{\mathbb{Z}_+}$ to be a shift

$$T((\omega_0, \omega_1, \omega_2, \dots)) = (\omega_1, \omega_2, \dots)$$

and $(X_0, \dots, X_n)$ will be distributed like $(f(\omega), f(T(\omega)), \dots, f(T^n(\omega)))$.

The only thing that remains to check is that T is measure preserving, i.e.

$$\mu(T^{-1}A) = \mu(A)$$

for $A \in \mathcal{A}$. Let $A_0, \dots, A_n \in \mathcal{B}(\mathbb{R})$. For a cylinder $C_{0,\dots,n}(A_0 \times \dots \times A_n)$ this is exactly a stationary condition, since

$$C_{0,\dots,n}(A_0 \times \dots \times A_n) = \{(\omega_j)_{j \in \mathbb{Z}_+} : \omega_0 \in A_0, \dots, \omega_n \in A_n\}$$

and

$$T^{-1}(A) = \{(\omega_j)_{j \in \mathbb{Z}_+} : \omega_1 \in A_0, \dots, \omega_{n+1} \in A_n\}.$$

But the whole σ -algebra is generated by the cylinders, so the claim is true for all $A \in \mathcal{A}$. ■

Theorem 4. *Let $(X_n)_{n \in \mathbb{Z}_+}$ be a stationary random process. Then there exists a stationary random process $(Y_n)_{n \in \mathbb{Z}}$ such that $(X_n)_{n \in \mathbb{Z}_+}$ and $(Y_n)_{n \in \mathbb{Z}_+}$ are equidistributed. (For every stationary random process with only non-negative times there exists a stationary extension to the "past".)*

The proof essentially repeats the proof of the previous theorem. For any $k \in \mathbb{Z}$ define a Borel measure $\mu_{k,k+1,\dots,k+n}$ by

$$\mu_{k,k+1,\dots,k+n}(A_0 \times \dots \times A_n) = \mathbb{P}(X_0 \in A_0, \dots, X_n \in A_n)$$

for $A_0 \dots A_n \in \mathcal{B}(\mathbb{R})$ and repeat the proof of the previous theorem.

Example 2. Let $(X_n)_{n \in \mathbb{Z}_+}$ be i.i.d. random variables. Then $(X_n)_{n \in \mathbb{Z}_+}$ is a stationary random process. In other words, the shift

$$T((\omega_0, \omega_1, \omega_2 \dots)) = (\omega_1, \omega_2, \dots)$$

is a measure preserving transformation on $\mathbb{R}^{\mathbb{Z}}$ equipped with the product measure

$$\mu = \prod_{j=0}^{\infty} \mu_j,$$

where μ_j is the distribution of X_j .

Example 3. (Rotation of the circle.)

Let $\Omega = \mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$. Let $\theta \in \mathbb{T}$. Let $T_\theta z = \theta \cdot z$.

Measure is the normalized measure on the circle (Haar measure), which is rotation invariant by definition, so transformation is measure preserving.

We can also define it on $\Omega' = \mathbb{R}/2\pi\mathbb{Z}$ by

$$T'_\tau x = x + \tau \mod 2\pi$$

This is, of course, different representations of the same thing: for the function $\varphi : \Omega' \rightarrow \Omega$, $\varphi(x) = e^{ix}$ we have a commutative diagram

$$\begin{array}{ccc} \Omega' & \xrightarrow{T'_\tau} & \Omega' \\ \varphi \downarrow & & \downarrow \varphi \\ \Omega & \xrightarrow{T'_{e^{i\tau}}} & \Omega \end{array}$$

Example 4. (Doubling in a circle.) Let $\Omega' = \mathbb{R}/2\pi\mathbb{Z}$.

$$Tx = 2x \mod 2\pi.$$

Easy to check that it is also a measure preserving transformation (and we can take any integer instead of 2.)

Definition 10 (ergodic transformation). *A measure preserving transformation T on the space $(\Omega, \mathcal{A}, \mu)$ is called ergodic if all invariant sets are almost trivial, namely,*

$$\forall A \in \mathcal{A} \text{ such that } A = T^{-1}A \text{ has a measure } \mu(A) \in \{0, 1\}.$$

Let's look at our previous examples.

1) Let T be a shift on $\mathbb{R}^{\mathbb{Z}_+}$ equipped by a product measure. Let A be an invariant set. Then $T^{-n}A = A$. Let $n \in \mathbb{N}$ and $\mathcal{A}_n$ is a sub

σ -algebra of $\mathcal{A}$ generated by the cylinders

$$C_{n,n+1,\dots,n+m}(A_m), \quad A_m \in \mathcal{B}(\mathbb{R}^{m+1})$$

Obviously, $\mathcal{A} = \mathcal{A}_0 \supset \mathcal{A}_1 \supset \mathcal{A}_2 \dots$ and invariant set lies in the intersection of all this σ -algebras, $A \in \bigcap_{n=0}^{\infty} \mathcal{A}_n$. According to Kolmogorov's zero-one law, this intersection is a trivial σ -algebra, so $\mu(A) = \{0, 1\}$.

2) Rotation of the circle is obviously not ergodic if it's periodic. But if τ is irrational rotation angle (i.e. $\tau \notin 2\pi\mathbb{Q}$), then T'_τ is ergodic. Let's prove this using Fourier analysis. Let $A \subset \mathbb{R}/2\pi\mathbb{Z}$, then

$$A = A - \tau \pmod{2\pi},$$

and so for any $n \in \mathbb{Z}$

$$\begin{aligned} \hat{1}_A(n) &= \hat{1}_{A-\tau}(n) = e^{-in\tau} \hat{1}_A(n), \\ (1 - e^{in\tau}) \hat{1}_A(n) &= 0. \end{aligned}$$

So for all $n \in \mathbb{Z} \setminus \{0\}$ $\hat{1}_A(n) = 0$ and the only possibly non-zero coefficient is with $n = 0$. Then Fourier series of the indicator function has only one term:

$$\mathbb{1}_A(x) = \sum_{n \in \mathbb{Z}} \hat{1}_A(n) e^{inx} = \hat{1}_A(0) \cdot e^{i0x} = \text{const.}$$

This is the equality of two L^2 -functions, i.e. indicator of the set A is constant almost everywhere, hence A has full or zero measure.

Next time we will discuss doubling in the circle example.

LECTURE 4. ERGODIC TRANSFORMATIONS

Recall the definition of the ergodic transformation.

Definition 11 (ergodic transformation). *A measure preserving transformation T on the space $(\Omega, \mathcal{A}, \mu)$ is called ergodic if all invariant sets are almost trivial, namely,*

$$\forall A \in \mathcal{A} \text{ such that } A = T^{-1}A \text{ has a measure } \mu(A) \in \{0, 1\}.$$

Example. Let's prove that doubling in the circle ($\Omega = [0, 1]$, $T\omega = 2\omega \pmod{1}$) is ergodic. For our convenience consider $\tilde{\Omega} = \mathbb{R}/(2\pi\mathbb{Z})$ instead of Ω .

Let $A \subset \tilde{\Omega}$ be invariant set, then $\forall n \in \mathbb{Z}$

$$\begin{aligned} \hat{1}_A(n) &= \hat{1}_{T^{-1}A}(n) = \int_0^{2\pi} \mathbb{1}_{T^{-1}A}(x) e^{-inx} dm(x) = \int_0^{2\pi} \mathbb{1}_A(Tx) e^{-inx} dm(x) = \\ &= (\text{substitution } y = Tx = 2x) = \frac{1}{2} \int_0^{4\pi} \mathbb{1}_A(y) e^{-iny/2} dm(y) \end{aligned}$$

Hence,

$$\hat{\mathbb{1}}_{T^{-1}A}(n) = \begin{cases} \hat{\mathbb{1}}_{T^{-1}A}(\frac{n}{2}) & \text{if } n \text{ is even;} \\ 0 & \text{if } n \text{ is odd.} \end{cases}$$

Thus, $\int_0^{2\pi} \mathbb{1}_{T^{-1}A}(x)e^{-inx}dm(x) = 0$ wherever $n \neq 0$. End of the proof is the same as in the example about rotation of the circle.

This example can be also done without Fourier analysis. (The second way will also provides us with the representation of the doubling which will be important in future.)

Let $\Lambda = \{0, 1\}^{\mathbb{N}}$ be a Cantor group and measure $\mu = \prod_{n=1}^{\infty} \mu_n$, where μ_n is a sequence of Bernulli measures

$$\mu_n(\{0\}) = \mu_n(\{1\}) = \frac{1}{2}.$$

Define transformation $S : \Lambda \rightarrow \Lambda$ by $S(\varepsilon_1, \varepsilon_2, \dots) = (\varepsilon_2, \varepsilon_3, \dots)$. And let $P : \Lambda \rightarrow \mathbb{R}/\mathbb{Z}$ be

$$P(\varepsilon_1, \varepsilon_2, \dots) = \sum_{n=1}^{\infty} \frac{\varepsilon_n}{2^n}.$$

Then $P\mu = \lambda$, push forward measure is exactly a Lebesgue measure on the segment. (We can check it in the following way: it coincides with the Lebesgue measure on binary intervals:

$$\lambda([0, 1/2]) = \mu(P^{-1}([0, 1/2])) = 1/2;$$

$$\lambda([1/4, 1/2]) = \mu(P^{-1}([1/4, 1/2])) = 1/4;$$

binary intervals generate all Borel sets, so measures coincide everywhere.)

Also, $P \circ S(\varepsilon_1, \dots) = P(\varepsilon_2, \dots) = \sum_{n=1}^{\infty} \frac{\varepsilon_{n+1}}{2^n} = 2P(\varepsilon_1, \dots) \pmod{1}$.

So, we have a commutative diagram

$$\begin{array}{ccc} \Lambda & \xrightarrow{S} & \Lambda \\ P \downarrow & & \downarrow P \\ \mathbb{R}/\mathbb{Z} & \xrightarrow{T} & \mathbb{R}/\mathbb{Z} \end{array}$$

Let $A \in \mathcal{B}(\mathbb{R}/\mathbb{Z})$ be such that $T^{-1}A = A$. Then

$$P^{-1}A = P^{-1}T^{-1}A = S^{-1}P^{-1}A.$$

Hence, $\mu(P^{-1}A) \in \{0, 1\}$ (S is ergodic). Therefore $\lambda(A) \in \{0, 1\}$. So T is ergodic.

Notion of ergodicity have come to math from physics, where ergodic transformation is essentially the transformation such that its mean by

space is equal to its mean by time. The following theorem formalizes this property.

Theorem 5 (Birkhoff-Khinchin ergodic theorem). *Let $(\Omega, \mathcal{A}, \mu)$ be a probability space, and let $T : \Omega \rightarrow \Omega$ be a measure preserving transformation. Let $f \in L_1(\mu)$. Set*

$$S_n f(\omega) = \frac{1}{n+1} \sum_{j=0}^n f(T^j \omega)$$

Then

- (1) $\{S_n f\}_{n=1}^\infty$ converges a.s.
- (2) $\{S_n f\}_{n=1}^\infty$ converges in $L_1(\mu)$
- (3) $\bar{f} = \bar{f} \circ T$
- (4) $\int f d\mu = \int \bar{f} d\mu$

where $\bar{f} = \lim_{n \rightarrow \infty} S_n f$.

There are several different proofs of the theorem. The first one was due Riesz, the most popular in the books due to Garsia (1972) (it's based on the "trick" so it is the shortest one). Also there are proofs of interest due to Wiener, von Neumann and Hartman (1948).

We will consider in class only Garsia's proof. And we need to start with the following theorem (note that it's almost the statement of the Hardy-Littlewood principle).

Theorem 6 (Wiener's maximal theorem). *We have a probability space $(\Omega, \mathcal{A}, \mu)$, measure preserving transformation $T : \Omega \rightarrow \Omega$ and a function $f \in L_1(\mu)$. Denote*

$$S_n f(x) = \frac{1}{n+1} \sum_{j=0}^n f(T^j \omega)$$

and

$$M_n f = \max(S_0 f, \dots, S_n f).$$

Then for all $t \geq 0$

$$(1) \quad t\mu(\{\omega : M_n f(\omega) \geq t\}) \leq \int_{\Omega} f d\mu.$$

Proof. Denote $E_t(f) = \{\omega : M_n f(\omega) \geq t\}$. We will prove that

$$t \cdot \mu(E_t(f)) \leq \int_{E_t(f)} f d\mu.$$

Let $g = f - t \cdot \mathbb{1}_{E_t(f)}$. Then inequality is equivalent to

$$\int_{E_0(g)} g d\mu \geq 0.$$

Note that

$$E_0(g) = \{\omega : \max_{k \leq n} \frac{1}{k+1} \sum_{j=0}^k g(T^j \omega) \geq 0\} = \{\omega : \max_{k \leq n} V_k g(\omega) \geq 0\},$$

where $V_k g(\omega) = \sum_{j=0}^k g(T^j \omega)$.

With a slight abuse of notation, denote $Tg := g \circ T$. For $\overline{M}_n g = \max_{k \leq n} V_k g$, we have

$$\begin{aligned} \overline{M}_{n-1} g &\leq \overline{M}_n g = \max(g, \max_{k \leq n-1} (g + V_k Tg)) = \\ &= \max(g, g + \overline{M}_{n-1} Tg) = g + (\overline{M}_{n-1} Tg)_+ \end{aligned}$$

Therefore,

$$\begin{aligned} \int_{E_0(g)} g d\mu &= \int_{E_0(g)} \overline{M}_n g d\mu - \int_{E_0(g)} (\overline{M}_{n-1} Tg)_+ d\mu \geq \\ &\geq \int_{\Omega} (\overline{M}_n g)_+ d\mu - \int_{\Omega} (\overline{M}_{n-1} Tg)_+ d\mu = \int_{\Omega} (\overline{M}_n g)_+ - \int_{\Omega} (T \overline{M}_{n-1} g)_+ d\mu = \\ &= \int_{\Omega} (\overline{M}_n g)_+ d\mu - \int_{\Omega} T(\overline{M}_{n-1} g)_+ d\mu = \int_{\Omega} (\overline{M}_n g)_+ d\mu - \int_{\Omega} (\overline{M}_{n-1} g)_+ d\mu \geq 0 \end{aligned}$$

(In the last equality we use measure preserving property of T .) $\square$

Corollary 1. *In the notation of Wiener's theorem, set*

$$Mf = \sup_{n \in \mathbb{N}} S_n f \quad \text{and let} \quad E_t(f) = \{\omega : Mf(\omega) \geq t\}.$$

Then $\forall t \geq 0$,

$$t\mu(E_t(f)) \leq \int_{E_t(f)} f d\mu$$

Proof. Note that $E_t(f) = \cup_{n=1}^{\infty} E_t^n(f)$, where

$$E_t^n(f) = \{\omega : M_n f(\omega) \geq t\}.$$

The sets $E_t^n(f)$ form an increasing sequence. Hence,

$$\mu(E_t(f)) = \lim_{n \rightarrow \infty} \mu(E_t^n(f)),$$

and since $f \in L_1(\mu)$, the Lebesgue dominated convergence theorem implies

$$\int_{E_t(f)} f d\mu = \lim_{n \rightarrow \infty} \int_{E_t^n(f)} f d\mu.$$

Inequality (1) completes the proof. $\square$

This Corollary is often called the Wiener maximal inequality.

Proof of a.s. convergence in Birkhoff theorem.

Let $\alpha \in \mathbb{R}$, $\varepsilon > 0$ and

$$F_{\alpha,\varepsilon}(f) = \{\omega : \liminf S_n f \leq \alpha - \varepsilon < \alpha + \varepsilon \leq \limsup S_n f\}.$$

It's enough to show that $\mu(F_{\alpha,\varepsilon}(f)) = 0$ for all α, ε . (Then consider a sequence of $\varepsilon_i \rightarrow 0$ and establish that a.s. $\liminf$ and $\limsup$ are coincide).

Assume that

$$\mu(F_{\alpha,\varepsilon}(f)) > 0.$$

Consider a shift $g = f - \alpha$, then

$$E := F_{0,\varepsilon}(g) = F_{\alpha,\varepsilon}(f) = \{\omega : \liminf S_n g < -\varepsilon < \varepsilon < \limsup S_n g\}$$

Then $\mu(E) > 0$. Note that $\omega \in E$ if and only if $T\omega \in E$ (limit does not depend on the first term). Consider natural restriction of the probability space to E

$$(E, \mathcal{A}|_E, \mu' = \frac{\mu}{\mu(E)}).$$

The function T is a measure preserving transformation on E . $\forall \omega \in E$

$$Mg(\omega) \geq \limsup S_n g(\omega) \geq \varepsilon.$$

By the Wiener Maximal theorem

$$\varepsilon \mu'(E) \leq \int_E g d\mu'$$

and $\mu'(E) = 1$ by definition. Similarly,

$$M(-g)(\omega) \geq \limsup S_n(-g)(\omega) = -\liminf S_n g(\omega) \geq \varepsilon,$$

$$\varepsilon \mu'(E) \leq \int_E (-g) d\mu'$$

hence,

$$\int_E g d\mu' \leq -\varepsilon.$$

This contradiction shows that $\mu(E) = 0$. Almost sure convergence is proven.

LECTURE 5. BIRKHOFF-KHINCHIN ERGODIC THEOREM

Theorem 7 (Birkhoff-Khinchin ergodic theorem). *Let $(\Omega, \mathcal{A}, \mu)$ be a probability space, and let $T : \Omega \rightarrow \Omega$ be a measure preserving transformation. Let $f \in L_1(\mu)$. Set*

$$S_n f(x) = \frac{1}{n+1} \sum_{j=0}^n f(T^j \omega)$$

Then

- (1) $\{S_n f\}_{n=1}^\infty$ converges a.s.
- (2) $\{S_n f\}_{n=1}^\infty$ converges in $L_1(\mu)$
- (3) $\bar{f} = f \circ T$
- (4) $\int f d\mu = \int \bar{f} d\mu$

where $\bar{f} = \lim_{n \rightarrow \infty} S_n f$.

Last time we've proven the first part (a.s. convergence). Now our aim is to derive L_1 -convergence. Standard way to do this is to find a dominating function, but it's not the case now. However, for probability spaces we have a criterion of L_1 -convergence given a.s. convergence: we need additional condition of uniform integrability.

Definition 12 (uniformly integrable set of functions). *A set $F : \Omega \rightarrow \mathbb{R}$ of measurable functions is called uniformly integrable if $\forall \varepsilon > 0 \exists M > 0 \forall f \in F$*

$$\int |f(\omega)| \cdot \mathbb{1}_{[M, +\infty)}(|f(\omega)|) d\mu < \varepsilon.$$

Example. Let $F = \{f : \Omega \rightarrow \mathbb{R} : |f| \leq g\}$, where $g \in L_1(\mu)$. Then this one element set F is uniformly integrable. Indeed,

$$\begin{aligned} & \int |f(\omega)| \cdot \mathbb{1}_{[M, +\infty)}(|f(\omega)|) d\mu(\omega) \leq \\ & \leq \int |g(\omega)| \cdot \mathbb{1}_{[M, +\infty)}(|g(\omega)|) d\mu(\omega) \rightarrow_{M \rightarrow \infty} 0. \end{aligned}$$

Theorem 8. *Let $\{f_n : \Omega \rightarrow \mathbb{R}\}_{n=1}^\infty$ be a sequence of $L_1(\mu)$ functions, such that $f_n \rightarrow f$ a.s. The following are equivalent:*

- (1) $\{f_n\}_{n=1}^\infty$ is uniformly integrable
- (2) $f_n \rightarrow f$ in $L_1(\mu)$
- (3) $\|f_n\|_1 \rightarrow \|f\|_1$.

Proof. (1) $\Rightarrow$ (2).

Let $\varepsilon > 0$. Choose M , such that $\forall n \in \mathbb{N}$

$$\int |f_n| \cdot \mathbb{1}_{[M, +\infty)}(|f_n|) d\mu(\omega) < \varepsilon,$$

$$\int |f| \cdot \mathbb{1}_{[M, +\infty)}(|f|) d\mu(\omega) < \varepsilon.$$

Set $g_n = \varphi(f_n)$, where

$$\varphi_n(x) = \begin{cases} M & \text{if } x \geq M; \\ x & \text{if } -M \leq x \leq M; \\ -M & \text{if } x \leq -M \end{cases}$$

Also set $h_n = f_n - g_n$, $g = \varphi(f)$, $h = f - g$.

(Note that we cannot set g_n to be a simple truncation $g_n = f_n \cdot \mathbb{1}_{(-M, M)}(f_n)$. Because for noncontinuous truncations we even do not know a.s. convergence to g , and therefore cannot apply dominated convergence theorem. Consider the counterexample $f_n = (1 - 1/n)M \cdot \mathbb{1}$. In this case $f_n \rightarrow M \cdot \mathbb{1}$, $g_n = f_n$, $g \equiv 0$.)

Then by triangle inequality

$$\|f_n - f\| \leq \|g_n - g\| + \|h_n\|_1 + \|h\|_1 \leq \|g_n - g\|_1 + 2\varepsilon$$

Therefore

$$\limsup \|f - f_n\| \leq \limsup \|g_n - g\|_1 + 2\varepsilon,$$

$g_n \rightarrow g$ in L_1 by dominated convergence theorem (they are bounded), so L_1 convergence for f_n is proven.

(2) $\Rightarrow$ (1)

Choose ψ as follows:

$$\psi_n(x) = \begin{cases} 0 & \text{if } x \leq -M - 1; \\ \text{linear function decreasing from 0 to } -M & \text{if } -M - 1 \leq x \leq -M; \\ x & \text{if } -M \leq x \leq M; \\ \text{linear function decreasing from } M \text{ to } 0 & \text{if } M \leq x \leq M + 1; \\ 0 & \text{if } x \geq M + 1 \end{cases}$$

Let $M > 0$, set $g_n = \psi(f_n)$, $h_n = f_n - g_n$, $g = \psi(f)$, $h = f - g$ (note that $h = 0$ a.s.). Then

$$\begin{aligned} \int |h_n| d\mu - \int |h| d\mu &\leq \int |h_n - h| d\mu \leq \int |h_n + g_n - h - g| d\mu + \int |g_n - g| d\mu = \\ &= \int |f_n - f| d\mu + \int |g_n - g| d\mu \rightarrow 0, \end{aligned}$$

as f_n are converging in L_1 and g_n are bounded and $g_n \rightarrow g$ a.s., therefore also converging in L_1 .

Hence,

$$\int |f_n| d\mu \cdot \mathbb{1}_{[M+1, \infty)}(|f_n|) d\mu \leq \int |h_n| d\mu \leq$$

$$\leq \int |h| d\mu + \int |f_n - f| d\mu + \int |g_n - g| d\mu \rightarrow 0.$$

(2) $\Rightarrow$ (3) is trivial.

(3) $\Rightarrow$ (2) Define $g_n = |f_n| + |f| - |f_n - f|$. Note that g_n are non-negative, therefore by Fatou's lemma

$$\liminf \int g_n d\mu \geq \int \liminf g_n d\mu$$

And we know that $\int |f_n - f| = 0$ as function under integral is equal to zero a.s, so inequality is equivalent to

$$\liminf \left(\int |f_n| d\mu + \int |f| d\mu - \int |f_n - f| d\mu \right) \geq \int |f_n| d\mu + \int |f| d\mu,$$

or (using the convergence of L_1 norms of f_n to norm of f)

$$2 \int |f| d\mu - \limsup \int |f_n - f| d\mu \geq 2 \int |f| d\mu,$$

$$\Updownarrow$$

$$\limsup \int |f_n - f| \leq 0,$$

$$\Updownarrow$$

$$f_n \rightarrow_{L_1} f.$$

Theorem 2 is proven. $\square$

Back to Birkhoff. Let us check the uniform integrability condition. Let $M > 0$, and truncate f at the level M :

$$f = g + h, \quad \text{where} \quad g = f \cdot \mathbb{1}_{(-M, M)}(f).$$

Then $S_n f = S_n g + S_n h$. Let

$$E_M = \{\omega \in \Omega : |S_n f(\omega)| \geq 2M\}.$$

We have to show that for any $\varepsilon > 0$, there exists $M > 0$ such that $\int_{E_M} |S_n f| d\mu < \varepsilon$.

Let $\omega \in E_M$. Then

$$2M \leq |S_n f(\omega)| \leq |S_n g(\omega)| + |S_n h(\omega)| \leq M + |S_n h(\omega)|.$$

This implies $|S_n h(\omega)| \geq M$, and so $|S_n f(\omega)| \leq 2|S_n h(\omega)|$. Therefore,

$$\int_{E_M} |S_n f| d\mu \leq 2 \int_{\Omega} |S_n h| d\mu \leq 2 \int_{\Omega} S_n |h| d\mu = \|h\|_1.$$

The last quantity is independent of n and can be made arbitrarily small by an appropriate choice of M .

(3) Let $\omega \in \Omega$ and assume that $S_n f(\omega)$ and $S_n(T\omega)$ converge. Then

$$|S_n f(\omega) - S_n f(T\omega)| = \frac{1}{n+1} |f(\omega) - f(T^{n+1}\omega)|.$$

Hence

$$\|S_n f - S_n(f \circ T)\|_1 \leq \frac{1}{n+1} (\|f\|_1 + \|f \circ T^{n+1}\|) \rightarrow 0$$

Since $S_n f \rightarrow \tilde{f}$ a.s. $S_n(f \circ T) \rightarrow \tilde{f} \circ T$ a.s., we have $\tilde{f} = \tilde{f} \circ T$ a.s.

(4)

$$\int \tilde{f} d\mu = \int \lim S_n f d\mu = \lim \int S_n f d\mu = \int f d\mu.$$

Birkhoff's theorem is proven. $\square$

Corollary.

Let $\{X_n\}_{n=0}^\infty$ be a sequence of i.i.d. L_1 random variables. Recall that $\Omega = \mathbb{R}^{\mathbb{Z}_+}$,

$$T(y_1, y_2, \dots) = (y_2, y_3, \dots)$$

Set $f(y_1, y_2, \dots) = y_1$ (first coordinate function).

Then

$$\|f\|_1 = \int_{\mathbb{R}^{\mathbb{N}}} |f(y_0, \dots)| d\mu = \int_{\mathbb{R}^{\mathbb{Z}_+}} |y_0| d\mu = \int_{\mathbb{R}} |y_0| d\mu = \mathbb{E}|X_1| < +\infty.$$

By Birkhoff's theorem

$$S_n f(y_0, \dots) \rightarrow \tilde{f} \text{ a.s.}$$

What is the distribution of $S_n f$?

$$\begin{aligned} S_n f(y_0 \dots) &= \frac{1}{n+1} (f(y_0 \dots) + f \circ T(y_0, \dots) + \dots + f \circ T^n(y_0 \dots)) = \\ &= \frac{1}{n+1} [y_0 + y_1 + \dots + y_n] \sim \frac{1}{n+1} [X_0 + \dots + X_n] \end{aligned}$$

And what is the limit function $\tilde{f}$? Recall that a set $E_a = \tilde{f}^{-1}((-\infty, a])$ is invariant for every $a \in \mathbb{R}$. Shift is ergodic, hence $\mu(E_a) \in \{0, 1\}$. This means that $\tilde{f} = \lambda \cdot \mathbb{1}$. And so,

$$\mathbb{E}X_0 = \int y_0 d\mu = \int f d\mu = \int \tilde{f} d\mu = \lambda.$$

Therefore, average of X_i converges almost surely to the expectation,

$$\frac{1}{n+1}[X_0 + \dots + X_n] \rightarrow \mathbb{E}X_0.$$

We obtained Kolmogorov's L_1 law of large numbers.

LECTURE 6. BIRKHOFF-KHINCHIN COROLLARIES

Definition 13. *The invariant σ -algebra is an algebra of all sets invariant with respect to transformation T , namely, the set*

$$\mathcal{I} = \{A \in \mathcal{A} : T^{-1}A = A\}.$$

It's easy to check that these sets form a σ -algebra. Let $A \in \mathcal{I}$. Then

$$\int_A \tilde{f} d\mu = \int_A \lim_{n \rightarrow \infty} S_n f d\mu = \lim_{n \rightarrow \infty} \int_A S_n f d\mu =$$

(we can change the order of limit and integral as $S_n f \mathbb{1} \rightarrow \tilde{f} \mathbb{1}$ a.s. as well as in L_1)

$$= \lim_{n \rightarrow \infty} \frac{1}{n+1} \sum_{j=0}^n \int_{\Omega} f(T^j(\omega)) \mathbb{1}_A(\omega) d\mu(\omega) = \int_A f d\mu$$

(Last equality is true because

$$\int_{\Omega} f(T^j(\omega)) \mathbb{1}_A(\omega) d\mu(\omega) = \int_{\Omega} f(T^j(\omega)) \mathbb{1}_{T^{-j}(A(\omega))} d\mu(\omega) = \int_{\Omega} f(\omega) \mathbb{1}_{A(\omega)} d\mu(\omega).)$$

And we can see that $\tilde{f} = \mathbb{E}[f|\mathcal{I}]$ (by Birkhoff's theorem $\tilde{f} = \tilde{f} \circ T$ is measurable with respect to $\mathcal{I}$). Hence, if T is ergodic, then

$$\tilde{f} = \mathbb{E}(f) \text{ a.s.}$$

Corollary 2 (Recurrence). *Let $T : \Omega \rightarrow \Omega$ be an ergodic transformation. Then for any $A \in \mathcal{A}$*

$$(2) \quad \lim_{n \rightarrow \infty} \frac{|\{j \leq n : T^j(\omega) \in A\}|}{n+1} = \mu(A) \text{ a.s.}$$

Proof. Apply Birkhoff's theorem to the function $f = \mathbb{1}_A$. $\square$

Example. Rotation of the circle. Let $\theta \in \mathbb{R} \setminus \mathbb{Q}$. $T : [0, 1] \rightarrow [0, 1]$. $T\omega = \omega + \theta \pmod{1}$. T is ergodic. For the set of intervals, we have recurrence property pointwise (not only a.s.)

Theorem 9. *For any $(a, b) \subset [0, 1]$ and for any $\omega \in [0, 1]$*

$$\lim_{n \rightarrow \infty} \frac{|\{j \leq n : T^j(\omega) \in (a, b)\}|}{n+1} = b - a$$

Proof. For $0 \leq p < q \leq 1$ set

$$E_{p,q} = \{\omega \in [0, 1] : (2) \text{ does not hold for } A = (p, q)\}$$

$$m(E_{p,q}) = 0$$

$$m\left(\bigcup_{(p,q) \subset ([0,1] \cap \mathbb{Q})} E_{p,q}\right) = 0.$$

Denote A to be complement to this union.

Let $\omega \in [0, 1]$. Let $(a, b) \subset [0, 1]$. Choose p, q so that $a < p < q < b$. Since $m(A) = 1$, we can choose $\omega' \in A$ such that

$$d(\omega, \omega') < \min(b - q, p - a)$$

Note that $d(T^j \omega, T^j \omega') = d(\omega, \omega')$ (this is kind of continuity condition that allows our to estimate for every point) and so if $T^j \omega' \in (p, q)$ then $T^j \omega \in (a, b)$.

By definition of A ,

$$\liminf_{n \rightarrow \infty} \frac{|\{j \leq n : T^j(\omega) \in (a, b)\}|}{n+1} \geq \liminf_{n \rightarrow \infty} \frac{|\{j \leq n : T^j(\omega') \in (p, q)\}|}{n+1} = q - p.$$

Since p, q are arbitrary, we have shown that

$$\liminf_{n \rightarrow \infty} \frac{|\{j \leq n : T^j(\omega) \in (a, b)\}|}{n+1} \geq b - a$$

To show the opposite inequality, choose p, q such that $a < p < q < b$. This will yield

$$\limsup_{n \rightarrow \infty} \frac{|\{j \leq n : T^j(\omega) \in (a, b)\}|}{n+1} \leq b - a. \square$$

Let $(X_n)_{n \in \mathbb{N}}$ be a random process with values in $\mathbb{R}^d$. Let $((\mathbb{R}^d)^{\mathbb{N}}, \mathcal{A}, \mu)$ be the probability space equipped with the measure generated by the process. Then the shift

$$T : (\mathbb{R}^d)^{\mathbb{N}} \rightarrow (\mathbb{R}^d)^{\mathbb{N}}$$

$$T(x_1, x_2, \dots) = (x_2, x_3, \dots)$$

is measure preserving ($x_i \in \mathbb{R}^d$).

The process $(X_n)_{n \in \mathbb{N}}$ is called *ergodic* if T is ergodic. For example, if X_n are independent then $(X_n)_{n \in \mathbb{N}}$ is ergodic.

Define a random walk in $\mathbb{R}^d$ by

$$Y_n = \sum_{j=1}^n X_j$$

Note that our definition allows it to be multidimensional, may not be on lattice, X_n may be dependent. Let's consider trajectory of a walk and ask how dense is it in the space (or, the same question, how fast it goes away from the beginning point)?

Theorem 10. *Let $(X_n)_{n \in \mathbb{N}}$ be a random process in $\mathbb{R}^d$ and $(Y_n)_{n \in \mathbb{N}}$ be the corresponding random walk. Set*

$$A = \{\omega \in \Omega : Y_n \neq 0 \ \forall n \in \mathbb{N}\}$$

(A is event that walk will never return to the origin). Then

$$\lim_{n \rightarrow \infty} \frac{|\{Y_1, \dots, Y_n\}|}{n} = \mathbb{P}(A) \text{ a.s.}$$

(Note that every visited point is counted only once.)

LECTURE 7. RANDOM WALKS (RETURN TO ORIGIN, CHUNG-FUCHS THM)

Definition 14. *Let $(X_n)_{n \in \mathbb{N}}$ be a stationary random process. It is called ergodic if the shift is an ergodic transformation on $(\mathbb{R}^d)^{\mathbb{N}}$ equipped with the measure generated by the process $(X_n)_n \in \mathbb{N}$.*

Clearly, it means that given an ergodic transformation $T : \Omega \rightarrow \Omega$, we can construct an ergodic process. If μ is some measure, $f : \Omega \rightarrow \mathbb{R}$ is an integrable function ($f \in L_1(\mu)$), then $X_n(\omega) = f(T^n \omega)$ is an ergodic process by definition.

The following example shows that the opposite is not true.

Example. Let transformation $T : \Omega \rightarrow \Omega$ be ergodic. Set

$$\Omega^{(2)} = \Omega \times \{1, 2\}.$$

Set $\tilde{T} : \Omega^{(2)} \rightarrow \Omega^{(2)}$

$$\tilde{T}\omega = (T\omega, j) \quad j \in \{1, 2\}.$$

This means that our space $\Omega^{(2)}$ consists of two disjoint copies of Ω , and the transformation $\tilde{T}$ acts on each copy as T . Define measure $\tilde{\mu}$ on $\Omega^{(2)}$ by

$$\tilde{\mu}(E) = \frac{1}{2}\mu(E \cap \Omega_1) + \frac{1}{2}\mu(E \cap \Omega_2)$$

Then $\tilde{T}$ is clearly measure preserving, but not ergodic (Ω_1 is an invariant set of measure 0.5).

However we can construct an ergodic process by the way described above. Let $f : \Omega \rightarrow \mathbb{R}, f \in L_1(\mu)$. Define $\tilde{f} : \Omega^{(2)} \rightarrow \mathbb{R}$ by

$$\tilde{f}((\omega, j)) = f(\omega) \quad j \in \{1, 2\}.$$

Set $X_n((\omega, j)) = \tilde{f}(\tilde{T}^n(\omega, j))$, $(\omega, j) \in \Omega^{(2)}$. Let's show that the process (X_n) is ergodic. Take $A \subset \mathbb{R}^{\mathbb{N}}$ – an invariant measurable set. Define $E \in \Omega^{(2)}$ by

$$\tilde{E} = \{(\omega, j) \in \Omega^{(2)} : (\tilde{f}(\tilde{T}(\omega, j)), \tilde{f}(\tilde{T}^2(\omega, j)), \dots) \in A\}$$

Set $E_j = \{\omega : (\omega, j) \in \tilde{E}\}$, $j = 1, 2$.

Then

$$E_j = \{\omega \in \Omega : (f(T\omega), f(T^2\omega), \dots) \in A\}$$

is an invariant set. Hence $\mu(E_j) \in \{0, 1\}$ (T is ergodic on each of the components). Actually, measures of E_1 and E_2 must coincide, as it's the same set by definition. Therefore, $\tilde{\mu}(\tilde{E}) \in \{0, 1\}$. So, the process is ergodic and it was obtained from the non-ergodic transformation.

Let $(X_n)_{n \in \mathbb{N}}$ be a random process with values in $\mathbb{R}^d$. Set

$$Y_n = \sum_{k=1}^n X_k.$$

Theorem 11. *Let $(X_n)_{n \in \mathbb{N}}$ be an ergodic random process with values in $\mathbb{R}^d$, and $(Y_n)_{n \in \mathbb{N}}$ be a corresponding random walk.*

Set $R_n = |\{Y_1, \dots, Y_n\}|$. Then

$$\lim_{n \rightarrow \infty} \frac{R_n}{n} = \mathbb{P}(A) \text{ a.s. ,}$$

where $A = \{\omega : \forall n \in \mathbb{N} Y_n(\omega) \neq 0\}$ (A is the event that walk never returns to the starting point.)

Proof. Note that

$$R_n(\omega) \geq \sum_{k=1}^n \mathbb{1}_A(T^k \omega),$$

where $T : (\mathbb{R}^d)^{\mathbb{N}} \rightarrow (\mathbb{R}^d)^{\mathbb{N}}$ is the shift.

$$\mathbb{1}_A(T^k \omega) = \begin{cases} 1, & \text{if } ((X_{k+1}(\omega), X_{k+2}(\omega), \dots) \in A \\ 0, & \text{otherwise} \end{cases}$$

(So $(X_{k+1}(\omega), \dots) \in A$ means that $(Y_{k+1} \dots)$ will never return to $Y_k(\omega)$. In the right hand side we compute each vertex only once in the moment of the last visit. Even if the last visit will happen after n -th step, we do not count the vertex before, that makes the inequality be strict sometimes.)

Then

$$\liminf \frac{R_n(\omega)}{n} \geq \lim \frac{1}{n} \sum_{k=1}^n \mathbb{1}_A(T^k \omega) = \mathbb{E}[\mathbb{1}_A | \mathcal{I}] \text{ a.s.}$$

by Birkhoff theorem and ergodicity.

We need to obtain reverse inequality. Let $m \in \mathbb{N}$. Set

$$A_m = \{\omega : \forall n < m \ Y_n(\omega) \neq 0\}.$$

Then for any $m < n$,

$$R_n(\omega) \leq \sum_{k=1}^{n-m} \mathbb{1}_{A_m}(T^k \omega) + m,$$

(last m iterations are counted in any case, even if we have loops there)

Using Birkhoff's theorem for the second equality we get (for all m)

$$\limsup \frac{R_n(\omega)}{n} \leq \limsup \frac{1}{n} \sum_{k=1}^{n-m} \mathbb{1}_{A_m}(T^k \omega) + 0 = \mathbb{P}(A_m) \text{ a.s.}$$

Therefore,

$$\limsup \frac{R_n(\omega)}{n} \leq \liminf_{m \rightarrow \infty} \mathbb{P}(A_m)$$

Recall that $A_1 \supset A_2 \supset \dots$ and $\bigcap_{n=1}^{\infty} A_n = A$ (if walk will not return after any finite number of steps, it means that it will never return to the origin).

Thus,

$$\lim_{m \rightarrow \infty} \mathbb{P}(A_m) = \mathbb{P}(A).$$

□

Theorem 12 ("Chung-Fuchs"). *Let $(X_n)_{n \in \mathbb{N}}$ be an ergodic integer-valued process. Let $(Y_n)_{n \in \mathbb{N}}$ be the corresponding random walk. If $\mathbb{E}X_1 = 0$ then*

$$\mathbb{P}(Y_n = 0 \text{ infinitely often}) = 1.$$

(In standard version of Chung-Fuchs theorem X_n are supposed to be independent. There is also easy version with X_n independent with values ± 1 , that can be done by elementary means.)

Proof.

(1) Let's prove that $\mathbb{P}(A) = 0$, where A is an event

$$A = \{\omega : \forall n \in \mathbb{N} \ Y_n \neq 0\}.$$

Since $\mathbb{E}X_1 = 0$, $\lim_{n \rightarrow \infty} \frac{Y_n}{n} = \mathbb{E}X_n = 0$ a.s. (use Birkhoff's theorem with the projection function $f((X_1, \dots)) = X_1$.)

Then, for $m < n$

$$\frac{\max_{k \leq n} |Y_k|}{n} \leq \frac{\max_{k \leq m} |Y_k|}{n} + \frac{\max_{m < k \leq n} |Y_k|}{n} \leq$$

$$\leq \frac{\max_{k \leq m} |Y_k|}{n} + \max_{k \leq n} \frac{|Y_k|}{n} \rightarrow 0 \text{ a.s.}$$

Since $(X_n)_{n \in \mathbb{N}}$ is integer valued and due to the crucial fact that our walk is 1-dimensional,

$$R_n \leq 2 \max_{k \leq n} |Y_k| + 1,$$

$$\mathbb{P}(A) = \lim_{n \rightarrow \infty} \frac{R_n}{n} = 0$$

by the previous theorem. It means that walk almost surely returns at 0 at least once.

- (2) Let E_k be the event that $(Y_n)_{n \in \mathbb{N}}$ returns to 0 at least k times. Let's prove by induction that probability of each of these events is 1. Part (1) is the base: $\mathbb{P}(E_1) = 1$. Assume that $\mathbb{P}(E_k) = 1$. Set

$$U_j = \{\omega : Y_l(\omega) \text{ is in 0 k-th time for } l = j\}$$

$$F_j = \{\omega \in U_j : Y_l \neq 0 \text{ for } l > j\}.$$

Obviously,

$$F_j \subset \{\omega : \sum_{l=j+1}^n X_l(\omega) \neq 0\}$$

and the probability

$$\mathbb{P}(F_j) \leq \mathbb{P}(\omega : \sum_{l=1}^{n-j} X_l(\omega) \neq 0 \forall n > j) = 0$$

(we use stationarity and then result of part (1)).

Since $E_{k+1} = \bigcup_{j \in \mathbb{N}} U_j \setminus F_j$, we conclude that

$$\mathbb{P}(E_{k+1}) = \mathbb{P}(\bigcup_{j \in \mathbb{N}} U_j) = \mathbb{P}(E_k) = 1.$$

LECTURE 8. RANDOM WALK RETURN TIMES

Theorem 13. *Let $T : \Omega \rightarrow \Omega$ be a measure preserving transformation. Let $A \in \mathcal{A}$ such that*

$$\mathbb{P}(\exists n > 0 T^n \omega \in A) = 1.$$

Set $R_j(\omega)$ to be the time of j -th return to A .

Then $(R_{j+1} - R_j)_{j \in \mathbb{Z}_+}$ is a stationary random process and

$$\mathbb{E}(R_{j+1}(\omega) - R_j(\omega) | \omega \in A) = \frac{1}{\mu(A)}$$

Note: by definition, $\mathbb{P}(B|\omega \in A) = \frac{\mu(A \cap B)}{\mu(A)}$.

Proof. Consider a random process

$$X_n(\omega) = T^n \omega,$$

$n \in \mathbb{Z}_+$. Extend it to a random process on $\mathbb{Z}$ defining $Y_n : \Omega \rightarrow \Omega$, as we did in Lecture 3 (a measure on $\Omega^{\mathbb{Z}}$ such that $(Y_n)_{n \in \mathbb{Z}_+}$ and $(X_n)_{n \in \mathbb{Z}_+}$ are equidistributed). The process $(Y_n)_{n \in \mathbb{Z}}$ is also stationary.

Set

$$T_j = R_{j+1} - R_j.$$

We have to show

$$\mathbb{P}(T_2 = n_1, T_3 = n_2, \dots, T_{k+1} = n_k) = \mathbb{P}(T_1 = n_1, T_2 = n_2, \dots, T_k = n_k),$$

or, for two random variables (and we can consider any number of moments with the same way),

$$\mathbb{P}(T_2 = n, T_3 = m) = \mathbb{P}(T_1 = n, T_2 = m).$$

Define

$$I(\omega) = \{n \in \mathbb{Z}, Y_n(\omega) \in A\}.$$

We want to show that

$$\begin{aligned} \mathbb{P}(\exists l \in \mathbb{N} : I(\omega) \cap [0, l + m + n] = \{0, l, l + n, l + m + n\}) &= \\ &= \mathbb{P}(I(\omega) \cap [0, m + n] = \{0, n, n + m\}) \end{aligned}$$

(recall that T_i are time differences between visits of A).

Stationarity of the process $(Y_n)_{n \in \mathbb{Z}}$ implies

$$\begin{aligned} \mathbb{P}(\exists l \in \mathbb{N} : I(\omega) \cap [0, l + m + n] = \{0, l, l + n, l + m + n\}) &= \\ &= \mathbb{P}(\exists l \in \mathbb{N} : I(\omega) \cap [-l, m + n] = \{-l, 0, n, m + n\}) = \\ &= \mathbb{P}((I(\omega) \cap [0, m + n] = \{0, n, m + n\}) \text{ and } (I(\omega) \cap [-l, -1] = \{-l\})) = \\ &= \mathbb{P}(E \cap \bigcup_{l \in \mathbb{N}} F_l), \end{aligned}$$

where $F_l = \{\omega : I(\omega) \cap [-l, -1] = \{-l\}\}$.

So our aim is to prove that

$$\mathbb{P}(\bigcup_{l \in \mathbb{N}} F_l) = 1.$$

The complement of the event $\bigcup_{l \in \mathbb{N}} F_l$ consists of those $\omega \in \Omega$ for which $Y_l(\omega) \notin A$ whenever $l < 0$.

Hence, for any $l \in \mathbb{N}$,

$$\left(\bigcup_{l \in \mathbb{N}} F_l \right)^c \subset V_l := \{\omega : I(\omega) \cap [-l, -1] = \emptyset\}.$$

Using stationarity again, we get

$$\mathbb{P}(V_l) = \mathbb{P}(I(\omega) \cap [1, l] = \emptyset).$$

By assumption of the theorem, $\mathbb{P}(I(\omega) \cap [1, l] = \emptyset) \rightarrow 0$, and so $\mathbb{P}((\bigcup_{l \in \mathbb{N}} F_l)^c) = 0$. Stationarity is proven.

Now, we want to prove the second claim, that

$$\mathbb{E}[T_j | \omega \in A] = \frac{1}{\mu(A)},$$

or

$$x_j = \mathbb{E}[T_j(\omega) \cdot \mathbb{1}_A(\omega)] = 1.$$

Note that x_j does not depend on j and set $j = 1$. We want

$$x_j = \mathbb{E}[R_1(\omega) \cdot \mathbb{1}_A(\omega)] = 1.$$

What do we know?

$$\begin{aligned} \mathbb{E}[R_1(\omega) \cdot \mathbb{1}_A(\omega)] &= \sum_{k \in \mathbb{N}} k \cdot \mathbb{P}(I(\omega) \cap [0, k] = \{0, k\}) = \\ &= \sum_{k \in \mathbb{N}} \sum_{j=1}^k \mathbb{P}(I(\omega) \cap [0, k] = \{0, k\}) = \sum_{j \in \mathbb{N}} \sum_{k=j}^{\infty} \mathbb{P}(I(\omega) \cap [0, k] = \{0, k\}) = \\ &= \sum_{j \in \mathbb{N}} \mathbb{P}(\exists k \geq j \ I(\omega) \cap [0, k] = \{0, k\}). \end{aligned}$$

The problem is that events in the sum are not disjoint. We will address this problem by using the same shifting trick, as in the first part of the proof. Note that

$$\begin{aligned} &\{\omega : \exists k \geq j \ I(\omega) \cap [0, k] = \{0, k\}\} \\ &= \{\omega : I(\omega) \cap [0, j-1] = \{0\}\} \cap \{\omega : I(\omega) \cap [j, +\infty) \neq \emptyset\}. \end{aligned}$$

The second event has probability 1 as we are given that path will eventually return to A (and j does not matter due to stationarity).

Therefore,

$$\mathbb{E}[R_1(\omega) \cdot \mathbb{1}_A(\omega)] = \sum_{j \in \mathbb{N}} \mathbb{P}(I(\omega) \cap [0, j-1] = \{0\}).$$

And now applying the shift, we can make all the events disjoint:

$$\mathbb{E}[R_1(\omega) \cdot \mathbb{1}_A(\omega)] = \sum_{j \in \mathbb{N}} \mathbb{P}(I(\omega) \cap [-j, -1] = \{-j\}) = 1,$$

where the last equality follows from the proof of the first part. Theorem is completely proven. $\square$

LECTURE 9. ERGODIC AND MIXING TRANSFORMATIONS

Birkhoff's theorem and ergodicity imply that if $f \in L_1(\mu)$

$$\frac{1}{n+1} \sum_{j=0}^n f(T^j \omega) \rightarrow \int f d\mu \text{ a.s. and in } L_1(\mu).$$

Definition 15 (mixing). *A measure preserving transformation $T : \Omega \rightarrow \Omega$ is mixing if*

$$\forall A, B \in \mathcal{A} \lim_{n \rightarrow \infty} \mu(A \cap T^{-n}B) = \mu(A)\mu(B)$$

Example. Let $(X_n)_{n \in \mathbb{Z}_+}$ be i.i.d. random variables, and let $\Omega = \mathbb{R}^{\mathbb{Z}_+}$ equipped with the product measure μ generated by this random process. Then the shift $T : \Omega \rightarrow \Omega$ is mixing. To see it, consider first Borel sets $E, F \subset \mathbb{R}$ and set

$$\begin{aligned} A &= \{\omega \in \Omega : \omega_0 \in E\}, \\ B &= \{\omega \in \Omega : \omega_0 \in F\}. \end{aligned}$$

Then

$$A \cap T^{-n}B = \{X_0 \in E, X_n \in F\}.$$

By independence and stationarity,

$$\mathbb{P}(X_0 \in E, X_n \in F) = \mathbb{P}(X_0 \in E)\mathbb{P}(X_n \in F) = \mathbb{P}(X_0 \in E)\mathbb{P}(X_0 \in F)$$

for all $n \in \mathbb{N}$. Hence,

$$\lim_{n \rightarrow \infty} \mathbb{P}(A \cap T^{-n}B) = \mathbb{P}(X_0 \in E)\mathbb{P}(X_0 \in F) = \mu(A) \cdot \mu(B).$$

This calculation can be easily extended to the case when A, B are cylinders. Since the cylinders generate the σ -algebra on Ω , it can be further extended to all measurable subsets of Ω by approximation.

Now we will show the general fact, that mixing property is stronger than ergodicity. Start with some definitions.

Definition 16 (Cesaro limit). *Let $\{a_n\}_{n \in \mathbb{N}} \in \mathbb{C}$. Define $C \lim_{n \rightarrow \infty} a_n$ by*

$$C \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n a_k,$$

wherever the last limit exists.

Cesaro limit is the extension of the limit notion to provide natural limits to sequences as $(-1)^{n-1}$. The following lemma is known from analysis.

Lemma 1. *If $\lim a_n$ exists, then $C \lim a_n$ exists as well and it's equal to $\lim a_n$.*

Theorem 14. *Let $T : \Omega \rightarrow \Omega$ be a measure preserving transformation. The following are equivalent:*

- (1) *T is ergodic;*
- (2) $\forall A, B \in \mathcal{A} \text{ } C\lim \mu(A \cap T^{-n}B) = \mu(A)\mu(B).$

See that we have an obvious corollary, that every mixing transformation is ergodic.

Proof. (1) $\Rightarrow$ (2)

$$\begin{aligned} C\lim \mu(A \cap T^{-1}B) &= \lim_{n \rightarrow \infty} \frac{1}{n+1} \sum_{k=1}^n \mu(A \cap T^{-k}B) = \\ &= \lim_{n \rightarrow \infty} \frac{1}{n+1} \sum_{k=1}^n \int_A \mathbb{1}_{T^{-k}B}(\omega) d\mu(\omega) = \lim_{n \rightarrow \infty} \int_A \frac{1}{n+1} \sum_{k=1}^n \mathbb{1}_B(T^k\omega) d\mu(\omega) \end{aligned}$$

Inner part of the integral converges to the $\int \mathbb{1}_B(\omega) d\mu(\omega) = \mu(B)\mathbb{1}$ a.s. (by Birkhoff's theorem) and so, by dominated convergence theorem,

$$\int_A \frac{1}{n+1} \sum_{k=1}^n \mathbb{1}_B(T^k\omega) d\mu(\omega) \rightarrow \int_A \mu(B) d\mu(\omega) = \mu(A)\mu(B)$$

(2) $\Rightarrow$ (1)

Let $A \in \mathcal{I}$ where

$$\mathcal{I} = \{E \in \mathcal{A} : T^{-1}E = E\}$$

is the invariant σ -algebra. Then $\forall n \in \mathbb{Z}_+$

$$\mu(A \cap T^{-n}A) = \mu(A),$$

$$C\lim \mu(A \cap T^{-n}A) = \mu(A),$$

$$\mu(A) = \mu(A)^2 = 0 \text{ or } 1.$$

□

How to check mixing property?

Let $T : \Omega \rightarrow \Omega$ be a measure preserving transformation for $n \in \mathbb{Z}_+$.

Let $\mathcal{A}_{-n}$ be the σ -algebra generated by the sets

$$T^{-n}A, \quad A \in \mathcal{A}.$$

So $\mathcal{A}_{-n} = T^{-n}\mathcal{A}$. $\mathcal{A} = \mathcal{A}_0 \supset \mathcal{A}_1 \supset \dots$

Definition 17 (tail σ -algebra). *Define the tail σ -algebra by*

$$\mathcal{T} = \bigcap_{n \in \mathbb{Z}_+} \mathcal{A}_{-n} (= \mathcal{A}_{-\infty})$$

Examples.

(1) Shift $(X_n)_{n \in \mathbb{Z}_+}$ is a random process on space $(\mathbb{R}^{\mathbb{Z}_+}, \mathcal{A}, \mu)$.
 Sigma-algebra $\mathcal{A}$ generated by (X_n) is generated by cylinders

$$\{\omega \in \mathbb{R}, \omega_0 \in A_0, \dots, \omega_k \in A_k\} = \mathcal{C}_{A_0, \dots, A_k}$$

Then

$$T^{-n}\mathcal{C}_{A_0, \dots, A_k} = \{\omega \in \mathbb{R}, \omega_n \in A_0, \dots, \omega_{n+k} \in A_k\}$$

The σ -algebra $\mathcal{A}_0$ is generated by $(X_0, X_1, \dots)$.

The σ -algebra $\mathcal{A}_1$ is generated by $(X_1, X_2, \dots)$.

...

If X_n are independent, then by Kolmogorov's zero-one law $\forall A \in \mathcal{T}$
 $\mu(A) \in \{0, 1\}$.

(2) Rotation of a circle. $\Omega = \mathbb{R}/\mathbb{Z}$, $\mathcal{A} = \mathcal{B}(0, 1)$.

$$T^{-1}\mathcal{A} = \{A - \theta : A \in \mathcal{A}\} = \mathcal{A}$$

Hence, $\mathcal{T} = \mathcal{A}$.

(3) Doubling. $T\omega = 2\omega \pmod{1}$. We will do it later.

Definition 18 (regular transformation). *A measure preserving transformation is called regular if corresponding tail sigma-algebra is trivial: $\forall A \in \mathcal{T} \mu(A) \in \{0, 1\}$. A random process is called regular if corresponding transformation is regular.*

Definition 19 (backwards martingale). *Let $\mathcal{A} = \mathcal{A}_0 \supset \mathcal{A}_{-1} \supset \dots$ be a sequence of sigma-algebras on a set Ω . Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable, $\mathbb{E}|X| < +\infty$. The sequence*

$$X_{-n} = \mathbb{E}[X | \mathcal{A}_{-n}]$$

is called a backwards martingale.

It's a martingale indeed:

$$X_0 = \mathbb{E}[X | \mathcal{A}_0] = X,$$

and $\forall n \in \mathbb{N}$

$$\mathbb{E}[X_{-n+1} | \mathcal{A}_{-n}] = \mathbb{E}[\mathbb{E}[X | \mathcal{A}_{-(n-1)}] | \mathcal{A}_{-n}] = \mathbb{E}[X | \mathcal{A}_{-n}] = X_{-n}.$$

LECTURE 10. MIXING AND REGULAR TRANSFORMATIONS

Theorem 15 (Doob, Levy). *Let $(X_{-n})_{n \in \mathbb{Z}_+}$ be a backwards martingale. Then*

$$X_n \rightarrow \mathbb{E}[X | \mathcal{A}_{-\infty}] \text{ a.s. and in } L_1(\mathbb{P}).$$

Proof. Take $n \in \mathbb{N}$. Define

$$Y_k = X_{k-n}, \quad k = 0, \dots, n,$$

$$\mathcal{A}'_k = \mathcal{A}_{k-n}.$$

Then $(Y_k)_{k \in \{0, \dots, n\}}$ is a martingale with respect to $(\mathcal{A}'_k)_{k=0, \dots, n}$.

Recall Doob's upcrossing inequality: if we have two levels $\alpha, \beta \in \mathbb{R}$, $\alpha < \beta$, and $U_{\alpha, \beta}^n$ is the number of upcrossings $[\alpha, \beta]$ by the martingale $(Y_k)_{k=0}^n$, then

$$(\beta - \alpha) \mathbb{E} U_{\alpha, \beta}^n \leq \mathbb{E}(Y_n - \alpha)_+ - \mathbb{E}(Y_0 - \alpha)_+.$$

For the proof of Doob-Levy, define

$$E_{\alpha, \beta} = \{\omega \in \Omega : \liminf X_{-n} < \alpha < \beta < \limsup X_{-n}(\omega)\}$$

and define $U_{\alpha, \beta}$ to be a number of (α, β) -upcrossings for the process $(X_{-n})_{n \in \mathbb{Z}^+}$. Then

$$U_{\alpha, \beta} = \sup_n U_{\alpha, \beta}^n, \quad \text{and } U_{\alpha, \beta}^1 < U_{\alpha, \beta}^2 < \dots$$

By definition of $E_{\alpha, \beta}$, $U_{\alpha, \beta}(\omega) = +\infty$ for any $\omega \in E_{\alpha, \beta}$. By Doob's upcrossing inequality

$$(\beta - \alpha) \mathbb{E} U_{\alpha, \beta}^n \leq \mathbb{E}(Y_n - \alpha)_+ - \mathbb{E}(Y_0 - \alpha)_+$$

Hence

$$\mathbb{E} U_{\alpha, \beta} \leq \frac{1}{\beta - \alpha} \mathbb{E}(X_0 - \alpha) < +\infty$$

and so, the random variable $U_{\alpha, \beta}$ is finite almost surely. Therefore, $\mathbb{P}(E_{\alpha, \beta}) = 0$.

To show L^1 -convergence, remind that if X is bounded, then a.s. convergence implies L^1 -convergence. So this will follow from the future HW problem.

Lemma 2. *Any backwards martingale is uniformly integrable (HW).*

It remains to show that $\tilde{X} = \lim_{n \rightarrow \infty} X_n$ is the conditional expectation with respect to $\mathcal{A}_{-\infty}$. Since we can write $\tilde{X} = \lim_{n \rightarrow \infty, n > k} X_n$, $\tilde{X}$ is $\mathcal{A}_{-k}$ measurable. Therefore $\tilde{X}$ is $\mathcal{A}_{-\infty}$ measurable. To check that $\tilde{X} = \mathbb{E}[X | \mathcal{A}_{-\infty}]$ it is enough to show that $\forall A \in \mathcal{A}_{-\infty}$

$$\mathbb{E}[X \mathbb{1}_A] = \mathbb{E}[\tilde{X} \mathbb{1}_A]$$

(as we know that it is the only function satisfying the equality).

We have

$$\mathbb{E}[\tilde{X} \mathbb{1}_A] = \mathbb{E}[\lim_{n \rightarrow \infty} \mathbb{E}[X | \mathcal{A}_{-n}] \mathbb{1}_A] = \lim_{n \rightarrow \infty} \mathbb{E}[\mathbb{E}[X | \mathcal{A}_{-n}] \mathbb{1}_A] = \mathbb{E}[X \mathbb{1}_A].$$

□

Mixing and regularity. Let $T : \Omega \rightarrow \Omega$, $\mathcal{A}_{-n} = T^{-n}\mathcal{A}$. Recall that the tail sigma-algebra is defined by

$$\mathcal{T} = \bigcap_{n=0}^{\infty} \mathcal{A}_{-\infty}.$$

Note, that $\mathcal{I} \subset \mathcal{T}$. Rotation by irrational angle shows that reverse inclusion isn't true.

Theorem 16. *Let T be a measure preserving transformation. TFAE*

- (1) T is regular;
- (2) $\lim_{n \rightarrow \infty} \sup_{B \in \mathcal{A}} \left(\mu(A \cap T^{-n}B) - \mu(A)\mu(B) \right) = 0$ for all $A \in \mathcal{A}$.

Note, that the second condition is strictly stronger than mixing (which is stronger than ergodicity).

Proof.

(1) $\Rightarrow$ (2) Let $A, B \in \mathcal{A}$. Then

$$\begin{aligned} |\mu(A \cap T^{-n}B) - \mu(A) \cdot \mu(B)| &\leq \left| \int_{T^{-n}B} (\mathbb{1}_A - \mu(A)\mathbb{1}) d\mu \right| = \\ &= \left| \int_{T^{-n}B} (\mathbb{E}[\mathbb{1}_A | \mathcal{A}_{-n}] - \mu(A)\mathbb{1}) d\mu \right| \leq \\ &\leq \int_{\Omega} |\mathbb{E}[\mathbb{1}_A | \mathcal{A}_{-n}] - \mu(A)\mathbb{1}| d\mu. \end{aligned}$$

The right-hand side is independent of B and converges to 0 by Doob–Levy and the dominated convergence theorems (as we know that $\mathbb{E}[\mathbb{1}_A | \mathcal{A}_{-\infty}] = \mu(A) \cdot \mathbb{1}$).

(2) $\Rightarrow$ (1)

Let $A \in \mathcal{T}$. Then for any $n \in \mathbb{N}$ $\exists B_n$ $T^{-n}B_n = A$,

$$\begin{aligned} \mu(A \cap T^{-n}B_n) &= \mu(A), \mu(B_n) = \mu(A), \\ \mu(A \cap T^{-n}B_n) - \mu(A)\mu(B_n) &\rightarrow 0 \\ \mu(A) - \mu^2(A) &\rightarrow 0 \\ \mu(A) &= \{0, 1\}. \end{aligned}$$

□

So, we have the following correspondence scheme:

Regularity $\Leftrightarrow$ (2)

$\Downarrow$

Mixing $\Leftrightarrow \forall A, B \in \mathcal{A} \lim_{n \rightarrow \infty} \sup_{B \in \mathcal{A}} (\mu(A \cap T^{-n}B) - \mu(A)\mu(B))$

$\Downarrow$

$$\text{Ergodicity} \Leftrightarrow \forall A, B \in \mathcal{A} \quad C - \lim_{n \rightarrow \infty} \sup_{B \in \mathcal{A}} (\mu(A \cap T^{-n}B) - \mu(A)\mu(B)) = 0$$

Our examples show that reverse implications are not true.

- (1) (ergodic, not mixing) **An irrational rotation** of the circle ergodic, but not mixing:

$\forall \varepsilon > 0, \forall m \in \mathbb{N} \exists n > m \quad n\theta \in (0, \varepsilon) \pmod{1}$. Let $A = (0, 1/2)$. Then

$$\mu(A \cap T^{-n}A) > \frac{1}{2} - \varepsilon > \frac{1}{4}.$$

- (2) (mixing, not regular) Let $(X_n)_{n \in \mathbb{Z}}$ be **independent symmetric Bernoulli** random variables. Set $\Omega = \{0, 1\}^{\mathbb{Z}}$, $\mu_n(\{0\}) = \mu_n(\{1\}) = \frac{1}{2}$ and

$$\mu = \prod_{n \in \mathbb{Z}} \mu_n.$$

The shift T

$$T((\varepsilon_n)_{n \in \mathbb{Z}}) = (\varepsilon_{n+1})_{n \in \mathbb{Z}}$$

is mixing. Why? The argument here is the same as in the example we considered in the last lecture. We start with checking the mixing condition for cylinders. Let $I, J \subset \mathbb{Z}$ be finite sets. Let $\varepsilon \in \{0, 1\}^I$, $\delta \in \{0, 1\}^J$.

$$\mathcal{C}_I(\varepsilon) = \{(x_n)_{n \in \mathbb{Z}} \in \{0, 1\}^{\mathbb{Z}} : x_n = \varepsilon_n \text{ for any } n \in I\}$$

$$\mathcal{C}_J(\delta) = \{(x_n)_{n \in \mathbb{Z}} \in \{0, 1\}^{\mathbb{Z}} : x_n = \delta_n \text{ for any } n \in J\}$$

Choose $n \in \mathbb{N}$ such that $(I + n) \cap J = \emptyset$. Then

$$\mu(T^{-n}\mathcal{C}_I(\varepsilon) \cap \mathcal{C}_J(\delta)) = 2^{-|I|-|J|} = \mu(\mathcal{C}_I(\varepsilon)) \cdot \mu(\mathcal{C}_J(\delta)).$$

The general case follows by approximation of measurable sets by cylinders.

But there are no regularity. Let's show contradiction with the second condition. Set $I = \{0\}$, $\varepsilon = \{0\}$, $J = \{-n\}$ ($n \in \mathbb{N}$), $\delta = \{1\}$. Then

$$\mathcal{C}_I(\varepsilon) \cap T^{-n}\mathcal{C}_J(\delta) = \emptyset.$$

(First elements are obviously different.)

MARKOV CHAINS

We consider countable spaces now.

Definition 20. An integer valued process $(X_n)_{n \in \mathbb{Z}_+}$ is called *Markov chain* if $\forall A \in \mathbb{Z} \forall n \in \mathbb{N}$

$$\mathbb{P}(X_n \in A | X_0, \dots, X_{n-1}) = \mathbb{P}(X_n \in A | X_{n-1}). (*)$$

Note that we use advantage of the countable space: every subset A is measurable. Moreover, condition $(*)$ is equivalent to

$$\mathbb{P}(X_n = j_n | X_0, \dots, X_{n-1}) = \mathbb{P}(X_n = j_n | X_{n-1}),$$

or

$$\mathbb{P}(X_n = j_n | X_0 = j_0, \dots, X_{n-1} = j_{n-1}) = \mathbb{P}(X_n = j_n | X_{n-1} = j_{n-1}).$$

Theorem 17 (trivial in countable case!). Let $(X_n)_{n \in \mathbb{Z}_+}$ be an integer valued random process. TFAE

- (1) it is a Markov process
- (2) there exists a sequence $\{p_k(i, j)\}_{i, j, k \in \mathbb{Z}}$ and $\{\pi(i)\}_{i \in \mathbb{Z}}$ satisfying

$$\sum_{i \in \mathbb{Z}} \pi(i) = 1, \quad \sum_{j \in \mathbb{Z}} p_k(i, j) = 1 \quad \forall k, i$$

such that

$$\mathbb{P}(X_0 = i) = \pi(i), \quad \text{and} \quad \mathbb{P}(X_n = j | X_{n-1} = i) = p_n(i, j).$$

LECTURE 11. MARKOV CHAINS

Definition 21. An integer valued process $(X_n)_{n \in \mathbb{Z}_+}$ is called a **Markov chain** if $\forall A \in \mathbb{Z}, \forall n \in \mathbb{N}$

$$\mathbb{P}(X_n \in A | \mathcal{A}_{n-1}) = \mathbb{P}(X_n \in A | X_{n-1}), (*)$$

where $\mathcal{A}_{n-1}$ is an algebra generated by $X_0, \dots, X_{n-1}$.

If the process $(X_n)_{n \in \mathbb{Z}_+}$ is integer valued, then $(*)$ is equivalent to the condition hat for all $x_{n+1} \in \mathbb{Z}$

$$\mathbb{P}(X_{n+1} = x_{n+1} | \mathcal{A}_n) = \mathbb{P}(X_{n+1} = x_{n+1} | X_n) = \sum_{j \in \mathbb{Z}} p_{n+1}(j, x_{n+1}) \mathbb{1}_{\{j\}}(X_n).$$

Hence,

$$\mathbb{P}(X_{n+1} = x_{n+1} | X_n = j) = p_{n+1}(j, x_{n+1}).$$

Finally, we have

$$\begin{aligned} \mathbb{P}(X_{n+1}, X_n = x_n, \dots, X_0 = x_0) &= \\ &= p_{n+1}(x_n, x_{n+1}) \cdot p_n(x_{n-1}, x_n) \cdot \dots \cdot \mathbb{P}(X_0 = x_0), \end{aligned}$$

that completely describes probability of a trajectory.

Note that it implies the last theorem from the previous class. However in real valued case transition probability will be a Borel (discrete)

measure, not the set of numbers, and similar theorem is quite complicated.

Definition 22. A Markov chain is called **stationary**, if

$$p_n(1, k) = p(j, k) \text{ for all } j, k \in \mathbb{Z}.$$

Note: it's not necessary a stationary process!

Note 2: we will consider only stationary Markov chains.

Denote $\mathbb{P}(A) = \mathbb{P}_\pi(A)$, where π is the distribution of X_0 . Let $(X_n)_{n \in \mathbb{Z}_+}$ be a stationary chain. Then

$$\begin{aligned} \mathbb{E}(\mathbb{1}_{\{x_{n+1}\}}(X_{n+1})|\mathcal{A}_n) &= \mathbb{E}(\mathbb{1}_{\{x_{n+1}\}}(X_{n+1})|X_n) = \sum_{j \in \mathbb{Z}} p(j, x_{n+1}) \mathbb{1}_{\{j\}}(X_n) = \\ &= \sum_{j \in \mathbb{Z}} \mathbb{E}_j(\mathbb{1}_{\{x_{n+1}\}}(X_1)) \mathbb{1}_{\{j\}}(X_n) = \mathbb{E}_{X_n}(\mathbb{1}_{\{x_{n+1}\}}(X_1)). \end{aligned}$$

(Last two equalities are essentially notations.) Or, equivalently,

$$\mathbb{E}(\mathbb{1}_{\{x_{n+1}\}}((T^n X)_1)|\mathcal{A}_n) = \mathbb{E}_{X_n}(\mathbb{1}_{\{x_{n+1}\}}(X_1)),$$

where $T : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ is a shift.

Lemma 3. For a function f being an indicator of a cylinder, $f(X) = \mathbb{1}_{\{y_0, \dots, y_k\}}(X_0, \dots, X_k)$, a similar equality holds:

$$\mathbb{E}(f(T^n X)|\mathcal{A}_n) = \mathbb{E}_{X_n}(f(X)). (**)$$

Indeed,

$$\begin{aligned} LHS &= \mathbb{E}(\mathbb{1}_{\{y_0, \dots, y_k\}}(X_n, \dots, X_{n+k})|\mathcal{A}_n) = \mathbb{P}(X_n = y_0, \dots, X_{n+k} = y_k|\mathcal{A}_n) \\ &= \mathbb{1}_{\{y_0\}}(X_n) \cdot p(y_1, y_2) \cdot p(y_2, y_3) \cdot \dots \cdot p(y_{k-1}, y_k), \\ &= \mathbb{P}_{X_n}(X_n = y_0, X_{n+1} = y_1, \dots, X_{n+k} = y_k) \\ &= RHS \end{aligned}$$

Lemma 4. Let $f : \mathbb{Z}^+ \rightarrow \mathbb{R}$ be a non-negative or bounded function. Then $(**)$ holds for f .

Indeed, Lemma 1 proves that $(**)$ holds for any $f = \mathbb{1}_C$ where

$$C = \{X : X_0 = y_0, \dots, X_k = y_k\}.$$

By the Monotone Convergence Theorem $(**)$ holds for the indicator of any cylinder, approximate measurable sets by cylinders (use MCT) again and $(**)$ holds for $f = \mathbb{1}_A$. Now extend it for simple functions, and then for any non-negative. Every bounded function is the difference of 2 non-negative functions.

We do not extend the argument for the whole L_1 because $(**)$ may not be defined for unbounded function (but when it is defined, conclusion holds too).

Definition 23. Sequence of sigma-algebras $\mathcal{A}_0 \subset \mathcal{A}_1 \subset \dots \subset \mathcal{A}$ is called a **filtration**. A random variable $N : \Omega \rightarrow \mathbb{Z}_+ \cup \{+\infty\}$ is called a **stopping time**, if

$$\forall n \in \mathbb{Z}_+ \{\omega : N = n\} \in \mathcal{A}_n.$$

If $\mathcal{A}_n$ is generated by $X_0, \dots, X_n$ then this means that the event $\{N = n\}$ is determined by the values of $X_0, \dots, X_n$.

Example 1. *Hitting time* is a stopping time. Let $A \subset \mathbb{Z}$

$$\tau_A = \min_{n \in \mathbb{N}} (X_n \in A) \text{ [} n \in \mathbb{N} \text{ is important]}$$

Example 2. The event $\tau = \min_{n \in \mathbb{N}} \{k \in \mathbb{N} : X_k \leq X_n\}$ – the time when X_k reaches its maximum value – is not a stopping time.

Definition 24. Let N be a stopping time. Define the stopping time sigma-algebra $\mathcal{A}_N$ by

$$\mathcal{A}_N = \{A \in \mathcal{A} : \forall n \in \mathbb{Z}_+ A \cap \{\omega : N = n\} \in \mathcal{A}_n\}.$$

Informally, $\mathcal{A}_N$ consists of all events which occur by the time N .

Theorem 18 (Strong Markov property). Let $(X_n)_{n \in \mathbb{Z}_+}$ be a Markov chain. Let $f : \mathbb{Z}^{\mathbb{Z}_+} \rightarrow \mathbb{R}$ be a non-negative or bounded function. Let N be a stopping time. Then on $\{N < +\infty\}$

$$\mathbb{E}(f(T^N X) | \mathcal{A}_N) = \mathbb{E}_{X_N}(f(X)).$$

Proof. Let $A \in \mathcal{A}_N$. We have to check that

$$\mathbb{E}[f(T^N X) \mathbb{1}_A \mathbb{1}_{N < +\infty}] = \mathbb{E}[\mathbb{E}_{X_N}(f(X) \mathbb{1}_A \mathbb{1}_{N < +\infty})].$$

Firstly, note that

$$\mathbb{1}_A \mathbb{1}_{N < +\infty} = \sum_{n=0}^{\infty} \mathbb{1}_A \mathbb{1}_{N=n} = \sum_{n=0}^{\infty} \mathbb{1}_{A \cap \{N=n\}}.$$

Hence,

$$\begin{aligned} LHS(A) &= \sum_{n=0}^{\infty} \mathbb{E}[f(T^N X) \mathbb{1}_{A \cap \{N=n\}}] = \sum_{n=0}^{\infty} \mathbb{E}[f(T^n X) \mathbb{1}_{A \cap \{N=n\}}] \\ &= \sum_{n=0}^{\infty} \mathbb{E}[\mathbb{E}(f(T^n X) \mathbb{1}_{A \cap \{N=n\}} | \mathcal{A}_n)] \end{aligned}$$

Since $\mathbb{1}_{A \cap \{N=n\}}$ is $\mathcal{A}_n$ -measurable, we can pull it out of conditional expectation. Then, using $(**)$, we can rewrite the previous expression as

$$\begin{aligned} &= \sum_{n=0}^{\infty} \mathbb{E}[\mathbb{E}_{X_n}(f(X)) \mathbb{1}_{A \cap \{N=n\}}] = \sum_{n=0}^{\infty} \mathbb{E}[\mathbb{E}_{X_N}(f(X) \mathbb{1}_A \mathbb{1}_{\{N=n\}})] \\ &= \mathbb{E}[\mathbb{E}_{X_N}(f(X) \mathbb{1}_A \mathbb{1}_{N < +\infty})] = RHS. \end{aligned}$$

□

LECTURE 12. CLASSIFICATION OF STATES

Define the first hitting time

$$T_{x,y} = \inf\{n : X_n = y, X_0 = x\}.$$

Define the k -th hitting time ($k \in \mathbb{N}$) of y by

$$\begin{aligned} T_{x,y}^1 &= T_{x,y} \\ T_{x,y}^{k+1} &= \inf\{n > T_{x,y}^k, X_n = y, X_0 = x\} \end{aligned}$$

Then for all k $T_{x,y}^k$ is a stopping time (last theorem from the previous class – strong Markov property – proves exactly this).

Lemma 5. *If $\rho_{x,y}^k = \mathbb{P}(T_{x,y}^k < +\infty)$ (probability of k hits). Then*

$$\rho_{x,y}^k = \rho_{x,y}^1 \cdot (\rho_{y,y}^1)^{k-1} \cdot (*)$$

Proof. Induction on k . $k=1$ – trivial. Assume that $(*)$ holds for k . Let $N = T_{xy}^k$. Besides, denote by A the event that the process returns to y :

$$A = \{(X_m)_{m \in \mathbb{Z}_+} : X_0 = y, \exists n \in \mathbb{N} X_n = y\}.$$

Then

$$\begin{aligned} \mathbb{P}_x(T_{xy}^{k+1} < +\infty) &= \mathbb{E}_x(\mathbb{1}_{\{T_{xy}^{k+1} < +\infty\}}(X) \cdot \mathbb{1}_{\{T_{xy}^k < +\infty\}}(X)) = \\ &= \mathbb{E}_x(\mathbb{1}_A(T^N X) \cdot \mathbb{1}_{\{N < +\infty\}}) = \mathbb{E}_x(\mathbb{E}[\mathbb{1}_A(T^N X) \mathbb{1}_{\{N < +\infty\}} | \mathcal{A}_N]) = \\ &= \mathbb{E}_x[\mathbb{E}_{X_N}(\mathbb{1}_A(X)) \mathbb{1}_{\{N < +\infty\}}]. \end{aligned}$$

We used strong Markov's property in the last step. Note that by definition of N ,

$$\mathbb{E}_{X_N}(\mathbb{1}_A(X)) = \mathbb{E}_y(\mathbb{1}_A(X)) = \mathbb{P}_y(A) = \rho_{yy}^1.$$

So, we proved that

$$\begin{aligned} \rho_{x,y}^{k+1} &= \mathbb{P}_x(T_{xy}^{k+1} < +\infty) = \mathbb{E}_x[\rho_{yy}^1 \cdot \mathbb{1}_{\{N < +\infty\}}] = \\ &= \rho_{yy}^1 \mathbb{P}_x(T_{xy}^k < +\infty) = \rho_{xy}^k \cdot \rho_{yy}^1. \end{aligned}$$

Applying inductive assumption we complete the proof.

□

To shorten the notation, we will write ρ_{xy} instead of ρ_{xy}^1 .

Let R_{xy} be the number of times the chain started at x hits y :

$$R_{xy} = \sum_{n=1}^{\infty} \mathbb{1}_{\{y\}}(X_n).$$

Then

$$\mathbb{E}R_{x,y} = \sum_{n=1}^{\infty} \mathbb{P}(X_0 = x, X_n = y). \quad (*)$$

This representation of $\mathbb{E}R_{x,y}$ will be used later. Now we will evaluate it differently. By definition,

$$\begin{aligned} \mathbb{E}R_{x,y} &= \sum_{k=1}^{\infty} k \cdot \mathbb{P}(R_{xy} = k) = \sum_{m=1}^{\infty} \sum_{k \geq m} \mathbb{P}(R_{xy} = k) = \sum_{m=1}^{\infty} \mathbb{P}(R_{xy} \geq m) = \\ &= \sum_{m=1}^{\infty} \mathbb{P}(T_{xy}^m < +\infty) = \sum_{m=1}^{\infty} \rho_{xy} \rho_{yy}^{m-1} = \frac{\rho_{xy}}{1 - \rho_{yy}} \\ \mathbb{E}R_{x,y} &= \frac{\rho_{xy}}{1 - \rho_{yy}}. \end{aligned}$$

That motivates the following

Definition 25. A state y is called **recurrent** if $\rho_{yy} = 1$ and **transient** otherwise.

Corollary 3. (1) A state y is transient iff $\mathbb{E}R_{y,y} < +\infty$
 (2) A state y is recurrent iff $\mathbb{P}(X_n = y \text{ infinitely often}) = 1$.

Indeed, if y is recurrent, then it's not transient and $\mathbb{E}R_{xy} = +\infty$, so $\rho_{yy} = 1$. This means that

$$\mathbb{P}(T_{yy}^1 < +\infty) = 1$$

Then for any $k \in \mathbb{N}$

$$\mathbb{P}(T_{xy}^k < +\infty) = 1.$$

Thus, taking countable intersection,

$$\begin{aligned} \mathbb{P}(\forall k \in \mathbb{N} \ T_{xy}^k < +\infty) &= 1, \text{ or} \\ \mathbb{P}(X_n = y \text{ infinitely often}) &= 1. \end{aligned}$$

□

Definition 26. A state x **communicates** with y if $\rho_{x,y} > 0$. (Notation: $x \rightarrow y$.)

Lemma 6. *Assume that x is recurrent and $x \rightarrow y$. Then y is also recurrent and $\rho_{xy} = \rho_{yx} = 1$.*

Proof. Suppose that $\rho_{yx} < 1$.

Since $\rho_{xy} > 0$, there exist $x_1, \dots, x_n$ such that

$$p(x, x_1) \cdot p(x_1, x_2) \dots p(x_{n-1}, x_n) \cdot p(x_n, y) > 0.$$

Now, x is recurrent, hence

$$\begin{aligned} 0 &= \mathbb{P}(T_{x,x} = \infty) \geq \mathbb{P}(T_{xx} = \infty, X_1 = x_1, \dots, X_n = x_n, X_{n+1} = y) = \\ &= \mathbb{P}(X_0 = x, X_1 = x_1, \dots, X_n = x_n, X_{n+1} = y) \cdot \\ &\quad \cdot \mathbb{P}(T_{xx} = +\infty | X_0 = x, X_1 = x_1, \dots, X_n = x_n, X_{n+1} = y) = \\ &= C \cdot \mathbb{P}(T_{yx} = +\infty) = C \cdot (1 - T_{yx}) > 0, \end{aligned}$$

since $C = \mathbb{P}(X_0 = x, X_1 = x_1, \dots, X_n = x_n, X_{n+1} = y)$ is positive. Contradiction proves the first claim: $\rho_{yx} = 1$.

So, there exist sequence of states $y_1, \dots, y_m$, such that $y_j \neq y$ and

$$\mathbb{P}(X_0 = y, X_1 = y_1, \dots, X_{m+1} = x) > 0.$$

We can bound the probability of returning to y below by considering only trajectories which go from y to x via $y_1, \dots, y_m$ and returning from x to y via $x_1, \dots, x_n$.

$$\begin{aligned} &\mathbb{P}(X_0 = y, X_{k+n+m+2} = y) \geq \\ &= \mathbb{P}(X_0 = y, \dots, X_m = y_m, X_{m+1} = x, X_{m+k+1} = x, \dots, X_{n+k+m+2} = y) = \\ &= p(y, y_1) \dots p(y_{m+1}, x) \cdot \mathbb{P}(X_{m+1} = x, X_{m+k+1} = x) \cdot p(x, x_1) \dots p(x_n, x) > \\ &> Const \cdot \mathbb{P}(X_0 = x, X_k = x). \end{aligned}$$

Using (*) we conclude that

$$\begin{aligned} \mathbb{E}R_{yy} &\geq \sum_{k=1}^{\infty} \mathbb{P}(X_n = y, X_{k+m+n+2} = y) \geq C \cdot \sum_{k=1}^{\infty} \mathbb{P}(X_0 = x, X_k = x) \\ &= C \mathbb{E}R_{xx} = +\infty. \end{aligned}$$

The last equality uses the assumption that x is recurrent. This shows that y is also recurrent, and as $\rho_{yx} = 1$, $y \rightarrow x$, we can use the first part of the proof to derive that $\rho_{xy} = 1$.

□

LECTURE 13. STATIONARY DISTRIBUTION AND ERGODICITY

Definition 27. A state x is a **positively recurrent**, if $\mathbb{E}_x(T_x) < +\infty$, where $T_x = T_x^1 = \inf\{n \in \mathbb{N} : X_n = x\}$

Definition 28. A subset $F \subset \mathbb{Z}$ is called **closed** if

$$\forall x \in F \forall y \ x \rightarrow y \Rightarrow y \in F$$

Definition 29. $F \subset S$ is **irreducible**, if $\forall E \subset F$ such that E is closed, $E = \emptyset$ or $E = F$. A Markov chain is called irreducible if its state space is irreducible.

Theorem 19 (decomposition of the set of recurrent states). Let $(X_n)_{n \in \mathbb{Z}_+}$ be a Markov chain. The set of all recurrent states can be decomposed in a disjoint union of closed irreducible subsets.

Proof. For the set R of recurrent states define a relation $x \sim y$ if $x \rightarrow y$. Easy to check that $\sim$ is an equivalence relation.

Definition 30. A probability measure π on S is called a **stationary distribution** for a Markov chain with the transition matrix p , if

$$\forall x \in S \ \pi(x) = \sum_{y \in S} \pi(y)p(y, x).$$

The stationary distribution is a left eigenvector of the transition matrix corresponding to the eigenvalue 1.

Theorem 20. Let $(X_n)_{n \in \mathbb{Z}_+}$ be an irreducible Markov chain (defined on an irreducible state space S) with stationary distribution π such that $\forall x \in S \ \pi(x) > 0$. Then the random process $(X_n)_{n \in \mathbb{Z}_+}$ with $\mathbb{P}(X_0 = x) = \pi(x)$ is ergodic.

Proof. As distribution is stationary, we can introduce a measure μ on $S^{\mathbb{Z}_+}$ such that

$$\mu\{(X_n)_{n \in \mathbb{Z}_+} : X_0 = x\} = \pi(x)$$

and it is invariant.

Let $A \subset S^{\mathbb{Z}_+}$ be an invariant set. Set

$$Y_n = \mathbb{E}[\mathbb{1}_A | \mathcal{A}_n].$$

Obviously, $(Y_n)_{n \in \mathbb{Z}_+}$ is a martingale. Moreover, it's bounded, therefore converges (by Martingale convergence theorem):

$$Y_n \rightarrow \mathbb{E}[\mathbb{1}_A | \mathcal{A}] = \mathbb{1}_A \text{ a.s.}$$

Now it's time to use that A is invariant:

$$Y_n = \mathbb{E}[\mathbb{1}_A(X) | \mathcal{A}_n] = \mathbb{E}[\mathbb{1}_{T^{-n}A}(X) | \mathcal{A}_n] = \mathbb{E}[\mathbb{1}_A(T^n X) | \mathcal{A}_n] = \mathbb{E}_{X_n}[\mathbb{1}_A(X)].$$

Denote $f(X) = Y_0 \cdot (1 - Y_0) = \mathbb{E}_{X_0}[\mathbb{1}_A(X)] \cdot (1 - \mathbb{E}_{X_0}[\mathbb{1}_A(X)])$. The equality above implies

$$\begin{aligned}\mathbb{E}[Y_n(1 - Y_n)] &= \mathbb{E}[\mathbb{E}_{X_n}[\mathbb{1}_A(X)] \cdot (1 - \mathbb{E}_{X_n}[\mathbb{1}_A(X)])] = \\ &= \mathbb{E}[f(T^n X)] = \mathbb{E}[f(X)] = \mathbb{E}[Y_0(1 - Y_0)].\end{aligned}$$

By the Dominated convergence theorem (Y_n is converging almost surely and $|Y_n| \leq 1$),

$$\mathbb{E}[Y_0(1 - Y_0)] = \mathbb{E} \lim_{n \rightarrow \infty} Y_n(1 - Y_n) = 0$$

This means that $Y_0(x) \in \{0, 1\}$ a.s.

Recall that

$$Y_0 = \mathbb{E}[\mathbb{1}_A(X)|\mathcal{A}_0] = \mathbb{E}[\mathbb{1}_A(X)|X_0].$$

Hence, there exists a set $F \subset S$ such that $Y_0 = \mathbb{1}_F(X_0)$ (A is the cylinder with base F). Now our aim is to show that F is closed.

$$\begin{aligned}Y_0 &= \mathbb{E}[\mathbb{1}_A(X)|\mathcal{A}_0] = \mathbb{E}[\mathbb{E}[\mathbb{1}_A(X)|\mathcal{A}_1]|\mathcal{A}_0] = \mathbb{E}[Y_1(X)|\mathcal{A}_0] = \\ &= \mathbb{E}[Y_0(TX)|\mathcal{A}_0] = \mathbb{E}[\mathbb{1}_F(X_1)|\mathcal{A}_0] = \mathbb{P}[X_1 \in F|X_0].\end{aligned}$$

We showed that

$$\mathbb{1}_F(X_0) = \mathbb{P}[X_1 \in F|X_0].$$

Let $x \in F$. Then the previous equality reads

$$1 = \mathbb{P}(X_1 \in F|X_0 = x) = \sum_{y \in F} p(x, y)$$

However,

$$\sum_{y \in S} p(x, y) = 1,$$

so for all $y \notin F$, $p(x, y) = 0$. Thus, F is closed, so it coincides with S or it's empty. Hence, the invariant set A has measure 0 or 1. $\square$

LECTURE 14. STATIONARY DISTRIBUTION AND POSITIVE RECURRENCE

Theorem 21. *Let $(X_n)_{n \in \mathbb{Z}_+}$ be an irreducible Markov chain. The following are equivalent*

- (1) *exists a stationary distribution π such that $\forall x \in S \pi(x) > 0$*
- (2) *$\pi(x) = \frac{1}{\mathbb{E}_x(T_x)}$ is a unique stationary distribution*
- (3) *there exists a positively recurrent state*
- (4) *any state is positively recurrent*

Proof. (1) $\Rightarrow$ (2) Recall that if $\theta : \Omega \rightarrow \Omega$ is a measure preserving ergodic transformation and $A \in \mathcal{F}$ is such that $\mu(\exists n \theta^n \omega \in A) = 1$, then for $T_A(\omega) = \inf\{n : \theta^n \omega \in A\}$

$$\mathbb{E}(T_A | \omega \in A) = \frac{1}{\mu(A)}.$$

Condition (1) and previous theorem imply that θ is ergodic. Set

$$A = \{(X_n)_{n \in \mathbb{Z}_+} : X_0 = x\}, \text{ then}$$

$$\mu(A) = \pi(x)$$

. Then $T_A(X) = T_x$ and so

$$\mathbb{E}_x(T_x) = \mathbb{E}(T_x | X_0 = x) = \frac{1}{\mu(A)}.$$

(2) $\Rightarrow$ (3) Follows from $\pi(S) = 1$.

(3) $\Rightarrow$ (4) Assume that x is positively recurrent and y is null-recurrent (it takes infinite time to return). Since the chain is irreducible, there exist trajectories $x, x_1, \dots, x_n, y$ and $y, y_1, \dots, x$ with positive transition probability. Hence,

$$\begin{aligned} \mathbb{E}_x(T_x) &\geq \mathbb{E}_X(T_x, X_1 = x_1, \dots, X_n = x_n, X_{n+1} = y, \dots, X_{T_x-m-1} = y, \\ &\quad X_{T_x-m} = y_1, \dots, X_{T_x} = x) \geq C \cdot \mathbb{E}_y(T_y) = +\infty. \end{aligned}$$

Contradiction proves (4).

(4) $\Rightarrow$ (1) It's enough to construct a stationary finite measure μ satisfying $\mu(x) > 0$ for all $x \in S$ (and then we can normalize it to obtain a stationary distribution). For any $x \in S$ define

$$\mu(y) = \mathbb{E}_x \left[\sum_{n=0}^{T_x-1} \mathbb{1}_{\{y\}}(X_n) \right]$$

(number of visits of y before returning to x). We have to prove that μ has both required properties.

- μ is a finite measure:

Note that since the chain is irreducible, $\mu(y) > 0$ for all y and

$$\begin{aligned} \sum_{y \in S} \mu(y) &= \sum_{y \in S} \sum_{n=0}^{\infty} \mathbb{P}_x(X_n = y, T_x > n) = \\ &= \sum_{n=0}^{\infty} \mathbb{P}_x(T_x > n) = \mathbb{E}_x[T_x] < +\infty \end{aligned}$$

- μ is stationary:

We have to show that $\forall z \in S$

$$\mu(z) = \sum_{y \in S} \mu(y) p(y, z).$$

Case 1. $z \neq x$

$$\begin{aligned} \sum_{y \in S} \mu(y) p(y, z) &= \sum_{y \in S} \sum_{n=0}^{\infty} \mathbb{P}_x(X_n = y, T_x > n) \cdot p(y, z) = \\ &= \sum_{n=0}^{\infty} \sum_{y \in S} \mathbb{P}_x(X_n = y, X_{n+1} = z, T_x > n+1) = \end{aligned}$$

(as $\mathbb{P}_x(X_0 = z, T_x > 0) = 0$ by assumption of the case 1, we can add this term in the last step)

$$= \sum_{m=1}^{\infty} \mathbb{P}_x(X_m = z, T_x > m) + \mathbb{P}_x(X_0 = z, T_x > 0) = \mu(z)$$

Case 2. $z = x$

$$\begin{aligned} \sum_{y \in S} \mu(y) p(y, z) &= \sum_{y \in S} \sum_{n=0}^{\infty} \mathbb{P}_x(X_n = y, T_x > n) \cdot p(y, x) = \\ &= \sum_{n=0}^{\infty} \sum_{y \in S} \mathbb{P}_x(X_n = y, X_{n+1} = x, T_x > n) = \\ &= \sum_{n=0}^{\infty} \sum_{y \in S} \mathbb{P}_x(X_n = y, X_{n+1} = x, T_x = n+1) = \\ &= \sum_{n=0}^{\infty} \mathbb{P}_x(T_x = n+1) = \mathbb{P}_x(T_x < +\infty) = 1 \end{aligned}$$

On the other hand, $\mu(z) = \mu(x) = 1$. Indeed,

$$\mathbb{P}_x(X_0 = x, T_x > 0) = 1$$

and for all $n > 0$

$$\mathbb{P}_x(X_n = x, T_x > n) = 0.$$

The theorem is completely proved. $\square$

The period of a state.

Let x be a state. Let

$$I_x = \{n \in \mathbb{N} \mid p^n(x, x) = \mathbb{P}_x(X_n = x) > 0\}.$$

Then I_x is a semigroup ($n \in I_x, m \in I_x$, then $n + m \in I_x$).

Definition 31 (period of a state). *The greatest common divisor of I_x is called the period of x .*

Lemma 7. *Let $(X_n)_{n \in \mathbb{Z}_+}$ be irreducible Markov chain. Then the periods of all states are the same.*

Proof. Assume that $\text{per}(x) = k$. Let $y \in S$. Since the chain is irreducible, $\exists m \ p^m(x, y) > 0$, $\exists l \ p^l(x, y) > 0$. Hence $m + l \in I_x$, so $\frac{m+l}{k} \in \mathbb{N}$. Let j be such that $p^j(y, y) > 0$. Then $m + j + l \in I_x$, so $\frac{m+j+l}{k} \in \mathbb{N}$ and thus $\frac{j}{k} \in \mathbb{N}$. We proved that $\frac{\text{per}(y)}{\text{per}(x)} \in \mathbb{N}$. Same is true for a reverse fraction, therefore $\text{per}(y) = \text{per}(x)$. $\square$

Definition 32. *The chain with period 1 is called aperiodic.*

LECTURE 15. EXCHANGEABLE EVENTS AND HEWITT-SAVAGE LAW

Let (V, U) be a measurable space, and let $(X_n : \Omega \rightarrow V)_{n \in \mathbb{Z}_+}$ be a sequence of random variables.

Definition 33 (exchangeable event). *A set $A \subset V^{\mathbb{Z}_+}$ is called exchangeable if for any finite permutation $\pi : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+ \ \forall \omega \in \Omega$*

$$(X_n)_{n \in \mathbb{Z}_+} \in A \Rightarrow (X_{\pi(n)})_{n \in \mathbb{Z}_+} \in A$$

Permutation π is called finite if $\pi : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+$ is a bijection such that $\exists N \in \mathbb{N} \ \forall n > N \ \pi(n) = n$.

Example. Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of independent random variables. The following events are exchangeable:

$$A = \{(X_n)_{n \in \mathbb{N}} : \lim_{n \rightarrow \infty} \sum_{k=1}^n X_k = +\infty\},$$

$$B = \{(X_n)_{n \in \mathbb{N}} : \limsup_{n \rightarrow \infty} \sum_{k=1}^n X_k \leq 7\}.$$

Note that A is a tail event, while B is not.

Lemma 8. *Let A be a tail event. Then A is exchangeable.*

Proof. Let $A \in \mathcal{T}$, then $(X_1, X_2, \dots) \in A \Leftrightarrow (X_{n+1}, X_{n+2}, \dots) \in T^n A$, where T is a shift. Let π be a finite permutation. Choose m such that $\forall n > m \ \pi(n) = n$. Then

$$\begin{aligned} (X_1, X_2, \dots) &\in A \\ &\Downarrow \\ (X_{m+1}, X_{m+2}, \dots) &\in T^m A \\ &\Downarrow \end{aligned}$$

$$\begin{aligned}
& (X_{\pi(m+1)}, X_{\pi(m+2)}, \dots) \in T^m A \\
& \quad \Updownarrow \\
& (X_{\pi(1)}, X_{\pi(2)} \dots) \in A.
\end{aligned}$$

Theorem 22 (Hewitt-Savage). *Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of i.i.d. random variables (taking values in a measurable space $(V, \mathcal{U})$). Let A be an exchangeable event. Then $\mathbb{P}(A) \in \{0, 1\}$.*

Proof. Let A be an exchangeable event. There exists a sequence of cylinders $\{C_k\}_{k=1}^\infty$ such that $\mathbb{P}(A \triangle C_k) \rightarrow_{k \rightarrow \infty} 0$. Fix $k \in \mathbb{N}$. Let

$$C_k = \{(X_n)_{n \in \mathbb{N}} : (X_1 \dots X_{m_k}) \in B_k, \text{ where } B_k \subset V^{m_k}\}.$$

Let $E \subset V^\mathbb{N}$ be an event. Then for any finite permutation π , $\mathbb{P}(E) = \mathbb{P}(\pi(E))$. Why? Because if

$$E = \{(X_n)_{n \in \mathbb{N}} : X_1 \in U_1, \dots, X_k \in U_k\},$$

then probabilities are clearly equal. Then by approximation argument it's true for all cylinders and for all measurable sets.

Consider a permutation π such that

- (1) $\forall n \leq m_k \pi(n) \in [m_k + 1, 2m_k]$
- (2) $\forall n \in [m_k + 1, 2m_k] \pi(n) \in [1, m_k]$
- (3) $\forall n > 2m_k \pi(n) = n$

This permutation swaps the intervals $[1, m_k]$ and $[m_k + 1, 2m_k]$. Then

$$\mathbb{P}(A \triangle C_k) = \mathbb{P}(\pi(A \triangle C_k)) = \mathbb{P}(\pi(A) \triangle \pi(C_k)) = \mathbb{P}(A \triangle \pi(C_k))$$

Set $C'_k = \pi(C_k)$. We have shown that

$$\mathbb{P}(A \triangle C_k) = \mathbb{P}(A \triangle C'_k).$$

Hence (triangle inequality),

$$\mathbb{P}(C_k \triangle C'_k) \leq \mathbb{P}(A \triangle C_k) + \mathbb{P}(A \triangle C'_k) \rightarrow_{k \rightarrow \infty} 0.$$

Also,

$$|\mathbb{P}(A) - \mathbb{P}(C_k)| \leq \mathbb{P}(A \triangle C_k) \rightarrow 0.$$

We have

$$\begin{aligned}
\mathbb{P}(C_k) &= \mathbb{P}(C_k \cap C'_k) + \mathbb{P}(C_k \setminus C'_k), \\
\mathbb{P}(C_k) &\rightarrow \mathbb{P}(A), \mathbb{P}(C_k \setminus C'_k) \rightarrow 0,
\end{aligned}$$

so

$$\mathbb{P}(C_k \cap C'_k) \rightarrow \mathbb{P}(A).$$

On the other hand, since the random variables $X_1, X_2, \dots$ are i.i.d.,

$$\begin{aligned}
\mathbb{P}(C_k \cap C'_k) &= \mathbb{P}((X_n)_{n \in \mathbb{N}} : (X_1, \dots, X_{m_k}) \in B_k \text{ and } (X_{\pi(1)}, \dots, X_{\pi(m_k)}) \in B_k) = \\
&= \mathbb{P}((X_n)_{n \in \mathbb{N}} : (X_1, \dots, X_{m_k}) \in B_k) \cdot \mathbb{P}((X_n)_{n \in \mathbb{N}} | (X_{\pi(1)}, \dots, X_{\pi(m_k)}) \in B_k) = \\
&= \mathbb{P}(C_k) \cdot \mathbb{P}(C'_k).
\end{aligned}$$

Finally,

$$\mathbb{P}(C'_k)\mathbb{P}(C'_k) \rightarrow \mathbb{P}(A),$$

and so (both multiples tend to $\mathbb{P}(A)$)

$$\mathbb{P}(A)^2 = \mathbb{P}(A).$$

□

Definition 34. Let $x \in S$. Define

$$I_x = \{n \in \mathbb{N} \mid \mathbb{P}_x(X_n = x) > 0\}$$

Period of the chain $\text{per}(X)$ is the greatest common divisor of I_x . If the chain has period 1, it is called aperiodic.

Lemma 9. Let $(X_n)_{n \in \mathbb{Z}_+}$ be an irreducible Markov chain. Then $\forall x, y \in S$ $p(x) = p(y)$.

(So we can talk about period of the chain)

Lemma 10. Assume that the chain is aperiodic, then

$$\forall x \in S \exists m \in \mathbb{N} \forall n > m \quad n \in I_x.$$

Proof. $\text{GCD}(I_x) = 1$, so there exists $\lambda_1, \dots, \lambda_k \in \mathbb{Z}, m_1, \dots, m_k \in I_x$ such that

$$\lambda_1 m_1 + \dots + \lambda_k m_k = 1.$$

WLOG, we assume that $\lambda_1, \dots, \lambda_j > 0, \lambda_{j+1}, \dots, \lambda_k < 0$. Set

$$l = \lambda_1 m_1 + \dots + \lambda_j m_j.$$

Then $l, (l-1) \in I_x$. Set $m = (l-1)^2$. Let $n > m$. Then $n = (l-1) \cdot a + b$, where $a \in \mathbb{N}, b < (l-1)$. Note that $a \geq (l-1)$. Hence,

$$n = (l-1)(a-b) + b(l-1) + b = (l-1)(a-b) + lb \in I_x.$$

□

Corollary 4. Let (X_n) be an irreducible Markov chain. For any $x, y \in S$ there exists $n \in \mathbb{N}$ such that

$$\mathbb{P}_x(X_n = x) > 0; \mathbb{P}_y(X_n = x) > 0.$$

Proof. Set

$$I_{yx} = \{n \in \mathbb{N} : \mathbb{P}_y(X_n = x) > 0\}.$$

If $p^k(y, x) > 0$, then $k + I_x \subset I_{yx}$, so the set I_{yx} contains all natural numbers greater than some $n_0 \in \mathbb{N}$. Since I_x possesses the same property, the intersection of these two sets is non-empty. □

LECTURE 16. CONDITION OF REGULARITY FOR MC

Theorem 23. *Let $(X_n)_{n \in \mathbb{Z}_+}$ be an aperiodic irreducible positively recurrent Markov chain. Let π be a stationary distribution, then the process $(X_n)_{n \in \mathbb{Z}_+}$ with the initial distribution π is regular.*

Proof. WLOG, we can assume that for any $x \in S$ and for any $\omega \in \Omega$ $\{n \in \mathbb{N} : X_n(\omega) = x\}$ is infinite. Let $A \subset \Omega$ be a tail event.

Step 1. Fix some $x \in S$. Let's show that $\mathbb{P}(A) \in \{0, 1\}$.

Let V be a set of all finite sequences $(s_0, \dots, s_{n+1}) \subset S$ defined by

$$V = \{(s_0, \dots, s_{n+1}) : s_0 = s_{n+1} = x, s_j \neq x, j = 1, \dots, n\}.$$

Define random variables $Y_0, Y_1, \dots : \Omega \rightarrow V$ by

$$Y_0 = (X_0, X_1, \dots, X_{T_x^1}),$$

$$Y_k = (X_{T_x^k}, X_{T_x^k+1}, \dots, X_{T_x^{k+1}}).$$

The random variables Y_k ($k \in \mathbb{N}$) are i.i.d. Why? Let us check that Y_0 and Y_1 are independent. The general case is treated in the same way.

Let $(s_0, \dots, s_{n+1}), (r_0, \dots, r_{m+1}) \in V$. Then $s_{n+1} = r_0 = x$, and thus,

$$\begin{aligned} & \mathbb{P}_x(Y_0 = (s_0, \dots, s_{n+1}), Y_1 = (r_0, \dots, r_{m+1})) = \\ &= \mathbb{P}_x(X_0 = s_0, \dots, X_n = s_n, X_{n+1} = s_{n+1}, X_{n+1} = r_0, \dots, X_{n+m+1} = r_{m+1}) = \\ &= \mathbb{P}_x(X_0 = s_0, \dots, X_{n+1} = s_{n+1}) \cdot \mathbb{P}_x(X_0 = r_0, X_{m+1} = r_{m+1}), \end{aligned}$$

so we have independence. And they are identically distributed by strong Markov property:

$$\begin{aligned} \mathbb{P}_x(Y_1 = (s_0, \dots, s_n)) &= \mathbb{E}_x[\mathbb{E}[\mathbb{1}_{(s_0, \dots, s_n)}(X_{T_x^1}, \dots, X_{T_x^1}) | \mathcal{A}_{T_x^1}]] = \\ &= \mathbb{E}_x[\mathbb{E}_{X_{T_x^1}} \mathbb{1}_{(s_0, \dots, s_n)}(X)] = \mathbb{E}_x[\mathbb{P}_x(X_0 = s_0, \dots, X_n = s_n)] \\ &= \mathbb{P}_x(Y_0 = (s_0, \dots, s_n)). \end{aligned}$$

Define an event $\tilde{A} \subset V^{\mathbb{Z}_+}$ by

$$\tilde{A} = \{(Y_n(\omega))_{n \in \mathbb{Z}_+} : \omega \in A\}.$$

Then, $\tilde{A}$ is exchangeable with respect to the sequence $\{Y_n\}_{n=0}^\infty$. Why?

Assume that $\omega \in A$. Let $U = \{(X_n(\omega))_{n \in \mathbb{Z}_+} : \omega \in A\}$. Any point $(s_n)_{n \in \mathbb{Z}_+} \in U$ can be mapped to a point of $\tilde{A}$ by considering the intervals between the successive returns to x . Fix $\omega \in A$ and let $s_n = X_n(\omega)$, so $(s_n)_{n \in \mathbb{Z}_+} \in U$. Since A is a tail event, for any finite permutation σ ,

$$(s_{\sigma(n)})_{n \in \mathbb{Z}_+} \in U.$$

Let m_1 be the smallest natural number m such that $s_m = x$, and let m_2 be the second smallest. Define $\sigma : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+$, swapping two number intervals:

$$\begin{aligned} [0, m_1] &\rightarrow [m_2 - m_1, m_2]; \\ [0, m_2 - m_1] &\rightarrow [m_1, m_2]. \end{aligned}$$

It corresponds to the transposition of Y_0 and Y_1 . Indeed, if $Y_0 = (s_0, \dots, s_{m_1+1})$, $Y_1 = (r_0, \dots, r_{m_2-m_1+1})$, then for the sequence $(s_{\sigma(n)})$, $Y_0 = (r_0, \dots, r_{m_2-m_1+1})$, $Y_1 = (s_0, \dots, s_{m_1})$. So, exchangeability with respect to Y_n follows from exchangeability with respect to X_n .

By Hewitt-Savage 0-1 Law, $\mathbb{P}(\tilde{A}) \in \{0, 1\}$. However, by construction of the random variables Y_n (and so, by construction of the measure on $V^{\mathbb{Z}_+}$), $\mathbb{P}(\tilde{A}) = \mathbb{P}_x(A)$, and thus $\mathbb{P}_x(A) \in \{0, 1\}$.

Step 2. We need to show, that $P_\pi(A) \in \{0, 1\}$. From Step 1 follows

$$f(x) = \mathbb{P}_x(A) = \mathbb{E}_\pi(\mathbb{1}_A | X_0 = x)$$

is an indicator function. This means

$$\mathbb{E}[\mathbb{1}_A(X) | \mathcal{A}_0] = \mathbb{1}_{U_0}(X_0).$$

for some $U_0 \subset S$. Hence

$$\mathbb{P}_\pi(A) = \pi(X_0 \subset U_0).$$

Suppose that $\pi(U_0) \in (0, 1)$, but $U_0 \neq \emptyset$ and $U_0 \neq S$.

Choose $x \in U_0$ and $y \notin U_0$.

Since the chain is aperiodic, there exists n such that

$$(3) \quad \mathbb{P}_\pi(X_0 = x, X_n = x) > 0, \mathbb{P}_\pi(X_0 = y, X_n = x) > 0.$$

As we have $\mathbb{P}_y(A) = 0$,

$$(4) \quad \mathbb{P}_y(A \text{ and } X_n = x) = 0.$$

Set $B = T^{-n}A$; then B is another tail event, probably, different from A . By Step 1,

$$\mathbb{P}_x(B) \in \{0, 1\}.$$

By the strong Markov property, for any $z \in S$,

$$\begin{aligned} \mathbb{P}_z(A \text{ and } X_n = x) &= \mathbb{E}_z(\mathbb{E}[\mathbb{1}_A(X) \cdot \mathbb{1}_x(X_n) | \mathcal{A}_n]) = \\ &= \mathbb{E}_z[\mathbb{E}[\mathbb{1}_{T^{-n}A}(T^n X) \cdot \mathbb{1}_x(X_n) | \mathcal{A}_n]] = \\ &= \mathbb{E}_z[\mathbb{E}_{X_n} \mathbb{1}_B(X) \mathbb{1}_x(X_n)] = \mathbb{E}_z[\mathbb{E}_x \mathbb{1}_B(X) \mathbb{1}_x(X_n)] \\ &= \mathbb{P}_x(B) \cdot \mathbb{P}_z(X_n = x). \end{aligned}$$

Assume that $\mathbb{P}_x(B) = 1$. Then, by the equation above and (3),

$$\mathbb{P}_y(A \text{ and } X_n = x) = \mathbb{P}_y(X_n = x) > 0,$$

which contradicts (4).

Assume now that $\mathbb{P}_x(B) = 0$. Then $\mathbb{P}_x(A \text{ and } X_n = x) = 0$. Since $\mathbb{P}_x(A) = 1$, we obtain

$$\mathbb{P}_x(X_n = x) \leq \mathbb{P}_x(A \text{ and } X_n = x) + \mathbb{P}_x(A^c) = 0,$$

which contradicts (3).

Theorem is proved. $\square$

LECTURE 17. MARKOV CHAIN CONVERGENCE, COUPLINGS

We've proved last time that if $(X_n)_{n \in \mathbb{Z}_+}$ is an aperiodic positively recurrent Markov chain with $\mathbb{P}(X_0 = x) = \pi(x)$, then (X_n) is a regular process. It means that for any measurable A

$$\sup_B |\mathbb{P}(X \in A, T^{-n}X \in B) - \mathbb{P}(X \in A)\mathbb{P}(X \in B)| \rightarrow 0.$$

Let $x, y \in S$. Let

$$\begin{aligned} A &= \{(X_n)_{n \in \mathbb{Z}_+} | X_0 = x\}; \\ B &= \{(X_n)_{n \in \mathbb{Z}_+} | X_0 = y\}. \end{aligned}$$

Then

$$\sup_{y \in S} |\mathbb{P}(X_0 = x, X_n = y) - \pi(x)\pi(y)| \rightarrow 0.$$

$$\mathbb{P}(X_0 = x, X_n = y) = \mathbb{P}(X_n = y | X_0 = x) \cdot \pi(x)$$

Since $\pi(x) > 0$,

$$\sup_{y \in S} |\mathbb{P}_x(X_n = y) - \pi(y)| \rightarrow 0.$$

Corollary 5. *Under the conditions of the Theorem, for any $s \in S$*

$$\lim_{n \rightarrow \infty} \sup_{y \in S} |\mathbb{P}_x(X_n = y) - \pi(y)| = 0.$$

So the theorem gives us uniform convergence condition. However for getting any information about rate of convergence, we need completely different tools.

From now we consider Markov chains with **finite state spaces** S . If μ is any distribution on S , then $\mu \in \mathbb{R}^n$

$$\forall x \in S \quad 0 \leq \mu(x) \leq 1,$$

$$\sum_{x \in S} \mu(x) = 1.$$

Let

$$\Delta_n = \{y \in \mathbb{R}^n : 0 \leq y_j \leq 1, \sum_{j=1}^n y_j = 1\}.$$

Theorem 24. *Let $(X_n)_{n \in \mathbb{Z}_+}$ be a finite Markov chain, then it has a stationary distribution.*

Proof. Let P be the transition matrix. By definition, the distribution $\pi \in \Delta_n$ is stationary if $\pi P = \pi$.

Let $F : \Delta_n \rightarrow \mathbb{R}^n$ be defined by $F(\mu) = \mu P$. Since P is stochastic, $F(\Delta_n) \subset \Delta_n$. By the Brouwer Fixed point theorem, F has a fixed point. $\square$

Note that we did not assume anything about the Markov chain here. For an irreducible Markov chain, we know that the chain should be positively recurrent. For irreducible finite chains, this condition is satisfied automatically. Indeed, since the chain is finite, there exists a recurrent state. By irreducibility, all states are recurrent. And again since the chain is finite, recurrence implies positive recurrence.

Theorem 25. *Let $(X_n)_{n \in \mathbb{Z}_+}$ be an aperiodic finite Markov chain with the initial distribution μ and stationary distribution π . Then*

$$\lim_{n \rightarrow \infty} \|\mu P^n - \pi\| = 0,$$

where the norm is $\|\nu\| = \sum_{x \in S} |\nu(x)|$.

Proof. Since the chain is finite, there exists $k \in \mathbb{N}$ such that for any $x, y \in S$, $P^k(x, y) > 0$. Let Π be an $|S| \times |S|$ matrix such that $\Pi(x, y) = \pi(y)$. Then,

$$\exists \delta > 0, P^k(x, y) \geq \delta \Pi(x, y)$$

for all $x, y \in S$. Set $\theta = 1 - \delta$. Define Q by

$$P^k = (1 - \theta)\Pi + \theta Q$$

Then Q is a stochastic matrix.

Claim 2.

$$\forall n \in \mathbb{N} P^{nk} = (1 - \theta^n)\Pi + \theta^n Q^n$$

Proof. We prove the Claim by induction. The base case follows from the definition of Q . Assume that the Claim holds for some n . Then,

$$\begin{aligned} P^{nk} \cdot P^k &= P^{(n+1)k} = (1 - \theta^n)\Pi P^k + \theta^n Q^n((1 - \theta)\Pi + \theta Q) = \\ &= (1 - \theta^n)\Pi + \theta^n(1 - \theta)\Pi + \theta^{n+1}Q^{n+1} = (1 - \theta^{n+1})\Pi + \theta^{n+1}Q^{n+1}, \end{aligned}$$

so the Claim is proved.

Let $0 \leq l < k$. Then

$$P^{nk+l} = P^l P^{nk} = (1 - \theta^n)\Pi + \theta^n P^l Q^n$$

Hence,

$$P^{nk+l} - \Pi = \theta^n (P^l Q^n - \Pi)$$

Therefore,

$$\|\mu P^{nk+l} - \pi\| = \|\mu P^{nk+l} - \mu \Pi\| \leq \theta^n \cdot (\|\mu P^l Q^n\| + \|\mu \Pi\|) \leq 2\theta^n \rightarrow 0$$

The theorem is proved. $\square$ The proof shows more than we claimed. We have shown that for a finite Markov chain, the rate of convergence is exponential.

Definition 35. Let $(\Omega_j, \mathcal{A}_j, \mathbb{P}_j)$, $j = 1, 2, 3$ be probability spaces and let $X_1 : \Omega_1 \rightarrow \mathbb{R}, X_2 : \Omega_2 \rightarrow \mathbb{R}$ be random variables. Then random variables $Y_1, Y_2 : \Omega_3 \rightarrow \mathbb{R}$ are called **a coupling** of X_1 and X_2 if $\forall B \in \mathcal{B}(\mathbb{R})$ (Borel σ -algebra on the line)

$$\mathbb{P}(X_i \in B) = \mathbb{P}(Y_i \in B), i = 1, 2.$$

Recall that, for a random variable $X : \Omega \rightarrow \mathbb{R}$, the distribution of X is a Borel measure on $\mathbb{R}$ defined by

$$\mu_X(B) = \mathbb{P}(X \in B), B \in \mathcal{B}(\mathbb{R}).$$

Theorem 26. Let $X_1 : \Omega \rightarrow \mathbb{R}, X_2 : \Omega_2 \rightarrow \mathbb{R}$ be random variables. Then

$$\|\mu_{X_1} - \mu_{X_2}\| = 2 \min \mathbb{P}(Y_1 \neq Y_2),$$

where the minimum is taken over all couplings (Y_1, Y_2) of X_1 and X_2 .

Norm on (non-negative) measures is defined as

$$\|\nu\| = \sup \left(\sum_{i=1}^{\infty} |\mu(A_j)| \right)$$

over all $\{A_j\} : \sqcup_{j=1}^{\infty} A_j = \mathbb{R}$. For a signed measure ν

$$\|\nu\| = \nu(A_+) - \nu(A_-) = |\nu(A_+)| + |\nu(A_-)|,$$

where $\mathbb{R} = A_+ \sqcup A_-$ is the Hahn decomposition.

Let $\nu = \mu_1 - \mu_2$, where μ_1 and μ_2 are probability measures. Then

$$\begin{aligned} \|\mu_1 - \mu_2\| &= (\mu_1 - \mu_2)(A_+) + 1 - 1 - (\mu_1 - \mu_2)(A_-) = \\ &= \mu_1(A_+) - \mu_2(A_+) + (\mu_1(A_+) + \mu_1(A_-)) - (\mu_2(A_+) + \mu_2(A_-)) - \\ &\quad - \mu_1(A_-) + \mu_2(A_-) = 2(\mu_1 - \mu_2)(A_+). \end{aligned}$$

In this case

$$\|\mu_1 - \mu_2\| = 2 \sup_{B \in \mathcal{B}(\mathbb{R})} (\mu_1 - \mu_2)(B) =: 2\|\mu_1 - \mu_2\|_{TV}.$$

Proof. Let (Y_1, Y_2) be a coupling of X_1 and X_2 . Then, for any $B \subset \mathbb{R}$

$$\mu_{X_1}(B) - \mu_{X_2}(B) = \mathbb{P}(Y_1 \in B) - \mathbb{P}(Y_2 \in B) \leq \mathbb{P}(Y_1 \in B, Y_2 \notin B) \leq \mathbb{P}(Y_1 \neq Y_2).$$

This establishes the inequality

$$\|\mu_{X_1} - \mu_{X_2}\| \leq 2 \inf \mathbb{P}(Y_1 \neq Y_2).$$

Next time we will discuss how to construct an optimal coupling, i.e. the coupling for which we have equality.

LECTURE 18. OPTIMAL COUPLING CONSTRUCTION, DISTANCE SHRINKING.

Definition 36 (Coupling). *Let $(\Omega_j, \mathcal{A}_j, \mathbb{P}_j)$, $j = 1, 2, 3$ be probability spaces and $X_1 : \Omega_1 \rightarrow \mathbb{R}, X_2 : \Omega_2 \rightarrow \mathbb{R}$ be random variables. Then random variables $Y_1, Y_2 : \Omega_3 \rightarrow \mathbb{R}$ are called a coupling of X_1 and X_2 if $\forall B \in \mathcal{B}(\mathbb{R})$*

$$\mathbb{P}(X_i \in B) = \mathbb{P}(Y_i \in B), i = 1, 2.$$

Note that one coupling always exists: independent coupling on product space

$$\begin{aligned} (Y_1, Y_2) : \Omega_1 \times \Omega_2 &\rightarrow \mathbb{R} : \\ (Y_1, Y_2)(\omega_1, \omega_2) &= (X_1(\omega_1), X_2(\omega_2)). \end{aligned}$$

Theorem 27. *Let $X_j : \Omega_j \rightarrow \mathbb{R}$ be random variables. Then*

$$\|\mu_{X_1} - \mu_{X_2}\| = 2 \min \mathbb{P}(Y_1 \neq Y_2)$$

over all couplings (Y_1, Y_2) .

Proof.

$$\|\mu_{X_1} - \mu_{X_2}\| \leq 2\mathbb{P}(Y_1 \neq Y_2)$$

for any coupling (done).

Let $\nu = \mu_{X_1} + \mu_{X_2}$. Then $\mu_{X_i} \ll \nu$ (absolutely continuous). Let densities be

$$\varphi_1 = \frac{d\mu_{X_1}}{d\nu}; \varphi_2 = \frac{d\mu_{X_2}}{d\nu}.$$

And set

$$\begin{aligned} f_1 &= \frac{\varphi_1 \wedge \varphi_2}{\|\varphi_1 \wedge \varphi_2\|_{L_1(\nu)}}, \\ f_2 &= \frac{(\varphi_1 - \varphi_2)_+}{\|(\varphi_1 - \varphi_2)_+\|_{L_1(\nu)}}, \\ f_3 &= \frac{(\varphi_1 - \varphi_2)_-}{\|(\varphi_1 - \varphi_2)_-\|_{L_1(\nu)}}. \end{aligned}$$

Define functions $Z_j : [0, 1] \rightarrow \mathbb{R}$ ($j = 1, 2, 3$) such that

$$\forall A \in \mathcal{B}(\mathbb{R}) \quad \mathbb{P}(Z_j \in A) = m(Z_j^{-1}(A)) = \int_A f_j d\nu$$

Let $\Omega = \{0, 1\} \times [0, 1]$,

$$\mathcal{A} = \{E \times A : E \subset \{0, 1\}, A \in \mathcal{B}([0, 1])\},$$

$$\mathbb{P}(\{1\} \times A) = p \cdot m(A),$$

$$\mathbb{P}(\{0\} \times A) = (1 - p) \cdot m(A),$$

where $p = \|\varphi_1 \wedge \varphi_2\|_{L_1(\nu)}$.

Set

$$Y_1(\delta, t) = \begin{cases} Z_1(t), & \text{if } \delta = 1 \\ Z_2(t), & \text{if } \delta = 0 \end{cases},$$

$$Y_2(\delta, t) = \begin{cases} Z_1(t), & \text{if } \delta = 1 \\ Z_3(t), & \text{if } \delta = 0 \end{cases}.$$

Let's prove that this is coupling and the inequality become equality on it.

Note that

$$\begin{aligned} \|(\varphi_1 - \varphi_2)_+\|_{L_1(\nu)} &= \int_{\mathbb{R}} (\varphi_1 - \varphi_2)_+ d\nu = \int_{\mathbb{R}} [(\varphi_1 - \varphi_2)_+ + \varphi_1 \wedge \varphi_2] d\nu - p = \\ &= \int_{\mathbb{R}} \varphi_1 d\nu - p = \mu_{X_1}(\mathbb{R}) - p = 1 - p. \end{aligned}$$

Similarly, $\|(\varphi_1 - \varphi_2)_-\|_{L_1(\nu)} = 1 - p$.

We have for any $A \in \mathcal{B}(\mathbb{R})$.

$$\begin{aligned} \mathbb{P}(Y_1 \in A) &= p \cdot \mathbb{P}(Z_1 \in A) + (1 - p) \mathbb{P}(Z_2 \in A) = \\ &= p \int_A f_1 d\nu + (1 - p) \int_A f_2 d\nu = \int_A \varphi_1 \wedge \varphi_2 d\nu + \int_A (\varphi_1 - \varphi_2)_+ d\nu = \\ &= \int_A \varphi_1 d\nu = \mu_{X_1}(A) = \mathbb{P}(X_1 \in A). \end{aligned}$$

We can write similar thing for Y_2 , so (Y_1, Y_2) is a coupling.

Now, let's check that constructed coupling is extremal.

$$\mathbb{P}(Y_1 \neq Y_2) = \mathbb{P}(Y_1(1, t) \neq Y_2(1, t)) \cdot p + \mathbb{P}(Y_1(0, t) \neq Y_2(0, t)) \cdot (1 - p).$$

By definition,

$$\begin{aligned} \mathbb{P}(Y_1(1, t) \neq Y_2(1, t)) &= 0, \\ \mathbb{P}(Y_1(0, t) \neq Y_2(0, t)) &= \mathbb{P}(Z_2(t) \neq Z_3(t)). \end{aligned}$$

Let $\mathbb{R} = A_+ \sqcup A_-$ be a Hahn decomposition for the measure $(f_2 - f_3)d\nu$. Then

$$\mathbb{P}(Z_2(t) = Z_3(t)) \leq \mathbb{P}(Z_2(t), Z_3(t) \in A_+) + \mathbb{P}(Z_2(t), Z_3(t) \in A_-) = 0.$$

Hence,

$$\mathbb{P}(Y_1(0, t) \neq Y_2(0, t)) = 1$$

and so,

$$\mathbb{P}(Y_1 \neq Y_2) = 1 - p.$$

On the other hand,

$$\|\mu_{X_1} - \mu_{X_2}\| = \int |\varphi_1 - \varphi_2| d\nu = \int [(\varphi_1 - \varphi_2)_+ + (\varphi_1 - \varphi_2)_-] d\nu = 2(1 - p).$$

The theorem is proved. $\square$

Recall

$$\|\mu_{X_1} - \mu_{X_2}\|_{TV} = \frac{1}{2} \|\mu_{X_1} - \mu_{X_2}\| = \sup_{A \in \mathcal{B}(\mathbb{R})} (\mu_{X_1}(A) - \mu_{X_2}(A)).$$

Consider a finite Markov chain. For initial distributions μ, ν set

$$\bar{d}_n(\mu, \nu) = \sup_{A \subset S} (\mathbb{P}_\mu(X_n \in A) - \mathbb{P}_\nu(X_n \in A)) = \|(\mu P^n) - (\nu P^n)\|_{TV}$$

and

$$\bar{D}_n = \max_{\mu, \nu \in \Delta_{|S|}} \bar{d}_n(\mu, \nu).$$

Note that $\bar{D}_n$ is attained on delta-measures $\mu = \delta_x$, $\nu = \delta_y$ for some $x, y \in S$.

Theorem 28. *For any $n, m \in \mathbb{N}$,*

$$\bar{D}_{m+n} \leq \bar{D}_n \cdot \bar{D}_m.$$

Proof. Let $x, y, z \in S$, $n, m \in \mathbb{N}$. Then

$$\mathbb{P}_x(X_{n+m} = z) - \mathbb{P}_y(X_{m+n} = z) = \mathbb{E}_x[\mathbb{P}_{X_m}(X_n = z)] - \mathbb{E}_y[\mathbb{P}_{Y_m}(Y_n = z)],$$

where (X_n) and (Y_n) are Markov chains with the same transitional probability P and initial distributions δ_x and δ_y .

Let (U, V) be the optimal coupling of X_m and Y_m . Then

$$\begin{aligned} \mathbb{P}_x(X_{m+n} = z) - \mathbb{P}_y(X_{m+n} = z) &= \mathbb{E}[P_U(X_n = z)] - \mathbb{E}[\mathbb{P}_V(X_n = z)] = \\ &= \mathbb{E}[\mathbb{P}_U(X_n = z) - \mathbb{P}_V(X_n = z)]. \end{aligned}$$

Let $A \subset S$. Define $\rho(A)$ by

$$\begin{aligned} \mathbb{P}_x(X_{m+n} \in A) - \mathbb{P}_y(X_{m+n} \in A) &= \\ &= \mathbb{E}[P_U(X_n \in A) - \mathbb{P}_V(X_n \in A)] = \rho(A). \end{aligned}$$

If $U = V$, then

$$P_U(X_n \in A) - \mathbb{P}_V(X_n \in A) = 0.$$

If $U \neq V$, then

$$P_U(X_n \in A) - \mathbb{P}_V(X_n \in A) \leq \bar{d}_n(U, V) \leq \bar{D}_n.$$

Hence,

$$\rho(A) \leq \bar{D}_n \cdot \mathbb{P}(U \neq V) = \bar{D}_n \cdot \bar{d}_m(\delta_x, \delta_y) \leq \bar{D}_n \cdot \bar{D}_m.$$

The equality above follows from the optimality of the coupling (U, V) .

Taking the supremum over $A \subset S$, we show that

$$\|(\delta_x P^{m+n}) - \delta_y P^{m+n}\|_{TV} \leq \bar{D}_n \cdot \bar{D}_m.$$

Taking another supremum over $x, y \in S$ completes the proof. $\square$

Definition 37. Let π be a stationary distribution of a finite state Markov chain. Set

$$D_n = \sup_{\mu} \bar{d}_n(\mu, \pi).$$

Clearly, $D_n \leq \bar{D}_n \leq 2D_n$.

Definition 38. Define the mixing time of the chain by

$$\tau_{mix} = \max_{x \in S} \tau_{mix}(x),$$

where

$$\tau_{mix}(x) = \inf_{n \in \mathbb{N}} \bar{d}_n(\delta_x, \pi) \leq \frac{1}{4}.$$

Corollary 6. Let μ be any initial distribution. Then for any $k \in \mathbb{N}$

$$\bar{d}_{k \cdot \tau_{mix}}(\mu, \pi) \leq 2^{-k},$$

and so,

$$D_{k \cdot \tau_{mix}} \leq 2^{-k}.$$

LECTURE 19. MARK. COUPLINGS, RANDOM MAPPING REPRESENTATION.

Recall definitions and facts from the previous lecture:

$$\bar{D}_n := \max_{\mu, \nu} \|\mu P^n - \nu P^n\|_{TV},$$

$$D_n := \max_{\mu} \|\mu P^n - \pi\|_{TV},$$

$$\bar{D}_{n+m} = \bar{D}_n \bar{D}_m,$$

$$\bar{D}_n \leq 2D_n \leq 2\bar{D}_n.$$

Also we defined mixing time

$$\tau_{mix}(\varepsilon) := \min\{n \in \mathbb{N} \mid D_n < \varepsilon\},$$

$$\tau_{mix} := \tau_{mix}\left(\frac{1}{4}\right)$$

$$n = \tau_{mix} \Rightarrow \forall k \in \mathbb{N} \bar{D}_{nk} \leq \bar{D}_n^k \leq 2^{-k},$$

$$\tau_{mix}(2^{-k}) \leq \tau_{mix} \cdot k.$$

Let $(X_n)_{n \in \mathbb{Z}_+}$, $(Y_n)_{n \in \mathbb{Z}_+}$ be finite Markov chains with the same transition matrix P . We want to construct coupled random processes $(\tilde{X}_n)_{n \in \mathbb{Z}_+}$, $(\tilde{Y}_n)_{n \in \mathbb{Z}_+}$, such that X and $\tilde{X}$ are equidistributed, Y and $\tilde{Y}$ are also equidistributed.

Let $((\tilde{X}_n, \tilde{Y}_n))_{n \in \mathbb{Z}_+}$ be a Markov chain on $S \times S$ with the transition matrix Q . Suppose that we want to have (for all $x_0, x_1, y_0 \in S$) the following condition:

$$\mathbb{P}(\tilde{X}_1 = x_1 | \tilde{X}_0 = x_0, \tilde{Y}_0 = y_0) = \mathbb{P}(\tilde{X}_1 = x_1 | \tilde{X}_0 = x_0) = P(x_0, x_1).$$

Let μ be the initial distribution of X_0 , and ν be the initial distribution of Y_0 . Then our condition implies

$$\begin{aligned} \mathbb{P}(\tilde{X}_1 = x_1) &= \sum_{x_0, y_0 \in S} \mathbb{P}(\tilde{X}_1 = x_1 | \tilde{X}_0 = x_0, \tilde{Y}_0 = y_0) \cdot \mathbb{P}(\tilde{X}_0 = x_0, \tilde{Y}_0 = y_0) = \\ &= \sum_{x_0, y_0 \in S} \mathbb{P}(\tilde{X}_1 = x_1 | \tilde{X}_0 = x_0) \mu(x_0) \nu(y_0) = \\ &= \sum_{x_0 \in S} \mathbb{P}(X_1 = x_1 | X_0 = x_0) \cdot \mu(x_0) = \mathbb{P}(X_1 = x_1). \end{aligned}$$

Note that this equality holds for any initial distributions μ and ν .

Similarly, for all $x_0, y_0, y_1 \in S$ we want to have

$$\mathbb{P}(\tilde{Y}_1 = y_1 | \tilde{X}_0 = x_0, \tilde{Y}_0 = y_0) = \mathbb{P}(y_0, y_1).$$

We will call couplings with these properties Markovian.

Definition 39. *The coupling is called Markovian, if*

- (1) $\forall x_0, x_1, y_0 \in S \quad \sum_{y_1 \in S} Q(x_0, y_0)(x_1, y_1) = P(x_0, x_1)$
- (2) $\forall x_0, y_0, y_1 \in S \quad \sum_{x_1 \in S} Q(x_0, y_0)(x_1, y_1) = P(y_0, y_1)$

Our analysis shows that Markovian coupling provides a coupling for any initial distributions.

We will also assume that

$$Q((x_0, x_0), (x_1, y_1)) = \begin{cases} P(x_0, x_1), & \text{if } y_1 = x_1, \\ 0, & \text{otherwise.} \end{cases}$$

This means that after the trajectories of $(X_n)_{n \in \mathbb{Z}_+}$ and $(Y_n)_{n \in \mathbb{Z}_+}$ meet, they evolve together.

Lemma 11. *If $(\tilde{X}_n)_{n \in \mathbb{Z}_+}, (\tilde{Y}_n)_{n \in \mathbb{Z}_+}$ is a Markovian coupling of $(X_n)_{n \in \mathbb{Z}_+}, (Y_n)_{n \in \mathbb{Z}_+}$ and $\tau_{couple} = \min\{n \in \mathbb{N} | \tilde{X}_0 = \tilde{Y}_0\}$, then*

$$\|\mu_{X_n} - \mu_{Y_n}\|_{TV} \leq \mathbb{P}(\tau_{couple} \geq n).$$

Proof.

$$\|\mu_{X_n} - \mu_{Y_n}\|_{TV} = \|\mu_{\tilde{X}_n} - \mu_{\tilde{Y}_n}\|_{TV} \leq \mathbb{P}(\tilde{X}_0 \neq \tilde{Y}_0) \leq \mathbb{P}(\tau_{couple} \geq n). \quad \square$$

But how to construct a nice Markovian coupling?

Example. Let $(Y_n)_{n \in \mathbb{Z}_+}$ be a simple random walk on $\mathbb{Z}$ started at 0.

$$P(n, m) = \begin{cases} p, & m = n + 1; \\ 1 - p, & m = n - 1; \\ 0, & \text{otherwise.} \end{cases}$$

Initial distribution – Dirac measure at 0.

Although we can view the simple random walk as a Markov chain with transition matrix P , there is a more convenient description. We can toss a biased coin at each step in order to choose a direction of the next movement. So, if $(Z_n)_{n \in \mathbb{N}}$ are i.i.d. ± 1 Bernoulli random variables with probabilities $p, 1 - p$, we can represent (Y_n) as

$$Y_0 = 0, \quad Y_{n+1} = Y_n + Z_{n+1}.$$

This is a **random mapping representation**. This leads to the following definition.

Definition 40. Let S be a finite or countable set, and let $f : S \times V \rightarrow S$ be a function. Let $(Z_n : \Omega \rightarrow V)_{n \in \mathbb{N}}$ be a sequence of i.i.d. random variables. Let Y_0 be an S -valued RV.

Define by induction $Y_n = f(Y_{n-1}, Z_n)$.

Then $(Y_n)_{n \in \mathbb{Z}_+}$ is a Markov chain on S with transitional probabilities

$$P(y_0, y_1) = \mathbb{P}(\omega \in \Omega \mid f(y_0, Z_0(\omega)) = y_1).$$

We say that this chain has a random mapping representation $(f, (Z_n : \Omega \rightarrow V)_{n \in \mathbb{N}})$.

Consider now a modification of the previous example. Let $(Y_n)_{n \in \mathbb{Z}_+}$ be a random walk on $\mathbb{Z}$ started at 0 with transition probabilities

$$P(n, m) = \begin{cases} p_n, & m = n + 1; \\ 1 - p_n, & m = n - 1; \\ 0, & \text{otherwise.} \end{cases}$$

for some sequence $\{p_n\}_{n \in \mathbb{Z}} \subset (0, 1)$. In this case, we cannot use i.i.d. Bernoulli random variables. Instead, we can take Z_n to be i.i.d. uniform RV on $[0, 1]$ and set

$$f(k, Z_0) = \begin{cases} k - 1, & \text{if } Z_0 < 1 - p_k, \\ k + 1, & \text{if } Z_0 \geq 1 - p_k. \end{cases}$$

And this is the general case:

Lemma 12. Let (X_n) be a finite Markov chain. Then it has a random mapping representation.

Proof. Let Z_n to be i.i.d. uniform RV on $[0, 1]$. Let $S = \{1 \cdots d\}$. For $1 \leq k, l \leq d$. Set

$$a_{l,k} = \sum_{j=1}^k P(l, j).$$

For $l \in S$ define

$$f(l, Z_0) = k, \text{ if } a_{l,k-1} \leq Z_0 < a_{l,k}.$$

Since $a_{l,d} = 1$, $f : \{1, \cdots d\} \times [0, 1] \rightarrow \{1, \cdots d\}$ is well-defined. Then

$$\mathbb{P}(f(l, Z_0) = k) = a_{l,k} - a_{l,k-1} = P(l, k) \quad \square$$

So, representation always exists. And it's clearly not unique (consider T , measure preserving transformation of $[0, 1]$, and composition $f \circ T$).

Our aim is to estimate τ_{mix} or $\mathbb{E}\tau_{couple}$ for the "good" coupling.

Example. Let G be a finite group. Let $(g_1, \cdots g_n)$ be a set of generators such that $g \in G$ implies $g^{-1} \in G$.

Define a transition matrix Q by

$$Q(u, v) = \begin{cases} \frac{1}{|E|}, & \text{if } v = gu, g \in E \\ 0, & \text{otherwise} \end{cases}$$

For example,

- 1) $G = \mathbb{Z}_d$. Random walk on discrete circle $E = \{+1, -1\}$.
- 2) Permutation group S_d , E be the set of transpositions in the random transposition shuffling. We can consider random transposition shuffling. (And the real-life shuffling too.)

To study mixing, we have to assume that the chain is aperiodic. This is not always the case. In particular, if d is even, then the random walk on $\mathbb{Z}_d$ is periodic. There is, however, a simple construction allowing to achieve aperiodicity.

If P be a transition matrix of a Markov chain then $Q = \frac{1}{2}(P + I)$ is called a transition matrix of the corresponding lazy chain. This chain is always aperiodic. If the original chain was mixing, then the mixing time of the lazy chain is larger. But this is not critical, since the mixing time increases at most twice.

LECTURE 20. RANDOM WALK ON DISCRETE TORUS.

Random walk on discrete torus $(\mathbb{Z}_n)^d$ is a Markov chain with the matrix

$$Q((k_1, \dots, k_d), (l_1, \dots, l_d)) = \begin{cases} \frac{1}{2d}, & \text{if } l_j = k_j + 1, l_i = k_i \text{ for } i \neq j; \\ \frac{1}{2d}, & \text{if } l_j = k_j - 1, l_i = k_i \text{ for } i \neq j; \\ 0, & \text{otherwise.} \end{cases},$$

Define lazy aperiodic random walk as before: $P = \frac{1}{2}(Q + I)$.

Theorem 29. *Let P be the transition matrix of a lazy random walk on $\mathbb{Z}_n^d$. Then*

$$\tau_{mix} \leq Cn^2 d \log d.$$

Proof. Case 1. $d = 1$

Let (U_m) be iid Bernoulli $\{0, 1\}$ random variables, and let (V_m) be independent iid Bernoulli $\{-1, 1\}$ random variables.

Consider $x, y \in \mathbb{Z}$ and let $(X_m), (Y_m)$ be lazy random walks, starting at x and y .

(1) If $X_m \neq Y_m$, then

- U_m selects which chain moves:

$$U_m = 1 \Rightarrow Y_{m+1} = Y_m,$$

$$U_m = 0 \Rightarrow X_{m+1} = X_m,$$

- V_m selects the direction:

$$U_m = 1, V_m = 1 \Rightarrow X_{m+1} = X_m + 1 \pmod{n},$$

$$U_m = 1, V_m = -1 \Rightarrow X_{m+1} = X_m - 1 \pmod{n}.$$

(2) If $X_m = Y_m$, then

- U_m determines whether X_m and Y_m move:

$$U_m = 0 \Rightarrow X_{m+1} = X_m = Y_m = Y_{m+1}$$

- V_m determines the direction:

$$U_m = 1, V_m = 1 \Rightarrow X_{m+1} = X_m + 1 \pmod{n},$$

$$U_m = 1, V_m = -1 \Rightarrow X_{m+1} = X_m - 1 \pmod{n}.$$

Let K_m be the distance between X_m and Y_m in the counterclockwise direction. Then $K_m \in \{0, \dots, n\}$. And (K_m) is a simple random walk on $\{0, \dots, n\}$ with absorbing barriers 0 and n .

$$\tau_{couple} = \min\{m \in \mathbb{N} : K_m \in \{0, n\}\}.$$

Consider (L_m) be a simple random walk on $\mathbb{Z}$, starting at $k \in \mathbb{Z}$:

$$L_0 = k, \quad L_m = \sum_{j=1}^m W_j + k$$

where W_j are Bernoulli ± 1 independent random variables. Note that L_m is a martingale and N , defined as the first time when $L_m \in \{0, n\}$,

is the stopping time. Now note that if $k \in \{0, \dots, n\}$ and $m \leq N$, then $L_m = K_m$. This means that N is our coupling time.

Moreover, we have $\mathbb{E}N < \infty$ (it's quite easy direct computation) and $|L_m - L_{m+1}| \leq 1$. Hence we can apply Doob's optional sampling theorem: the expectation of the stopped martingale

$$\mathbb{E}L_N = \mathbb{E}L_0 = k.$$

On the other hand,

$$\mathbb{E}L_N = n\mathbb{P}(L_N = n) + 0 \cdot \mathbb{P}(L_N = 0).$$

Hence,

$$\mathbb{P}(L_N = n) = \frac{k}{n}.$$

Define a random process (R_m) by

$$R_m = L_m^2 - m.$$

Then (R_m) is also a martingale:

$$\begin{aligned} \mathbb{E}(R_{m+1}|\mathcal{A}_m) &= \mathbb{E}((L_m + W_{m+1})^2 - m - 1|\mathcal{A}_m) = \\ &= L_m^2 + 2\mathbb{E}[L_m W_{m+1}|\mathcal{A}_m] + \mathbb{E}[W_{m+1}^2|\mathcal{A}_m] - m - 1 = R_m. \end{aligned}$$

And, by Doob's theorem,

$$\mathbb{E}R_N = \mathbb{E}R_0 = k^2.$$

On the other hand,

$$\mathbb{E}R_N = \mathbb{E}(L_N^2 - N) = n^2\mathbb{P}(L_N = n) + 0^2\mathbb{P}(L_N = 0) - \mathbb{E}N = n^2\frac{k}{n} - \mathbb{E}N.$$

So, $\mathbb{E}N = nk - k^2$.

We proved that

$$\mathbb{E}\tau_{couple} \leq \frac{n^2}{4},$$

so by Chebyshev's inequality

$$\mathbb{P}(\tau_{couple} > n^2) \leq \frac{1}{4},$$

and thus

$$\tau_{mix} \leq n^2.$$

Case 2. $d \in \mathbb{N}$

Let (J_m) be iid uniformly distributed in $\{1, \dots, d\}$. Let $(X_m)_{m \in \mathbb{Z}_+}$, $(Y_m)_{m \in \mathbb{Z}_+}$ be lazy random walks on $\mathbb{Z}_n^d$. As before,

- (1) if $X_m \neq Y_m$, then J_m determines the coordinate, U_m the walk which moves and V_m the direction;

- (2) if $X_m = Y_m$, then J_m determines the coordinate, U_m determines whether both walks move, V_m determines the direction.

Let $j \in \{1, \dots, d\}$. Let τ_{couple}^j be the first time when $X_{m,j} = Y_{m,j}$. Then

$$\tau_{couple} = \max_{j=1 \dots n} \tau_{couple}^j$$

Fix j . Let $\tilde{K}_{m,j}$ be the distance between $X_{m,j}$ and $Y_{m,j}$. The $(\tilde{K}_{m,j})$ is a (very) lazy random walk, absorbed at 0 and n . Let

$$\tilde{L}_{m,j} = \sum_{j=1}^m \tilde{W}_j$$

where $\tilde{W}_j \in \{0, 1, -1\}$, such that

$$\mathbb{P}(\tilde{W}_i = 1) = \mathbb{P}(\tilde{W}_i = -1) = \frac{1}{2d}$$

$$\mathbb{P}(\tilde{W}_j = 0) = 1 - \frac{1}{d}.$$

Let $\tilde{N}$ be the time of hitting $\{0, n\}$. Then $(\tilde{L}_{m,j})_{m \in \mathbb{Z}_+}$ is a martingale. Hence,

$$\begin{aligned} \mathbb{E}\tilde{L}_{\tilde{N},j} &= \mathbb{E}\tilde{L}_{0,j} = k \\ \mathbb{E}\tilde{L}_{\tilde{N},j} &= n \cdot \mathbb{P}(\tilde{L}_{\tilde{N},j} = n) \end{aligned}$$

and

$$\mathbb{P}(\tilde{L}_{\tilde{N},j} = n) = \frac{k}{n}$$

Set

$$\tilde{R}_{n,j} = \tilde{L}_{m,j}^2 - \frac{n}{d}$$

Then $(\tilde{R}_{m,j})$ is also a martingale:

$$\begin{aligned} \mathbb{E}[\tilde{R}_{m+1,j} | \mathcal{A}_m] &= \tilde{L}_{m,j}^2 - 2\mathbb{E}[\tilde{L}_{m,j} \cdot \tilde{W}_{m+1} | \mathcal{A}_m] + \mathbb{E}[\tilde{W}_{m+1}^2 | \mathcal{A}_m] - \\ &\quad - \frac{m+1}{d} - \tilde{L}_{m,j}^2 + \frac{1}{d} - \frac{m+1}{d} = \tilde{R}_{m,j}. \end{aligned}$$

So, applying Doob's theorem again,

$$\mathbb{E}\tilde{R}_{\tilde{N},j} = \mathbb{E}\tilde{R}_{0,j} = k^2.$$

$$\mathbb{E}\tilde{R}_{\tilde{N},j} = \mathbb{E}\tilde{L}_{\tilde{N},j}^2 - \frac{\mathbb{E}\tilde{N}}{d} = n^2 \cdot \mathbb{P}(\tilde{L}_{\tilde{N},j} = n) - \frac{\mathbb{E}\tilde{N}}{d} = nk - \frac{\mathbb{E}\tilde{N}}{d}.$$

We proved that

$$\mathbb{E}\tilde{N} = d(nk - k^2) \leq dn^2/4.$$

So

$$\mathbb{E}\tau_{couple}^j \leq \frac{dn^2}{4},$$

and

$$\mathbb{P}(\tau_{couple}^j > dn^2) \leq \frac{1}{4}.$$

Next time we will finish the proof.

LECTURE 21. RANDOM WALK ON TORUS-2, GRAPH COLORINGS.

Random walk on discrete torus $(\mathbb{Z}_n)^d$ is a Markov chain with the matrix

$$Q((k_1, \dots, k_d), (l_1, \dots, l_d)) = \begin{cases} \frac{1}{2d}, & \text{if } l_j = k_j + 1, l_i = k_i \text{ for } i \neq j; \\ \frac{1}{2d}, & \text{if } l_j = k_j - 1, l_i = k_i \text{ for } i \neq j; \\ 0, & \text{otherwise.} \end{cases},$$

Define lazy aperiodic random walk as before: $P = \frac{1}{2}(Q + I)$.

Theorem 30. *Let P be the transition matrix of a lazy random walk on $\mathbb{Z}_n^d$. Then*

$$\tau_{mix} \leq Cn^2 d \log d.$$

Proof.(continuation) Let $X_m = (X_{m,1}, \dots, X_{m,d})$ and τ_{couple}^j is the coupling time of $(X_{m,j})_{m \in \mathbb{Z}_+}$. We proved that

$$\mathbb{E}(\tau_{couple}^j) \leq \frac{dn^2}{4}.$$

Hence

$$\mathbb{P}\left(\tau_{couple}^j \geq \frac{dn^2}{2}\right) \leq \frac{1}{2}$$

The time τ_{couple}^j was the absorption time for the lazy random walk $\tilde{K}_{m,j}$ on $\{0, \dots, n\}$, starting at time $k = \{0, \dots, n\}$. Denote $Y_m = \tilde{K}_{m,j}$. Let N be the coupling time,

$$N = \min\{m \in \mathbb{N} : Y_n \in \{0, n\}\}.$$

So,

$$\mathbb{P}(N \geq \overline{m}) \leq \frac{1}{2},$$

where $\overline{m} = dn^2/2$.

Lemma 13.

$$(5) \quad \mathbb{P}(N > k\overline{m}) \leq \left(\frac{1}{2}\right)^k.$$

Proof of lemma. Let $\mathcal{I} = \{1, \dots, n-1\}$. Then

$$\mathbb{P}(N > k\bar{m}) = \mathbb{E}\mathbb{1}_{\mathcal{I}}(Y_{k\bar{m}}).$$

By induction on k , assume that (5) holds for k . Then

$$\begin{aligned} \mathbb{E}\mathbb{1}_{\mathcal{I}}(Y_{(k+1)\bar{m}}) &= \mathbb{E}(\mathbb{E}[\mathbb{1}_{\mathcal{I}}(Y_{(k+1)\bar{m}}) | \mathcal{A}_{k\bar{m}}]) = \\ &= \mathbb{E}(\mathbb{E}[\mathbb{1}_{\mathcal{I}}(Y_{(k+1)\bar{m}}) \mathbb{1}_{\mathcal{I}}(Y_{k\bar{m}}) | \mathcal{A}_{k\bar{m}}]) = \mathbb{E}(\mathbb{E}[\mathbb{1}_{\mathcal{I}}(Y_{(k+1)\bar{m}}) | \mathcal{A}_{k\bar{m}}] \mathbb{1}_{\mathcal{I}}(Y_{k\bar{m}})) = \\ &\text{(strong Markov)} \\ &= \mathbb{E}[\mathbb{E}_{Y_{k\bar{m}}} \mathbb{1}_{\mathcal{I}}(Y_{\bar{m}}) \mathbb{1}_{\mathcal{I}}(Y_{k\bar{m}})] \leq \\ &\text{(by (5) with } k=1) \\ &\leq \mathbb{E}[\frac{1}{2} \mathbb{1}_{\mathcal{I}}(Y_{k\bar{m}})] = \frac{1}{2} \mathbb{P}(N > k\bar{m}). \end{aligned}$$

Lemma is proved.

The Lemma implies

$$\mathbb{P}(\tau_{couple}^j > k\bar{m}) \leq \left(\frac{1}{2}\right)^k.$$

Take $k = \log_2(4d)$. Then

$$\mathbb{P}(\tau_{couple}^j > k\bar{m}) \leq \frac{1}{4d}.$$

Recall that $\tau_{couple} = \max_{j=1, \dots, d} \tau_{couple}^j$. Hence,

$$\mathbb{P}(\tau_{couple} > k\bar{m}) \leq \sum_{j=1}^d \mathbb{P}(\tau_{couple}^j > k\bar{m}) \leq \frac{1}{4},$$

and so,

$$\tau_{mix} \leq k\bar{m} \leq \log_2(4d) \frac{n^2 d}{2}.$$

Theorem is proved. $\square$

Counting of graph colorings.

Definition 41. Let $G = (V, E)$ be a graph. A mapping $g : V \rightarrow \{1 \dots q\}$ is called a proper coloring, if

$$\forall x, y \in V (x, y) \in E \Rightarrow \varphi(x) \neq \varphi(y).$$

Theorem 31 (Valiant). Counting of proper colorings is $\sharp$ -P-complete problem. (It means, in particular, that if there is a polynomial time algorithm for counting the number of proper colorings, then $P = NP$.)

We will discuss the idea of Jerrum, Sinclair for estimation of the number of proper colorings. Their approach is based on Markov chain Monte Carlo (MCMC) method originally introduced by von Neumann.

- (1) Generate a random proper coloring
- (2) Use it to estimate the number of proper colorings

Metropolis chain.

Take a proper color, choose a random vertex x uniformly from V . Choose a color j uniformly from $\{1 \dots q\}$. Replace the color of x by j , if new coloring is proper, otherwise retain the old one.

Lemma 14. *Let d be the max degree of a vertex. If $q > 2d$, then the stationary distribution for Metropolis chain is uniform on proper colorings.*

Proof. Let $\tilde{G}$ be the graph of all proper colorings. The vertices φ, ψ are connected by the edge, if

$$|\{x \in V : \varphi(x) \neq \psi(x)\}| = 1$$

Claim 3. $\tilde{G}$ is connected.

Define a Hamming metric on $\tilde{G}$ by

$$\rho(\varphi, \psi) = |\{x \in V : \varphi(x) \neq \psi(x)\}|.$$

Let $\tilde{G}_1$ and $\tilde{G}_2$ are two nonempty connected components of $\tilde{G}$. Take $\varphi \in \tilde{G}_1$ and $\psi \in \tilde{G}_2$ such that

$$\rho(\varphi, \psi) = \text{dist}(\tilde{G}_1, \tilde{G}_2).$$

Let $x \in V$ be such that $\varphi(x) \neq \psi(x)$. Since $q > 2d$, there is a color $j \in \{1, \dots, q\}$, that is proper for x for both φ and ψ . Recolor x to j , namely, let

$$\varphi'(y) = \begin{cases} j, & \text{if } y = x, \\ \varphi(y), & \text{otherwise,} \end{cases}$$

$$\psi'(y) = \begin{cases} j, & \text{if } y = x, \\ \psi(y), & \text{otherwise.} \end{cases}$$

Then $\varphi \leftrightarrow \varphi'$ so $\varphi' \in \tilde{G}_1$, $\psi \leftrightarrow \psi'$, so $\psi' \in \tilde{G}_2$ and

$$\rho(\varphi', \psi') = \rho(\varphi, \psi) - 1.$$

Contradiction (we took minimizing pair of colorings.) Claim is proved.

Since $\tilde{G}$ is connected, Metropolis chain is irreducible. Let π be its stationary distribution. For a vertex $x \in V$ and for a color $j \in \{1, \dots, q\}$, denote by $\varphi_{x,j}$ the coloring φ with the vertex x recolored with the color j . Let Q_x be the set of colors j for which $\varphi_{x,j}$ is a proper coloring.

The equality $\pi = \pi P$ can be rewritten as

$$\pi(\varphi) = \frac{1}{|V|} \sum_{x \in V} \left(\frac{q - |Q_x|}{q} \pi(\varphi) + \sum_{j \in Q_x} \frac{1}{q} \pi(\varphi_{x,j}) \right),$$

and so,

$$(6) \quad \pi(\varphi) = \frac{1}{|V|} \sum_{x \in V} \frac{1}{|Q_x|} \sum_{j \in Q_x} \pi(\varphi_{x,j}).$$

This means that $\pi(\varphi)$ is an average of the values of π at the neighboring proper colorings $\varphi_{x,j}$. This is enough to show that $\pi(\varphi) = \text{const.}$ Indeed, assume that π attains its maximal value at the coloring φ . Then by (6), $\pi(\varphi) = \pi(\varphi_{x,j})$ for all neighbors $\varphi_{x,j}$ of φ . Since the graph $\tilde{G}$ is connected, $\pi(\psi) = \pi(\varphi)$ for any coloring $\psi \in \tilde{G}$. $\square$

LECTURE 22. COUNTING OF GRAPH COLORINGS.

Definition 42. Let $G = (V, E)$ be a graph. A mapping $g : V \rightarrow \{1 \dots q\}$ is called a proper coloring, if

$$\forall x, y \in V (x, y) \in E \Rightarrow \varphi(x) \neq \varphi(y).$$

Theorem 32 (Valiant). Counting of proper colorings is $\sharp$ -P-complete problem

Metropolis chain. Let φ be a proper coloring, choose a random vertex x uniformly from V . Choose a color k uniformly from $\{1 \dots q\}$. Let $\psi : V \rightarrow N$

$$\psi(y) = \begin{cases} k, & \text{if } y = x, \\ \varphi(y), & \text{otherwise.} \end{cases}$$

We will call the coloring ψ *allowed* if for all y $y \leftrightarrow x \Rightarrow \psi(y) \neq k$. (Note that here we are interested only in the vertex x , so this notion can be applied to improper colorings).

If ψ is an allowed coloring and $X_n = \varphi$, set $X_{n+1} = \psi$. Otherwise, set $X_{n+1} = \varphi$. So, we constructed a Markov chain of colorings. Note that all proper colorings are its recurrent states, and improper colorings are the transient states.

Theorem 33. Let $(X_n)_{n \in \mathbb{Z}_+}$ be a Metropolis chain of proper colorings of a proper $G = (V, E)$ with $q > 3d$. Then for any $\varepsilon > 0$,

$$\tau_{\text{mix}}(\varepsilon) \leq cv \log \frac{v}{\varepsilon},$$

where $c = c(q, d)$, v denotes total number of vertices, q – total number of colors, d – maximal degree of a vertex.

Take any $X_0 = \varphi$, define $(X_n)_{n \in \mathbb{Z}_+}$ by i.i.d. random variables $(V_n)_{n \in \mathbb{Z}_+}$ (vertices), $(q_n)_{n \in \mathbb{Z}_+}$ (colors). Recall that Hamming distance between colorings was defined as

$$\rho(\varphi, \psi) = |\{x \in V : \varphi(x) \neq \psi(x)\}|.$$

Case 1.

Assume that $\rho(\varphi, \psi) = 1$ and $X_0 = \varphi$, $Y_0 = \psi$. Then for some $c' > 0$

$$\mathbb{E}\rho(X_1, Y_1) \leq 1 - \frac{c'}{v}.$$

Indeed, suppose that $\tilde{v}$ is the unique different vertex, $\varphi(\tilde{v}) \neq \psi(\tilde{v})$. There are 3 possible cases:

- If $V_1 \neq \tilde{v}$ (V_1 is the first vertex of possible change), $V_1 \leftrightarrow \tilde{V}$, then $\rho(X_1, Y_1) = \rho(X_0, Y_0) = 1$.
- If $V_1 = \tilde{V}$, then $\mathbb{P}(\rho(X_1, Y_1) = 0 \mid V_1) \geq \frac{q-d}{q}$.
- If $V_1 \leftrightarrow \tilde{V}$, then $\mathbb{P}(\rho(X_1, Y_1) = 2 \mid V_1) \leq \frac{2}{q}$, because q_1 should be either $\varphi(\tilde{v})$ or $\psi(\tilde{v})$

therefore

$$\begin{aligned} \mathbb{E}(\rho(X_1, Y_1) - 1) &= (-1) \cdot \mathbb{P}(\rho(X_1, Y_1) = 0 \mid V_1 = \tilde{v}) \cdot \mathbb{P}(V_1 = \tilde{v}) \\ &\quad + \mathbb{P}(\rho(X_1, Y_1) = 0 \mid V_1 \leftrightarrow \tilde{v}) \cdot \mathbb{P}(V_1 \leftrightarrow \tilde{v}) \\ &\leq (-1) \frac{1}{v} \frac{q-d}{q} + \frac{d}{v} \frac{2}{q} = -\frac{q-3d}{q} \frac{1}{v}. \end{aligned}$$

Case 2.

Assume that $\rho(\varphi, \psi) = r$. Then

$$\mathbb{E}\rho(X_1, Y_1) \leq r \cdot \left(1 - \frac{c'}{v}\right)$$

There exist colorings $\varphi_0, \varphi_1, \dots, \varphi_r$, such that $\varphi_r = \psi$, $\varphi_0 = \varphi$, distance between every two successive colorings is one. (Note that some φ_i may be improper.)

Let $(X_n^k)_{n \in \mathbb{Z}_+}$ be a coupled Metropolis chain with $X_0^k = \varphi_k$. Since ρ is a metric,

$$\begin{aligned} \mathbb{E}\rho(X_1, Y_1) &= \mathbb{E}\rho(X_1^0, X_1^r) \leq \sum_{j=0}^{r-1} \mathbb{E}\rho(X_1^j, X_1^{j+1}) \leq \\ &\leq \sum_{j=0}^{r-1} \left(1 - \frac{c'}{v}\right) = r \left(1 - \frac{c'}{v}\right). \end{aligned}$$

This means that

$$\mathbb{E}\rho(X_1, Y_1) \leq \rho(X_0, Y_0) \cdot \left(1 - \frac{c'}{v}\right).$$

Lemma 15. *We have geometric shrinking: if $X_0 = \varphi, Y_0 = \psi$,*

$$\mathbb{E}\rho(X_n, Y_n) \leq \rho(X_0, Y_0) \left(1 - \frac{c'}{v}\right)^n.$$

By induction,

$$\begin{aligned} \mathbb{E}\rho(X_{n+1}, Y_{n+1}) &= \mathbb{E}[\mathbb{E}[\rho(X_{n+1}, Y_{n+1}) | \mathcal{A}_n]] = \mathbb{E}[\mathbb{E}_{(X_n, Y_n)} \rho(X_1, Y_1)] \leq \\ &\leq \mathbb{E} \left[\rho(X_n, Y_n) \left(1 - \frac{c'}{v}\right) \right]. \end{aligned}$$

Lemma is proved.

Let φ, ψ be proper colorings, $(X_n)_{n \in \mathbb{Z}_+}, (Y_n)_{n \in \mathbb{Z}_+}$ be Metropolis chains with $X_0 = \varphi, Y_0 = \psi$. Then

$$\begin{aligned} \mathbb{P}(\tau_{\text{couple}} > n) &= \mathbb{P}(X_n \neq Y_n) = \mathbb{P}(\rho(X_n, Y_n) \geq 1) \leq \\ &(\text{Chebyshev inequality}) \end{aligned}$$

$$\leq \mathbb{E}\rho(X_n, Y_n) \leq \rho(\varphi, \psi) \cdot \left(1 - \frac{c'}{v}\right)^n \leq v \cdot \left(1 - \frac{c'}{v}\right)^n.$$

If for any proper φ, ψ $\mathbb{P}(\tau_{\text{couple}} > n) < \varepsilon$, then

$$\tau_{\text{mix}}(\varepsilon) \leq n.$$

If we take

$$n \geq \frac{v}{c'} \log \left(\frac{v}{\varepsilon} \right),$$

then

$$v \cdot \left(1 - \frac{c'}{v}\right)^n < \varepsilon.$$

So, $\tau_{\text{mix}}(\varepsilon) \leq cv \log \frac{v}{\varepsilon}$ and theorem is proved. $\square$

Now we want to estimate the size of the set of proper colorings $|S_v|$.

Enumerate the vertices of G . Select one proper coloring φ and fix it. For $1 \leq k \leq v$, let S_k be the set of all proper colorings φ such that for all $j > k$

$$\psi(j) = \varphi(j).$$

Then

$$\begin{aligned} |S_v| &= \prod_{k=1}^v \frac{|S_k|}{|S_{k-1}|}, \\ |S_0| &= 1 \end{aligned}$$

Idea: run the Metropolis algorithm on S_k starting from φ . Let π_k be the stationary distribution on S_k , which is uniform distribution. Then

$$\pi_k(S_{k-1}) = \frac{|S_{k-1}|}{|S_k|}.$$

So we need to approximate π_k . If $n \geq \tau_{mix}$ for this chain, then

$$|\mathbb{P}(X_n^k \in S_{k-1}) - \pi_k(S_{k-1})| \leq \|\mu_{X_n^k} - \pi\|_{TV} < \varepsilon.$$

Theorem 34 (Jerrum, Sinclair, 1996). *Assume that $q > 3d$. Then we can construct an estimator Y for the number of proper colorings $\text{col}(G)$ such that*

$$\mathbb{P}[(1 - \varepsilon)\text{col}(G) \leq Y \leq (1 + \varepsilon)\text{col}(G)] \geq 1 - \delta,$$

where Y depends on $cv^3 \log v$ random samples (it means that $Y = f(Z_1, \dots, Z_n)$, where Z_j are i.i.d).

LECTURE 23. COUNTING OF GRAPH COLORINGS - ALGORITHM.

Theorem 35 (Jerrum, Sinclair, 1996). *Assume that $q > 3d$. Then we can construct an estimator Y for the number of proper colorings $\text{col}(G)$ such that*

$$\mathbb{P}[(1 - \varepsilon)\text{col}(G) \leq Y \leq (1 + \varepsilon)\text{col}(G)] \geq 1 - \delta,$$

where Y depends on $cv^3 \log v$ random samples (it means that $Y = f(Z_1, \dots, Z_n)$, where Z_j are i.i.d).

Proof. Let π_A denote a uniform distribution on A . Take a chunk $A_1 \subset A$.

$$\pi_A(Z \in A_1) = \pi_A(A_1) = \frac{|A_1|}{|A|}.$$

Let $Z_1^1, Z_2^1, \dots$ be i.i.d copies of Z . Set

$$W_1 = \frac{1}{m_1} \sum_{j=1}^1 m \mathbb{1}_{A_1}(Z_j^1).$$

Then $W_1 \rightarrow \mathbb{E}W_1 = \pi_A(A_1)$ a.s. (Law of large numbers)

On the next step, define π_{A_1} – uniform distribution on A_1 .

$$\pi_{A_1}(A_2) = \frac{|A_2|}{|A_1|}.$$

Let $Z_1^2, Z_2^2, \dots$ be i.i.d copies of Z , taken uniformly on A_1 . Set

$$W_2 = \frac{1}{m_2} \sum_{j=1}^{m_2} \mathbb{1}_{A_2}(Z_j^2).$$

Then $W_2 \rightarrow \mathbb{E}W_2 = \pi_{A_1}(A_2)$ a.s.

Eventually, we have an estimation:

$$|A_v| = 1,$$

$$|A| = |A_0| = \prod_{j=1}^n \frac{1}{W_j} = \frac{1}{W}.$$

Realization of this algorithm will involve errors: expectations are limits, so they can be computed only approximately. Let's estimate the errors.

We want to have

$$\begin{aligned}
 (1 - \varepsilon)|A_0| &\leq \frac{1}{W} \leq |A_0|(1 + \varepsilon) \\
 &\Leftrightarrow \\
 \left| \frac{1}{|A_0|} \cdot \frac{1}{W} - 1 \right| &< \varepsilon \\
 &\Leftrightarrow \\
 \left| \log \frac{1}{|A_0|W} \right| &< \varepsilon \\
 &\Leftrightarrow \\
 \left| \log \frac{W}{1/|A_0|} \right| &< \varepsilon \\
 \log \frac{W}{1/|A_0|} &= \log \frac{\prod_{j=1}^v W_j}{\prod_{j=1}^v \frac{|A_{j+1}|}{|A_j|}} = \sum_{j=1}^v \log \frac{W_j}{\pi_{A_{j-1}}(A_j)} = \\
 (7) \quad &= \sum_j \left[\log \frac{W_j}{\mathbb{E}W_j} + \log \frac{\mathbb{E}W_j}{\pi_{A_{j-1}}(A_j)} \right].
 \end{aligned}$$

If (X_n) is a Metropolis chain of proper colorings on $G = (V, E)$, $|V| = v$, $q > 3d$ (q – number of colors, d – number of vertices), then

$$t_{mix}(\varepsilon) \leq Cn \log \frac{n}{\varepsilon}.$$

Let $0 \leq k \leq v$,

$$S_k = \{\psi - \text{proper coloring such that } \psi(i) = \varphi(i) \forall i > k\},$$

where φ is some fixed proper coloring. $|S_0| = 1$, $S_0 \subset S_1 \subset \dots \subset S_v$, S_v is the set of all proper colorings.

Let $m_k, n_k \in \mathbb{N}$. Let (X_n) be a Metropolis chain on the set S_k , starting from φ . Set

$$Z^k = \mathbb{1}_{S_{k-1}}(X_{n_k}).$$

Let $Z_1^k, Z_2^k, \dots, Z_{m_k}^k$ be i.i.d. copies of Z^k . Set

$$W_k = \frac{1}{m_k} \sum_{j=1}^{m_k} Z_j^k.$$

$$\mathbb{E}W_k = \mathbb{E}\mathbb{1}_{S_{k-1}}(X_{n_k}) = \mathbb{P}(X_{n_k} \in S_{k-1}),$$

$$|\mathbb{P}(X_{n_k} \in S_{k-1}) - \pi_{S_k}(S_{k-1})| \leq \|\mu_{X_{n_k}} - \pi_{S_k}\|_{TV} < \varepsilon_k,$$

whenever $n_k \geq t_{mix}(\varepsilon_k)$.

Hence,

$$(8) \quad \left| \frac{\mathbb{E}W_k}{\pi_{S_k}(S_{k-1})} - 1 \right| < \frac{\varepsilon_k}{\pi_{S_k}(S_{k-1})} \leq q\varepsilon_k,$$

as by definition

$$(9) \quad \frac{1}{\pi_{S_k}(S_{k-1})} = \frac{|S_k|}{|S_{k-1}|} \leq q.$$

Therefore,

$$\left| \sum_{k=1}^v \log \frac{\mathbb{E}W_k}{\pi_{S_k}(S_{k-1})} \right| \leq c \cdot \sum_{k=1}^v \left| \frac{\mathbb{E}W_k}{\pi_{S_k}(S_{k-1})} - 1 \right| < c \cdot \sum_{k=1}^v q\varepsilon_k < \varepsilon,$$

if we choose $\varepsilon_k = \frac{C\varepsilon}{k^2}$ with appropriate C .

This means that we have to run the k -th chain for

$$t_{mix}(\varepsilon_k) \leq Cv \log \frac{v}{\varepsilon_k} \leq Cv \log \frac{v k^2}{c \varepsilon} \leq C'v \log \frac{v}{\varepsilon}$$

steps to bound for the second term of (7) by ε .

The total number of random variables we need to generate is at most $N = \sum_{k=1}^v m_k C'v \log \frac{v}{\varepsilon}$. And m_k needs to be determined such that we'll get appropriate bound for the first term in (7).

Recall that

$$W_k = \frac{1}{m_k} \sum_{j=1}^{m_k} Z_j^k,$$

where W_k are i.i.d. Bernoulli random variables. By (8) and (9),

$$\mathbb{E}W_k = \mathbb{E}Z_j^1 \geq \frac{1}{2}\pi_{S_k}(S_{k-1}) \geq \frac{1}{2q}.$$

Bernstein's inequality implies

Lemma 16. *Let $Y_1, \dots, Y_m$ be i.i.d. Bernoulli. Then for $0 \leq t \leq 1$.*

$$\mathbb{P} \left[\sum_{j=1}^m (Y_j - \mathbb{E}Y_j) > t \cdot \mathbb{E}Y_1 \right] \leq 2 \exp(-t^2 m^2 \mathbb{E}Y_1).$$

By the Lemma,

$$\mathbb{P}(|W_k - \mathbb{E}W_k| \geq t_k \mathbb{E}W_k) \leq 2 \exp(-cm_k^2 t_k^2 \mathbb{E}W_k) \leq 2 \exp(-\frac{c}{2q} m_k^2 t_k^2).$$

We need

$$2 \sum_{k=1}^v \exp(-c' m_k^2 t_k^2) < \varepsilon.$$

It is enough to choose

$$m_k = \frac{c}{t_k} \log k,$$

Then, with high probability

$$\begin{aligned} |W_k - \mathbb{E}W_k| &\leq t_k \mathbb{E}W_k. \\ \Downarrow \\ \left| \frac{W_k}{\mathbb{E}W_k} - 1 \right| &\leq t_k \\ \Downarrow \\ \left| \sum_{k=1}^v \log \frac{W_k}{\mathbb{E}W_k} \right| &\leq c \cdot \sum_{k=1}^v \left| \frac{W_k}{\mathbb{E}W_k} \right| < c \cdot \sum_{k=1}^v t_k < \varepsilon. \end{aligned}$$

Let us choose t_k so that the last expression is less than ε . We can choose

$$t_k = \frac{C\varepsilon}{k \log^2 k}.$$

With this choice,

$$\left| \sum_{k=1}^v \log \frac{W_k}{\mathbb{E}W_k} \right| < \varepsilon.$$

Combining two estimations, we have (7):

$$\left| \frac{\prod_{k=1}^v W_k}{\prod_{k=1}^v \pi_{S_k}(S_{k-1})} - 1 \right| \leq 2\varepsilon.$$

How many random variables do we need to generate?

$$\begin{aligned} N &= \sum_{k=1}^v m_k C v \log \frac{v}{\varepsilon} = \sum_{k=1}^v \frac{c'}{t_k} \log k v \log \frac{v}{\varepsilon} = \\ &= \sum_{k=1}^v \frac{c}{\varepsilon} k \log^3 k v \log \frac{v}{\varepsilon} \leq \frac{c}{\varepsilon} v^3 \log^4 \frac{v}{\varepsilon}. \end{aligned}$$

The algorithm satisfies the conditions of Jerrum-Sinclair theorem.

LECTURE 24. PERFECT MATCHINGS, BENNETT'S INEQUALITY.

Let $G = (V, E)$ be a graph with $|V| = 2n$ number of vertices.

Definition 43. A perfect matching is a decomposition $V = V_1 \sqcup V_2$ such that $|V_1| = |V_2| = n$ and a bijection $\varphi : V_1 \rightarrow V_2$ such that $\forall v \in V_1$ $\varphi(v) \leftrightarrow v$.

It's also an exponential time problem. For example, consider an $n \times n$ bipartite graph. For a complete graph number of perfect matchings is $n! \sim (\frac{n}{e})^n$ – superexponential.

So, people were trying to find efficient probabilistic algorithm. Gerum, Sinclair, Vigoda – algorithm with Markov chains with complexity $O(n^{10})$.

We will discuss Barvinok's algorithm – no Markov chains, but considerably better time.

Consider adjacency matrix of the graph A ($n \times n$ bipartite graph). Then number of perfect matchings is

$$M(G) = \sum_{\pi \in S_n} \prod_{i=1}^n a_{i, \pi(i)}.$$

It looks almost like determinant (but signs are different). Let W be $n \times n$ random matrix with entries

$$W_{i,j} = g_{i,j} a_{i,j},$$

where $g_{i,j} \sim N(0, 1)$ i.i.d. random variables. Then

$$\det W = \sum_{\pi \in S_n} \text{sign}(\pi) \prod_{i=1}^n g_{i, \pi(i)} a_{i, \pi(i)}.$$

And if we consider square of determinant

$$\mathbb{E}(\det W)^2 = \sum_{\pi, \pi' \in S_n} \text{sign}(\pi) \text{sign}(\pi') \mathbb{E} \prod_{i=1}^n g_{i, \pi(i)} g_{i, \pi'(i)} a_{i, \pi(i)} a_{i, \pi'(i)} = M(G)$$

(note that a term in the sum is nonzero iff $\pi = \pi'$, so all nonzero terms are positive!)

Calculating the determinant requires less than n^3 operations. The question is how far it deviates from the expectation? There is some measure concentration argument that we do not discuss. Answer: with high probability $(\geq 1 - \frac{\epsilon}{n})$

$$e^{-\alpha} M(G) \leq (\det W)^2 \leq 2M(G).$$

And for all reasonable graphs ("expander" graphs) error is considerably less:

$$e^{-\beta n^{1/3}} M(G) \leq (\det W)^2 \leq 2M(G).$$

For the general (not bipartite) graphs any estimate for mixing time is unknown.

Bennett's inequality (60's, stronger version of Bernstein's inequality)

Usual notation:

$$\|a\|_2 = \left(\sum a_k^2\right)^{1/2},$$

$$\|a\|_\infty = \max |a_k|.$$

Theorem 36. *Let $X_1, \dots, X_n$ be i.i.d. random variables, such that*

- (1) $|X_k| \leq 1$ a.s.
- (2) $\mathbb{E}X_k = 0$, $\mathbb{E}X_k^2 = \delta$.

Then for any $a \in \mathbb{R}^n$

$$\mathbb{P}\left(\left|\sum_{j=1}^n a_j X_j\right| > t\right) \leq \begin{cases} 2 \exp\left(-\frac{t^2}{2e\delta\|a\|_2^2}\right), & \text{if } t \leq t_* \\ 2 \exp\left(-\frac{t}{4\|a\|_\infty} \log \frac{t\|a\|_\infty}{\delta\|a\|_2^2}\right), & \text{if } t > t_*, \end{cases}$$

where $\frac{t_*\|a\|_\infty}{\delta\|a\|_2^2} = e$.

Although the tail estimate looks complicated, its shape is natural. We can actually guess it from the classical limit theorems.

For simplicity, consider the case $a_1 = a_2 = \dots = a_n = 1/\sqrt{n}$, so $\|a\|_2 = 1$. If n is large, and δ is not too small, then by the Central Limit Theorem the tail of $\frac{1}{\sqrt{n}} \sum_{j=1}^n X_j$ should be close to that of a normal random variable with expectation 0 and variance δ , which decays as $\exp\left(-\frac{t^2}{2\delta}\right)$.

On the other hand, if δ is very small, the Central Limit Theorem does not provide a good approximation. In such regime, Poissonian Limit Theorem takes over. For example, if $X_1, \dots, X_n$ are i.i.d. centered Bernoulli random variables with variance $\delta = \lambda/n$, then the asymptotics of the tail of $\sum_{j=1}^n X_j$ should be close to a Poisson random variable with parameter λ . The tail of a Poisson random variable

$$\sum_{n \geq t} \frac{e^{-\lambda} \lambda^n}{n!}$$

behaves like the first term, since the terms in the series decay faster than a geometric progression. The order of the first term is

$$\frac{e^{-\lambda} \lambda^t}{t!} \sim \left(\frac{\lambda e}{t}\right)^t \sim \exp\left(-t \log \frac{t}{e\delta n}\right)$$

(we used Stirling's formula here). To compare this to Bennett's inequality, note that in this case, the sequence of coefficients is $a_1 = \dots = a_n = 1$, so $\|a\|_\infty = 1$ and $\|a\|_2 = \sqrt{n}$.

Bennett's inequality asserts that the only possible types of the tail behavior are those resembling the Gaussian and Poissonian tail distribution.

Proof. WLOG, assume that $\|a\|_\infty = 1$. Let $\lambda > 0$. Set

$$\varphi(\lambda) = \mathbb{E}(\exp(\lambda \sum_{j=1}^n a_j X_j)) = \prod_{j=1}^n \mathbb{E}(\exp(\lambda a_j X_j)).$$

For any $x \in \mathbb{R}$,

$$e^x \leq 1 + x + \frac{x^2}{2} e^{|x|}$$

(Taylor's formula with the Lagrange remainder yields $e^x \leq 1 + x + \frac{e^\theta}{2} x^2$ for some $\theta \in (0, x)$.) So, we have

$$\mathbb{E} \exp(\lambda a_j X_j) \leq 1 + \mathbb{E}(\lambda a_j X_j) + \mathbb{E} \left[\frac{\lambda^2 a_j^2 X_j^2}{2} \exp(\lambda |a_j| |X_j|) \right]$$

Then, (using $\mathbb{E} X_k = 0$)

$$\mathbb{E} \exp(\lambda a_j X_j) \leq 1 + \mathbb{E} \left[\frac{\lambda^2 a_j^2 X_j^2}{2} e^\lambda \right] = 1 + \frac{\lambda^2 a_j^2 \delta}{2} e^\lambda.$$

Hence,

$$\begin{aligned} \varphi(\lambda) &\leq \prod_{j=1}^n \left(1 + \frac{\lambda^2 a_j^2 \delta}{2} e^\lambda\right) \leq \prod_{i=1}^n \exp\left(\frac{\lambda^2 a_j^2 \delta}{2} e^\lambda\right) = \\ &= \exp\left(\frac{\lambda^2 e^\lambda \delta}{2} \|a\|_2^2\right). \end{aligned}$$

By Chebyshev's inequality,

$$\mathbb{P}\left(\sum_{j=1}^n a_j X_j > t\right) = \mathbb{P}(e^{\lambda \sum_{j=1}^n a_j X_j} > e^{\lambda t}) \leq \frac{\varphi(\lambda)}{e^{\lambda t}} = \exp\left(-\lambda t + \frac{\lambda^2 e^\lambda \delta}{2} \|a\|_2^2\right).$$

And now we want to choose λ , minimizing the expression.

Case 1. Assume that $0 < \lambda < 1$. Then

$$RHS \leq \exp\left(-\lambda t + \frac{\lambda^2 e^\lambda \delta}{2} \|a\|_2^2\right).$$

This expression attains the minimal value for

$$\lambda_* := \frac{t}{e\delta\|a\|_2^2}.$$

The condition $\lambda_* < 1$ implies the restriction $t < t_*$.

Case 2. If $\lambda > 1$, (use $\lambda < e^\lambda$)

$$RHS \leq \exp(\lambda t + \frac{\lambda e^{2\lambda}\delta}{2}\|a\|_2^2) = \exp(-\lambda[t - \frac{e^{2\lambda}\delta\|a\|_2^2}{2}]).$$

The choice

$$\lambda_* := \frac{1}{2} \log \frac{t}{\delta\|a\|_2^2}$$

gives us

$$RHS \leq \exp(-\lambda_* \frac{t}{2}) = \exp(-\frac{t}{4} \log \frac{t}{\delta\|a\|_2^2}).$$

This proves the estimate for $\mathbb{P}(\sum_{j=1}^n a_j X_j > t)$. To derive the estimate for $\mathbb{P}(|\sum_{j=1}^n a_j X_j| > t)$, we apply the previous result to $\mathbb{P}(\sum_{j=1}^n (-a_j) X_j > t)$. Combining these two bounds results in the coefficient 2 in front of the exponential terms. $\square$

Note that we can modify the condition in Case 1 by requiring $\lambda < \lambda_0$ for some $\lambda_0 > 0$. This will result in different numerical coefficients in the inequality, and a different switching point t_* . Case 2 actually works for any $\lambda > 0$.

LECTURE 25. PERMUTATION SHUFFLING.

Let J be a random variable, uniformly distributed on $\{1, \dots, n-1\}$ (n cards in a deck, J gives position of a card). Let $\sigma \in S_n$ (card order in our deck). If $J = j$, define $f(\sigma, J) = \sigma'$, where

$$\sigma'(k) = \begin{cases} \sigma(j+1), & \text{if } k = j \\ \sigma(j), & \text{if } k = j+1 \\ \sigma(k), & \text{if } k \neq j, j+1 \end{cases}$$

(transposition of 2 consecutive cards). This is a random mapping, add a random variable I , that "decides" whether chain moves according to the rule, or not ($\mathbb{P}(I=0) = \mathbb{P}(I=1) = \frac{1}{2}$). Together they define a lazy Markov chain.

Let $(X_n)_{n \in \mathbb{Z}_+}$ and $(Y_n)_{n \in \mathbb{Z}_+}$ be two chains with the transition probabilities defined by random mapping above, and different initial distributions. We will use the random mapping representation with random variables J_m, I_m , $m \in \mathbb{N}$ to couple these chains.

The random variable I_m defines the behavior of X_m : if $I_m = 1$, X_m moves, otherwise X_m stays. If $J_n = j$, the move consists of swapping the cards in positions j and $j + 1$. Assume that

$$\begin{aligned} X_{m-1}(j) &= u, X_{m-1}(j+1) = v, \\ Y_{m-1}(j) &= u', Y_{m-1}(j+1) = v'. \end{aligned}$$

If $u \neq v'$ and $v \neq u'$, Y_m does the same thing, as X_m , otherwise Y_m does the opposite.

Let $a \in \{1, \dots, n\}$ be a card. Let D_n^a be the distance between the position of a in X_n and in Y_n . Note that our coupling is constructed in such a way that $D_{m+1}^a = 0$, whenever $D_m^a = 0$.

Set

$$\tau_{couple}^a = \inf\{n \in \mathbb{N} : D_n^a = 0\},$$

then

$$\tau_{couple} = \max_a \tau_{couple}^a.$$

Fix a card $a \in \{1, \dots, n\}$. To shorten the notation, we will abbreviate D_m^a as D_m . Suppose that the positions of a card a in the decks X_m and Y_m are k_1 and k_2 .

Case 1. ($k_2 > k_1 + 1$)

$$\text{If } k_1 > 1, \mathbb{P}(D_m = D_{m-1} + 1 | J_m = k_1 - 1) = \frac{1}{2}.$$

$$\text{If } k_2 < n, \mathbb{P}(D_m = D_{m-1} + 1 | J_m = k_2) = \frac{1}{2}.$$

$$\mathbb{P}(D_m = D_{m-1} - 1 | J_m = k_1) = \frac{1}{2}.$$

$$\mathbb{P}(D_m = D_{m-1} - 1 | J_m = k_2 - 1) = \frac{1}{2}.$$

So

$$(*) \begin{cases} \mathbb{P}(D_m = D_{m-1} - 1) = \frac{1}{n-1}, \\ \mathbb{P}(D_m = D_{m-1} + 1 | D_{m-1}) \leq \frac{1}{n-1}, & \text{if } D_{m-1} < n-1, \\ \mathbb{P}(D_m = D_{m-1} + 1 | D_{m-1}) = 0, & \text{otherwise.} \end{cases}$$

Case 2. ($k_2 = k_1 + 1$.)

$$\mathbb{P}(D_m = D_{m-1} + 1 | J_m \in \{k_1 - 1, k_2\}) = \frac{1}{2}, \text{ if } k_1 > 1, k_2 < n,$$

$$\mathbb{P}(D_m = D_{m-1} - 1 | J_m = k_1) = 1.$$

This means that in Case 2 the random process $(D_m)_{m \in \mathbb{N}}$ obeys the same relations (*).

Our aim is to bound the coupling time τ_{couple}^a , which is the time when D_m is absorbed at 0. But D_m is not a random walk (it depends on previous history). So, we define a new random walk $(W_m)_{m \in \mathbb{Z}_+}$ and show that its absorption time bounds the absorption time for D_m .

Let $(W_m)_{m \in \mathbb{Z}_+}$ be a random walk on $\{0, \dots, n-1\}$ with

$$(10) \quad \begin{cases} \mathbb{P}(W_m = W_{m-1} + 1 | W_{m-1}) = \frac{1}{n-1}, & \text{if } W_{m-1} < n-1, \\ \mathbb{P}(W_m = W_{m-1} + 1 | W_{m-1}) = 0, & \text{if } W_{m-1} = n-1, \\ \mathbb{P}(W_m = W_{m-1} - 1 | W_{m-1}) = \frac{1}{n-1}, & \text{if } W_{m-1} > 0, \\ \mathbb{P}(W_m = W_{m-1} - 1 | W_{m-1}) = 0, & \text{if } W_{m-1} = 0. \end{cases}$$

Intuitively, the random process D_m should be absorbed faster than W_m , because the probability of increasing for D_m is smaller, and the probability of decreasing for both processes is the same. To make this intuition precise, we will redefine the process $(W_m)_{m \in \mathbb{Z}_+}$ using the same random variables J_m and I_m , as we used before. This will create a coupling of D_m and W_m .

Set $W_0 = D_0$.

Assume that $k_1 > 1$, $k_2 < n$, $k_2 > k_1 + 1$.

If $J_m \notin \{k_1 - 1, k_1, k_2 - 1, k_2\}$, set $W_m = W_{m-1}$.

If $J_m \in \{k_1 - 1, k_2\}$ and I_m is such that $D_m = D_{m-1} + 1$, set $W_m = W_{m-1} + 1$.

Then $\mathbb{P}(W_m = W_{m-1} + 1 | J_m \in \{k_1 - 1, k_2\}) = \frac{1}{2}$, and so in this case

$$\mathbb{P}(W_m = W_{m-1} + 1 | W_{m-1}) = \frac{1}{n-1},$$

as required in (10).

Similarly, if $J_m \in \{k_1, k_2 - 1\}$ and I_m is such that $D_m = D_{m-1} - 1$, set $W_m = W_{m-1} - 1$.

Then $\mathbb{P}(W_m = W_{m-1} - 1 | J_m \in \{k_1, k_2 - 1\}) = \frac{1}{2}$, and (10) is satisfied again. Note that in this case $W_m - W_{m-1} = D_m - D_{m-1}$.

Assume now that $k_1 = 1$, $k_2 < n$.

Define W_m as above, except for the case $J_m = k_2$. In this case we set $W_m = W_{m-1} + 1$ regardless of the value of I_m . Then

$$\mathbb{P}(W_m = W_{m-1} + 1 | W_{m-1}) = \frac{1}{n-1},$$

and so (10) holds. Also, in this case, $W_m - W_{m-1} \geq D_m - D_{m-1}$.

We can similarly define W_m in the remaining cases

- $k_1 > 1, k_2 = n$;
- $k_1 = 1, k_2 = n$;

- $k_2 = k_1 + 1$.

The process W_m defined this way satisfies the relations (10). Also, since $W_0 = D_0$ and always $W_m - W_{m-1} \geq D_m - D_{m-1}$, we have $D_m \leq W_m$ for all $m \in \mathbb{Z}_+$. Thus, $W_m = 0 \Rightarrow D_m = 0$, so we can estimate true random walk W_m instead of the random process D_m which does not have the Markov property.

Symmetry trick: let's double all the states of the finite chain W_m symmetrically and identify pairs (recall the picture from class).

The new random walk (V_m) has $2n$ states $(\{0, \dots, 2n-1\})$ and two absorbing barriers at 0 and $2n-1$. By definition,

$$\mathbb{P}(W_m = 0) = \mathbb{P}(V_m \in \{0, 2n-1\}).$$

Let $\tau = \inf\{m : W_m = 0\}$. Then (see lecture 20)

$$\mathbb{E}\tau = \frac{(2n-1)^2}{4 \frac{1}{n-1}} \leq n^3.$$

Hence,

$$\mathbb{E}\tau_{couple}^a \leq \mathbb{E}\tau \leq n^3.$$

Therefore, (Chebyshev)

$$\mathbb{P}(\tau_{couple}^a \geq 2n^3) \leq \frac{1}{2}.$$

But if the two walks do not merge after $4n^3$ steps, next $4n^3$ steps will be independent and have the same estimate, so, for any k

$$\mathbb{P}(\tau_{couple}^a \geq k \cdot 2n^3) \leq \left(\frac{1}{2}\right)^k.$$

Set $k = C \log n$, then

$$\mathbb{P}(\tau_{couple}^a \geq 2Cn^3 \log n) \leq \frac{1}{4n}.$$

By definition,

$$\tau_{couple} = \max_a \tau_{couple}^a,$$

$$\mathbb{P}(\tau_{couple} \geq 2Cn^3 \log n) \leq \sum_{a=1}^n \mathbb{P}(\tau_{couple}^a \geq 2Cn^3 \log n) \leq \frac{1}{4}.$$

This means that

$$\tau_{mix} \leq C'n^3 \log n.$$

Surprisingly, although our estimates were quite rough, the final estimation is precise up to a constant.